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Par

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RÉSUMÉ EN FRANÇAIS

Le personnage principal de cette thèse est la sphère de Riemann, qui apparaît sous de nombreux rôles différents : comme base d'un fibré vectoriel, comme fibre d'une surface de Hirzebruch, comme diviseur exceptionnel après un éclatement. La sphère de Riemann, définie précisément à la section 1.1, n'est rien d'autre qu'une sphère dont les points sont tous des nombres complexes, à l'exception du pôle nord, qui est un nouveau point que l'on ajoute artificiellement, appelé ∞ . On peut imaginer fermer le plan complexe sur lui-même, comme un paquet, en ajoutant le point à l'infini au sommet. Maintenant, imaginez que l'on colle un espace vectoriel complexe à chaque point, comme si l'on attachait une feuille de papier à chaque point. Les mathématicien.nes appellent un tel objet un *fibré vectoriel*. Ensuite, imaginez que l'on dessine une flèche sur chaque feuille de papier pour indiquer une direction. On ne peut pas le faire de manière complètement arbitraire : il faut respecter certaines « règles de régularité », ce qui signifie que si deux points sont très très proches, les flèches correspondantes ne peuvent pas pointer dans des directions radicalement différentes. C'est ce que les mathématicien.nes appellent une *section continue* du fibré vectoriel. Dans cette thèse, les règles de régularité que l'on demande sont strictes. Cela réduit la liberté dont nous disposons pour dessiner les flèches, ce qui peut sembler regrettable. D'un autre côté, en nous concentrant sur certaines sections hautement régulières, nous sommes en mesure d'approfondir l'étude. C'est ce que les mathématicien.nes appellent des *sections holomorphes*.

Comme de nombreux problèmes physiques ont été résolus avec succès grâce au calcul différentiel, les mathématiciens continuent de s'intéresser vivement aux dérivées et à l'étude des questions qui s'y rapportent. Je ne fais pas exception, et dans cette thèse, j'étudie une notion particulière de dérivation pour les sections holomorphes. La définition d'une dérivée dans ce contexte est plus subtile que celle introduite au lycée. La sphère de Riemann est courbée, et l'un de ses points est l'infini. Les mathématicien.nes appellent une telle dérivée généralisée une *connexion*. Elle mesure à quel point deux flèches voisines diffèrent l'une de l'autre. Les connexions qui m'intéressent sont appelées *connexions méromorphes*. Celles-ci ne sont pas bien définies en un nombre fini de

points de la sphère, appelés singularités. À mesure que l'on s'approche d'une singularité ou on lui tourne autour, la dérivée n'est plus maîtrisée. Cela produit un comportement caractéristique appelé *monodromie*.

L'objectif de ma thèse est de déformer ces connexions, tout en conservant la monodromie constante tout au long de la déformation.

En termes mathématiques modernes, cette thèse étudie la compactification du feuilletage isomonodromique induit par les solutions de l'équation Painlevé V. Ce feuilletage est un feuilletage analytique singulier de dimension un à l'intérieur de la 3-variété algébrique $\mathbb{C}on_{\Theta}^V$, qui est la compactification de l'espace de modules d'une certaine classe de connexions irrégulières de rang deux sur la sphère de Riemann, appelées connexions PV, à équivalence de jauge méromorphe près (voir la section 1.3.1). Cette relation d'équivalence est naturelle dans notre contexte, puisque les transformations de jauge méromorphes préservent la monodromie. En 1979, K. Okamoto [47] a introduit les espaces algébriques des conditions initiales pour toutes les équations de Painlevé. Ces espaces s'obtiennent par une suite d'éclatements explicites de $\mathbb{P}^1 \times \mathbb{P}^1$ ou de $\mathbb{F}_2$, la deuxième surface de Hirzebruch. Une avancée supplémentaire a été réalisée par P. Boalch [7], qui a étudié, dans un cadre analytique, les espaces de modules de connexions méromorphes irrégulières et leurs déformations isomonodromiques, en mettant l'accent sur leurs structure symplectique. En 2006, M. Inaba, K. Iwasaki et M-H. Saito [29] ont utilisé les résultats d'Okamoto pour étudier la géométrie de l'équation Painlevé VI dans un cadre entièrement algébrique. À l'aide de la théorie des invariants géométriques, ils ont construit l'espace de modules fin de connexions paraboliques stables sur $\mathbb{P}^1$ avec des pôles logarithmiques et ont étudié en profondeur l'application de Riemann-Hilbert.

Soit $t \in \mathbb{C}$ le paramètre temporel de l'équation Painlevé V. L'objectif de cette thèse est d'étendre les résultats antérieurs en étudiant la feuilletage isomonodromique lorsque $t = 0$ ou $t = \infty$.

On peut en effet démontrer qu'il existe une unique classe d'isomorphisme de connexions PV pour tout choix de certains paramètres que l'on appellera $(t, q, \hat{p}) \in (\mathbb{C} \setminus \{0\}) \times (\mathbb{P}^1 \setminus \{0, 1, \infty\}) \times \mathbb{C}$. Cependant, cette construction ne capture pas toutes ces connexions. Néanmoins, cela suggère que l'espace $U := (\mathbb{C} \setminus \{0\}) \times (\mathbb{P}^1 \setminus \{0, 1, \infty\}) \times \mathbb{C}$ fournit un point de départ naturel pour construire l'espace de modules que nous recherchons et sa compactification. Notons que les connexions appartenant à U sont donc précisément les connexions admettant la forme normale 2.1 trouvée par Diarra et Loray [19].

Les connexions avec $q = 0, 1, \infty$ ont été étudiées par Okamoto [47]. La principale nouveauté et motivation de cette thèse réside plutôt dans l'étude des connexions correspondant aux valeurs $t = 0$ et $t = \infty$. Dans tous les cas, nous entreprenons une étude complète de la compactification de l'espace des modules, y compris la construction des composantes de bord d'Okamoto.

Afin de construire et de compactifier l'espace des modules, nous avons besoin d'une interprétation géométrique des paramètres $(t, q, \hat{p})$. Sous l'action d'une transformation de Möbius sur une connexion PV, on note que le paramètre complexe $\hat{p}$ reste inchangé, tandis que le paramètre q se transforme comme un point de $\mathbb{P}^1$, et que t se transforme comme un élément de l'espace tangent de $\mathbb{P}^1$ en $x = 1$.

Nous appelons *courbes irrégulières rationnelles* (voir Section 3.2) des $\mathbb{P}^1$ avec quatre points $(0, 1, \infty, q)$ et un vecteur tangent $t \in T_1\mathbb{P}^1$ marqués ; elles peuvent être considérées comme une généralisation de la sphère à cinq trous où l'un des points est devenu un vecteur tangent. Dans cette perspective, une connexion PV peut être vue comme les données d'une courbe irrégulière rationnelle associées à un paramètre complexe supplémentaire, noté $\hat{p}$.

La première étape consiste donc à construire l'espace de modules, noté $\mathcal{M}$, des courbes irrégulières rationnelles à transformations de Möbius prés. On obtient alors que $U = \mathcal{M} \times \mathbb{C}$ a la structure d'un fibré trivial. Ensuite, nous compactifions $\mathcal{M}$ et étendons cette structure de fibré en droite à la compactification $\overline{\mathcal{M}}$. Enfin, nous devons comprendre comment la feuilletage isomonodromique induit par la cinquième équation de Painlevé se comporte le long des composantes de bord.

Le premier résultat est donc le suivant.

Theorem. *La compactification $\overline{\mathcal{M}}$ de l'espace des modules des courbes rationnelles irrégulières est isomorphe à la surface singulière obtenue par la contraction de l'unique $[-2]$ -courbe dans la surface del Pezzo faible de degré cinq, comme montré dans la Figure 5. De plus, la singularité obtenue est de type A_1 .*

La compactification de $\mathcal{M}$ qu'on obtient est donc singulière. On appellera $\widetilde{\mathcal{M}}$ sa version lisse, qui est la del Pezzo faible de degré 5. On peut alors en déduire l'extension du fibré en droites.

Theorem. *Le fibré triviale $\mathcal{M} \times \mathbb{C}$ s'étend à $\widetilde{\mathcal{M}}$ comme $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)$, où $D \in \text{Div}(\widetilde{\mathcal{M}})$ est le diviseur*

$$D = \text{div}(\hat{p}) \equiv A_0 + 2Q^1 + 2B_0^1 - B_\infty^0 - B_\infty^\infty,$$

où A_0 , Q^1 et B_j^i sont comme dans les Figures 3 and 5. De plus, le fibré en droites $\mathcal{O}_{\widetilde{M}}(D)$ est trivial si on se restreint aux composants de bord Q^i , et à la $[-2]$ -courbe A_∞ .

Grâce à ce résultat et à une étude approfondie des connexions présentes sur les composants de bord, on déduit le théorème suivant.

Theorem. Soit $X := \mathbb{P}(\mathcal{O} \oplus \mathcal{O}_{\widetilde{M}}(D))$ la compactification du fibré en droites, et $\mathcal{W} := \mathbb{P}(\mathcal{O}) \subseteq X$ la section à l'infini. L'espace des modules $\overline{\mathfrak{Con}}_{\Theta}^V$, montré en Figure 6, est construit comme suit: on appelle $\widetilde{X}$ l'éclatement de X le long the special sections in the Q^i s, et $\widetilde{Q}^i$ et $\widetilde{\mathcal{W}}$ les transformés strictes. Alors

$$\overline{\mathfrak{Con}}_{\Theta}^V = \widetilde{X} \setminus \left(\bigcup_{i \in \{0,1,\infty\}} \widetilde{Q}^i \cup \widetilde{\mathcal{W}} \right)$$

De plus, pour tout $t_0 \in \mathbb{C}^*$, l'hypersurface $\overline{\mathfrak{Con}}_{\Theta}^V \cap \{t = t_0\}$ est isomorphe à un espace d'Okamoto.

En étudiant l'extension du feuilletage isomonodromique, on obtient le résultat suivant :

Theorem. On dénote avec $\mathcal{F}^V$ le feuilletage isomonodromique associé à l'équation Painlevé V. Les hypersurfaces $\widetilde{Q}^i$ et $\widetilde{\mathcal{W}}$ sont invariantes pour $\mathcal{F}^V$, et la fonction t est une intégrale première pour le feuilletage en restriction. En plus, le feuilletage $\mathcal{F}^V$ est transverse à la fibration $\mathfrak{Con}_{\Theta}^V \xrightarrow{t} \mathbb{C}^*$ et il est donc un système local

$$(\mathfrak{Con}_{\Theta}^V, \mathcal{F}^V) \xrightarrow{t} \mathbb{C}^*.$$

Les composants de bord $\mathcal{A}_0 \cup \mathcal{B}_0^1$ et $\mathcal{B}_\infty \cup \mathcal{A}_\infty \cup \mathcal{B}_\infty^0$, qui correspondent respectivement aux connexions avec $t = 0$ et $t = \infty$, sont invariants pour $\mathcal{F}^V$. Les valeurs spectrales résiduelles au noeud de la courbe stable irrégulière nodale sont des intégrales premières du champ de vecteurs hamiltonien en restriction aux composants de bord correspondantes. En particulier, le feuilletage isomonodromique est un pinceau de coniques en restriction à $\mathcal{A}_0, \mathcal{B}_0^1, \mathcal{B}_\infty^\infty$ et $\mathcal{B}_\infty^0$, et c'est un pinceau de courbes de genre un en restriction à $\mathcal{A}_\infty$.

INTRODUCTION

The main character of this thesis is the Riemann sphere, which appears in many different roles: as the base of a vector bundle, as a fiber in a Hirzebruch surface, as an exceptional divisor after a blow-up. The Riemann sphere, properly defined in Section 1.1, is nothing more than a sphere whose points are almost all complex numbers, except for the north pole, which is a new point we artificially add, called ∞ . To visualize the Riemann sphere, one may imagine closing the complex plane onto itself, like a bindle, by adding the infinity point at the top.

Now, imagine gluing a complex vector space to each point, as if attaching a sheet of paper at every point. Mathematicians call such an object a *vector bundle*. Next, imagine drawing an arrow on each sheet of paper as to indicate a direction. One can not just do it arbitrarily: we must respect some "regularity rules", meaning that if two points are close, the corresponding arrows cannot point wildly different directions. That is what mathematicians call a *continuous section* of the vector bundle.

In this thesis, the regularity rules are very strict. It reduces the freedom we have in drawing arrows, that may seem sad. On the other hand, by focusing on some highly regular sections we are able to go deeper in the study. These are what mathematicians call *holomorphic sections*.

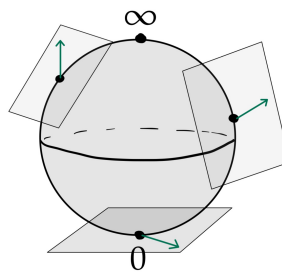


Figure 1 – The Riemann Sphere and a holomorphic section.

Since many physical problems have been successfully solved using differential calculus, mathematicians still retain a strong interest in computing derivatives and studying related questions. I am no exception, and in this thesis I investigate a particular notion of derivation for holomorphic sections. A derivative in this context has a more

subtle definition than the one the reader may have encountered at school. The Riemann sphere is indeed curved, and one of this point is the infinity. Mathematicians call such a generalized derivative a *connection*. It measures how two nearby arrows differ from one another. The particular connections that interest me are called *meromorphic connections*. These are not well defined at finitely many points of the sphere, called *singularities*. As one approaches a singularity or goes round it, the derivative is no more under control. This produces a characteristic behavior that is called *monodromy*.

The goal of my thesis is to deform such connections, while keeping the monodromy constant throughout the deformation.

In modern mathematical terms, this thesis studies the compactification of the isomonodromic foliation induced by the solutions of the fifth Painlevé equation. This foliation is a one-dimensional singular analytic foliation inside the three-dimensional algebraic variety $\overline{\mathcal{CM}}_{\Theta}^V$, which is the compactification of the moduli space of a certain class of rank-two irregular connections on the Riemann sphere, called PV connections, up to meromorphic gauge equivalence (see Section 1.3.1). This equivalence relation is natural in our context, since meromorphic gauge transformations preserve monodromy. In 1979, K. Okamoto [47] introduced the algebraic spaces of initial conditions for all the Painlevé equations. These spaces are obtained through a sequence of explicit blow-ups of $\mathbb{P}^1 \times \mathbb{P}^1$ of $\mathbb{F}_2$, the second Hirzebruch surface. A step further was taken by P. Boalch [7], who studied, in an analytic framework, the moduli spaces of irregular meromorphic connection and their isomonodromic deformations, with an emphasis on symplectic structures. In 2006, M. Inaba, K. Iwasaki, and M-H. Saito [29] used Okamoto's results to study the geometry of the sixth Painlevé equation in a completely algebraic setting. Using geometric invariant theory, they constructed the fine moduli space of stable parabolic connection on $\mathbb{P}^1$ with logarithmic poles and deeply studied the Riemann-Hilbert map.

Let $t \in \mathbb{C}$ denote the time parameter of the fifth Painlevé equation. The goal of this thesis is to extend the previous results by understanding the isomonodromic foliation when $t = 0$ or $t = \infty$. As the normal form (2.1, [19]) shows, t also appears as a parameter of PV connections. Then, in more geometrical terms, the goal is to extend the Okamoto compactification of the moduli space of PV connections by introducing boundary components encoding connections with $t = 0, \infty$, and study the behavior of the isomonodromic foliation along these boundary components.

Our work thus fits into a framework similar to that of H. Chiba [13], and E. Paul and J.P.

Ramis [21], with the difference that they used weighted projective spaces to construct a compact moduli space.

The connections we deform while keeping their monodromy constant are the so-called *PV connections*, introduced in Chapter 2. They are rank-two meromorphic connections on the Riemann sphere with two regular singularities at 0 and ∞ , and one irregular singularity of order two at 1. Local computations (see Example 1.4) show that a necessary condition for monodromy to remain constant along a deformation is that certain *residual spectral data* (see 1.3.2) remain fixed.

Roughly speaking, the residual spectral datum at a singularity is defined as follows. First of all recall that a connection ∇ can be written in a local chart as the differential operator $d + \Omega$, for a suitable matrix of one forms Ω . Consider the power series expansion of this connection matrix near the singularity. Note that it depends on the chosen local trivialization.

$$\Omega = \underbrace{\sum_{j=1}^r A_j \frac{dx}{x^j}}_{\text{Principal part}} + \underbrace{\sum_{i=0}^{\infty} B_i x^i dx}_{\text{Holomorphic part}}.$$

Up to meromorphic gauge transformation, one can suppose the principal part to be diagonal. The residual spectral datum is given by the difference of the eigenvalues of the residual matrix A_1 . Care is required to ensure that this quantity is independent of the chosen trivialization and that it is (up to a mild ambiguity; see Proposition 1.3.8) invariant under meromorphic gauge transformations.

Since we want to study isomonodromic deformations, we will consider PV connections with fixed residual spectral data at the poles. Moreover, we assume these data to be generic in order to avoid pathological phenomena.

In a recent article, Diarra and Loray [19] proved that, up to meromorphic gauge transformations, rank-two meromorphic connections on the Riemann sphere can be written in companion form, and they provided an explicit formula for a dense open subset of them. This *normal form* (see 2.1) yields a convenient representative for each equivalence class under meromorphic gauge transformations, which, as mentioned above, preserve monodromy. In this form, the residual spectral data at 0, 1, and ∞ become explicit, and certain free parameters naturally appear; these parameters denoted $(t, q, \hat{p})$, will play a key role in the construction of the moduli space $\mathcal{C}on_{\Theta}^V$.

During the normalization process, an additional logarithmic singularity appears at a

point $x = q$. By construction, this new logarithmic singularity has trivial monodromy and is therefore called an *apparent singularity*. Its residual matrix has integral eigenvalues, and the eigenvector corresponding to the nonzero eigenvalue depends on a complex parameter $\hat{p}$. Finally, another crucial parameter—traditionally denoted by t , as it represents the time variable in the fifth Painlevé equation—arises in the matrix associated with the irregular singularity.

One can prove that for any choice of $(t, q, \hat{p}) \in (\mathbb{C} \setminus \{0\}) \times (\mathbb{P}^1 \setminus \{0, 1, \infty\}) \times \mathbb{C}$ there exists a unique isomorphism class of PV connections with this complex parameters. However, this construction does not capture all such connections, since the approach of Diarra and Loray mentioned before requires $q \neq 0, 1, \infty$, whereas this condition is not always satisfied (see Section 2.3). Nevertheless, this suggests that the space $U := (\mathbb{C} \setminus \{0\}) \times (\mathbb{P}^1 \setminus \{0, 1, \infty\}) \times \mathbb{C}$ provides a natural starting point for constructing the moduli space we seek and its compactification. Note that connections lying in U are hence precisely the connections admitting the normal form 2.1 found by Diarra and Loray.

Connections with $q = 0, 1, \infty$ have been thoroughly studied by Okamoto [47]. The main novelty and motivation of this thesis is instead the study of connections corresponding to the values $t = 0$ and $t = \infty$. In any case, we undertake a complete study of the moduli space, including the construction of the Okamoto boundary components.

To construct and compactify the moduli space, we need a geometric interpretation of the parameters $(t, q, \hat{p})$. In Lemma 3.1.1 we show that, under the action of a Möbius transformation on the base $\mathbb{P}^1$, the complex parameter $\hat{p}$ remains unchanged; the parameter q , as expected, transforms as a point of $\mathbb{P}^1$; while t transforms as an element of the tangent space of $\mathbb{P}^1$ at $x = 1$.

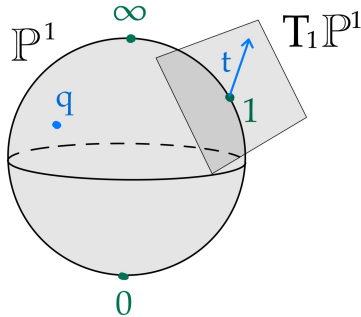


Figure 2 – Rational irregular curve.

We call such objects *rational irregular curves* (see Section 3.2); they may be regarded as a generalization of the five-punctured sphere. In this perspective, a PV connection can be viewed as the data of a rational irregular curve together with an additional complex parameter $\hat{p}$.

The first main step in the construction of the moduli space is therefore to build the moduli space $\mathcal{M}$ of rational irregular curves, up to Moebius transformations. We then obtain that $U = \mathcal{M} \times \mathbb{C}$ has the structure of a trivial holomorphic line bundle. The next step is to compactify $\mathcal{M}$ and extend this line bundle structure to the compactification $\overline{\mathcal{M}}$. The final step is to understand how the isomonodromic foliation induced by the fifth Painlevé equation behaves along the boundary components.

We resume in the following sections the main ideas, tools and results that we use for any of the steps we mentioned before.

Compactification of The Moduli Space of Irregular Rational Curves

We have already mentioned that the moduli space of smooth irregular rational curves, up to Möbius transformations, is $\mathcal{M} = (\mathbb{C} \setminus \{0\}) \times (\mathbb{P}^1 \setminus \{0, 1, \infty\})$ as illustrated in Figure 3. A geometric explanation of this description relies on the 3-transitivity of the automorphism group of $\mathbb{P}^1$, which allows any three distinct points to be sent to 0, 1, and ∞ and then the free parameters left are the point q and the tangent vector t .

In Figure 3 we show how $\mathcal{M}$ looks like. We also give a name to the omitted lines, they will play a central role in the compactification process.

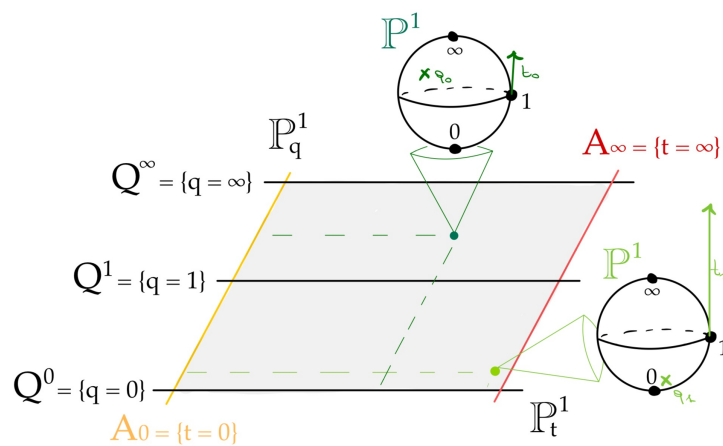


Figure 3 – The moduli space $\mathcal{M} \subseteq \mathbb{P}^1 \times \mathbb{P}^1$.

Compactifying $\mathcal{M}$ amounts to understanding what happens when q approaches

one of the other marked points, or when t tends to 0 or ∞ . The answer turns out to be subtler than one might initially expect.

Following the idea of M. Kapranov [32], we interpret an irregular rational curve as a parametrized smooth conic in $\mathbb{P}^2$ passing through four fixed points in general position and tangent at one of them with a prescribed slope, also in general position—meaning that the chosen tangent direction does not pass through any of the other marked points. The classical result that there exists exactly one smooth conic in $\mathbb{P}^2$ passing through five points in general position naturally extends to the situation in which one point is replaced by a tangent vector. Morally, it corresponds to choosing the fifth point infinitesimally close to one of the others. Removing one of the "simple" points (not the "double" point decorated with the tangent vector) allows us to view the family of all irregular rational curves as a pencil of marked conics in $\mathbb{P}^2$, considering the *markings* as the four common points/directions and the choice of the fifth one. Note that this implies that a single conic may represent several different marked conics. They are indeed not equivalent because the action of the automorphism group of $\mathbb{P}^2$ is four-transitive. This yields an alternative description, as in Figure 4, of the moduli space of irregular rational curves. Since the marking must be chosen in general position with the other four points, it is the complement of four lines in $\mathbb{P}^2$, as also shown in Figure 4.

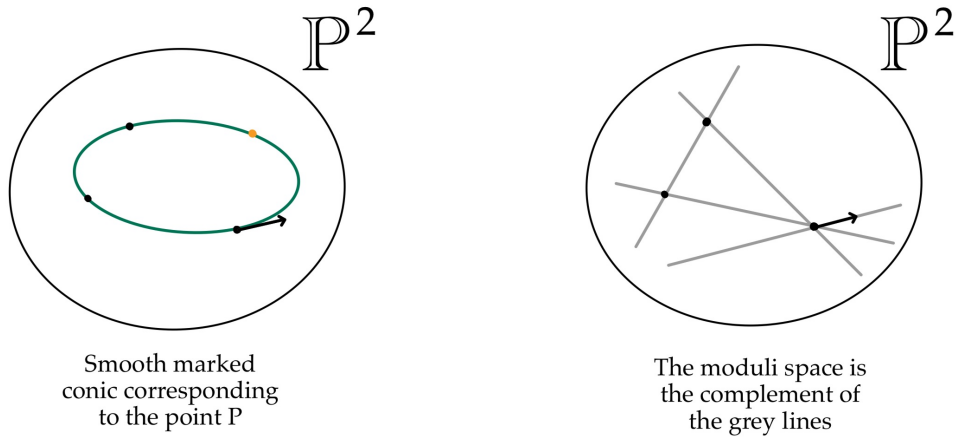


Figure 4 – The moduli space modeled on $\mathbb{P}^2$.

Starting from a birational morphism $\mathbb{P}^1 \times \mathbb{P}^1 \dashrightarrow \mathbb{P}^2$, in Section 3.2.1 an explicit isomorphism between these two descriptions is constructed. The original compactification problem then can be translated in $\mathbb{P}^2$, where one must determine which kind of conics corresponds to the choice of a fifth point that is not in general position with the others.

It is quite easy to see (Section 3.2.2) that picking such a marking gives a singular conic, that is, in a Deligne-Mumford [17] flavor, an irregular stable nodal curve.

The next step is to analyze what happens when the marking collapses into one of the other points. Here, Identifying as before very close points with the datum of a tangent direction, turns out to be useful: when the fifth point collapses into an another, then it indicates a prescribed tangent direction at this point. The model we get comes therefore out by blowing up once $\mathbb{P}^2$ in the three simple points and twice the point decorated with the tangent direction. We thus arrive at the following result.

Theorem A. *The compactification $\overline{\mathcal{M}}$ of the moduli space of rational irregular curves is isomorphic to the singular surface obtained by the contraction of the unique $[-2]$ -curve in the weak del Pezzo surface of degree five, as shown in Figure 5. Moreover the singularity obtained is of A_1 -type.*

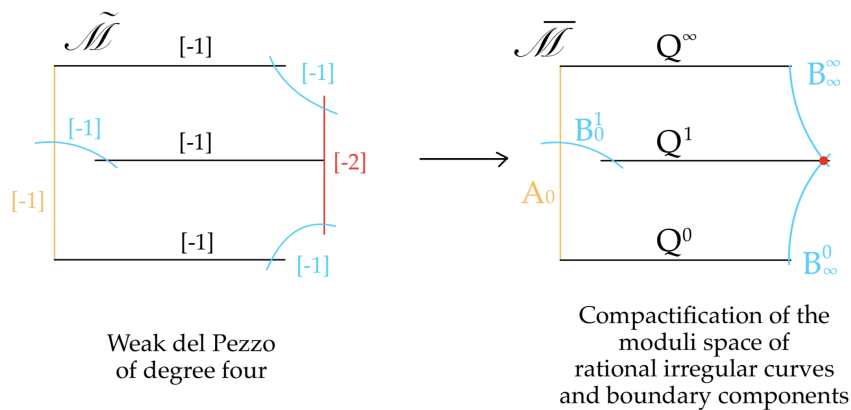


Figure 5 – The moduli space $\overline{\mathcal{M}}$ and its boundary components.

This result marks a significant difference with the compactification of the space $\mathcal{M}_{0,5}$ that is smooth.

We conclude with a description of the universal curve, defining irregular stable nodal curves. The main idea is the following: the confluence of q with another point, say 0, is just a particular viewpoint on the story. Since we are considering these curves up to automorphisms of $\mathbb{P}^1$, we could apply a Moebius transformation sending q far away from 0 and see what happens to the other points. It turns out that they collapse together. In other words, the confluence of q to 0 is Moebius equivalent to the confluence of 1 to ∞ and t to 0, as precisely shown in Section 3.2.3.

The Compactification of the Moduli Space of PV connections

We have already introduced $U := (\mathbb{C} \setminus \{0\}) \times (\mathbb{P}^1 \setminus \{0, 1, \infty\}) \times \mathbb{C}$, the parameter space describing almost all PV connections. We can now interpret it as $\mathcal{M} \times \mathbb{C}$, meaning that such connections correspond to (smooth) rational irregular curves together with an extra complex parameter $\hat{p} \in \mathbb{C}$. This construction endows the space U with the structure of the trivial line bundle $\mathcal{M} \times \mathbb{C}$, with global coordinates $(t, q, \hat{p})$.

The first step to compactify U and describe all PV connections, is to investigate how this line bundle extends over $\overline{\mathcal{M}}$. At this point, a choice is required. Since $\overline{\mathcal{M}}$ is singular, we instead work with $\widetilde{\mathcal{M}}$, the smooth surface shown on the left in Figure 5, before contracting the $[-2]$ -curve. It is indeed simpler to consider line bundles on smooth manifolds.

To study the geometry of the line bundle extension and its compactification, we first analyze how the connections lying on the boundary arise. We have a powerful tool at our disposal: the normal form 2.1 allows us to compute explicitly the limits for $q \rightarrow 0, 1, \infty$ and $t \rightarrow 0, \infty$. Connections on the boundary turn out to be connections on the corresponding stable nodal curve. In particular, such a connection splits into one connection on each smooth component, each of which has a pole in the node. In Section 3.3.1, we explicitly compute the limits, and we observe that, in order for them to converge, we are required to modify the parameter $\hat{p}$, which is the coordinate on the fibers.

Next, to understand the geometry of the extension of the line bundle over $\widetilde{\mathcal{M}}$, we analyze the poles and the zeros of a generic section. More precisely, we consider the global trivialising section $\hat{p} = 1$ of $\mathcal{M} \times \mathbb{C}$ and examine the behavior of its extension along the boundary components, as explained in Section 3.3.1. We obtain new local expressions for $\hat{p}$, that lead to the following result.

Theorem B. *The trivial line bundle $\mathcal{M} \times \mathbb{C}$ extends to $\widetilde{\mathcal{M}}$ as $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)$, where $D \in \text{Div}(\widetilde{\mathcal{M}})$ is the divisor*

$$D = \text{div}(\hat{p}) \equiv A_0 + 2Q^1 + 2B_0^1 - B_\infty^0 - B_\infty^\infty,$$

where A_0 , Q^1 and B_j^i are as in Figures 3 and 5. Moreover, the line bundle $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)$ is trivial when restricted to the boundary components Q^i s, and to the $[-2]$ -curve A_∞ .

The triviality of the line bundle over the Q^i s allows us to define some special global holomorphic sections whose equations are determined by the residual spectral data of the connection. For instance, denoting by P^0 the vertical coordinate on

$Q^0 := \mathcal{O}_{\widetilde{M}}(D)|_{Q^0} \cong Q^0 \times \mathbb{C}$, we have two special sections $P^0 = 0$ and $P^0 = \kappa_0$, where κ_0 is the residual spectral datum of the connections at the simple pole $x = 0$. We can do the same at Q^∞ , while for Q^1 the double pole imposes to be more careful.

Theorem C. *Let $X := \mathbb{P}(\mathcal{O} \oplus \mathcal{O}_{\widetilde{M}}(D))$ be the compactification of the line bundle found in Theorem B, and $\mathcal{W} := \mathbb{P}(\mathcal{O}) \subseteq X$ the infinity section. The moduli space $\overline{\mathfrak{Con}}_\Theta^V$, shown in Figure 6, is constructed as follows: denote by $\widetilde{X}$ the blow-up of X along the special sections in the Q^i 's, and by $\widetilde{Q}^i$'s and $\widetilde{\mathcal{W}}$ the strict transforms. Then*

$$\overline{\mathfrak{Con}}_\Theta^V = \widetilde{X} \setminus \left(\bigcup_{i \in \{0,1,\infty\}} \widetilde{Q}^i \cup \widetilde{\mathcal{W}} \right)$$

Moreover, for any $t_0 \in \mathbb{C}^*$, the hypersurface $\overline{\mathfrak{Con}}_\Theta^V \cap \{t = t_0\}$ is isomorphic to an Okamoto space.

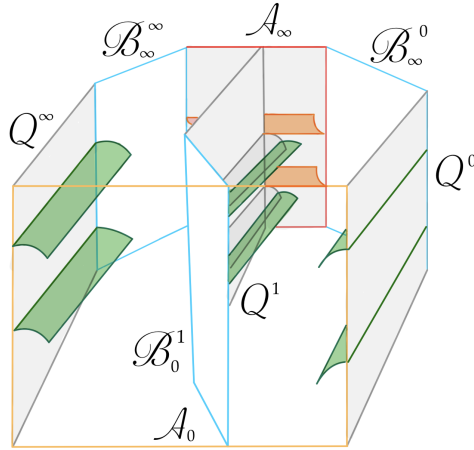


Figure 6 – The Compactification of the Moduli Space of PV Connections.

We hence find back the well known fact that the moduli space of PV connections is a family of Okamoto spaces indexed by $t \in \mathbb{C}^*$, that is

$$\mathfrak{Con}_\Theta^V = \bigcup_{t \in \mathbb{C}^*} \text{Ok}_t,$$

inducing a fibration $\overline{\mathfrak{Con}}_\Theta^V \xrightarrow{t} \mathbb{C}^*$ whose fibers are the Okamoto spaces.

What we add in $\overline{\mathfrak{Con}}_\Theta^V$ is the description of what happens for $t = 0, \infty$, that is, for the boundary components $\mathcal{A}_0 \cup \mathcal{B}_0^1$ and $\mathcal{B}_\infty \cup \mathcal{A}_\infty \cup \mathcal{B}_\infty^0$, as explained in Theorem 3.3.19.

The Isomonodromic Foliation

The fifth Painlevé equation 4.1 can be expressed in the variables $(t, q, \hat{p})$, which are the global coordinates of $\mathcal{M} \times \mathbb{C}$. In Section 4.2.1, we study the global Hamiltonian vector field, as it appears in the work of Y. Ohyama [46], whose integral curves are precisely the solutions of the fifth Painlevé equation, or, equivalently, the fibers of the Riemann-Hilbert map. Since the geometric description of $\overline{\text{Con}}_{\Theta}^V$ comes with a precise system of coordinates, we are able to explicitly express the extension of the vector field to the whole compactification, leading to the following result, already known by Okamoto [47].

Theorem D. *Let us denote by $\mathcal{F}^V$ the isomonodromic foliation induced by the fifth Painlevé equation. The hypersurfaces $\widetilde{Q}^i$'s and $\widetilde{W}$ are invariant for $\mathcal{F}^V$, and the function t is a first integral for the restricted foliation. Moreover, the foliation $\mathcal{F}^V$ is transverse to the fibration $\text{Con}_{\Theta}^V \xrightarrow{t} \mathbb{C}^*$ and makes it into a local system*

$$(\text{Con}_{\Theta}^V, \mathcal{F}^V) \xrightarrow{t} \mathbb{C}^*.$$

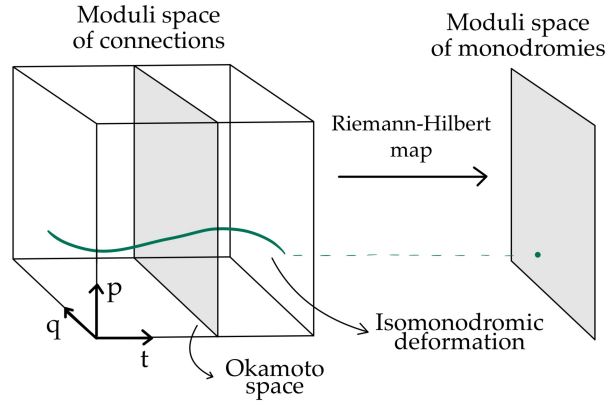


Figure 7 – Isomonodromic Deformation.

Since the Okamoto spaces coincide with the spaces of initial conditions for the fifth Painlevé equation, Theorem D is a geometric counterpart of the Painlevé property.

To describe the integral curves and understand the isomonodromic deformations we need to determine some first integrals of the Hamiltonian vector field. Consider performing the limit for $t \rightarrow 0, \infty$. During the deformation of the connection, the ra-

tional irregular curve is also deformed until it becomes nodal. We already mentioned that the node becomes a pole of the limit connection.

Theorem E. *The boundary components $\mathcal{A}_0 \cup \mathcal{B}_0^1$ and $\mathcal{B}_\infty^\infty \cup \mathcal{A}_\infty \cup \mathcal{B}_\infty^0$, respectively corresponding to connections with $t = 0$ and $t = \infty$, are invariant for $\mathcal{F}^V$. Moreover, residual spectral data at the nodes of the irregular stable nodal curves are first integrals of the hamiltonian vector field restricted to the corresponding boundary component. In particular the isomonodromic foliation is a pencil of conics when restricted to $\mathcal{A}_0, \mathcal{B}_0^1, \mathcal{B}_\infty^\infty$ and $\mathcal{B}_\infty^0$, while it is a pencil of genus one curves when restricted to $\mathcal{A}_\infty$.*

In Section 4.2 we describe the vector fields and the first integrals appearing in any boundary component.

Example. Let us consider the boundary component $\mathcal{B}_\infty^0$. It corresponds to the locus of irregular stable nodal curves with $q = 0$ and $t = \infty$, where q and t are as in Figure 2. For simplicity, we represent $\mathbb{P}^1$ as a line. The corresponding irregular stable curves then appear as in Figure 8, for different values of $R \in \mathbb{P}^1$.

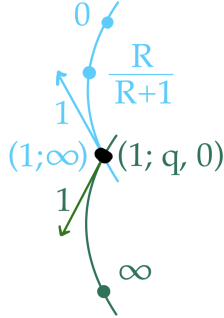


Figure 8 – Irregular stable nodal curve in $\mathcal{B}_\infty^0$.

The limit connection then splits as a confluent hypergeometric equation in the green smooth component and in a confluent Heun equation in the blue one. Consider local coordinates (T, R, P^0) of the moduli space $\overline{\mathcal{C}on}_\Theta^V$ centered in $\mathcal{B}_\infty^0 \cap \mathcal{Q}^0$. In these coordinates, $\mathcal{B}_\infty^0$ corresponds to the hypersurface $\{T = 0\}$, where $T = 1/t$. The residual spectral datum in the node is

$$2P^0 - \kappa_0 - 1 + \frac{2P^0(P^0 - \kappa_0)}{R},$$

and its level sets represent the pencil of conics we mentioned in Theorem E.

Plan of the Thesis

In Chapter 1 we collect all the notions that are needed to understand what follows. In Sections 1.1 and 1.2, particular care is taken to define the Riemann sphere and some basic theory of holomorphic vector bundles of rank two. Next, in Section 1.3, we introduce meromorphic connections paying attention to precisely define the residual spectral data and their behavior under meromorphic gauge transformations. With all this machinery, we can give a proof of the regular Riemann-Hilbert correspondence in Section 1.4. We conclude this section by presenting the example of the Euler systems, the simplest meromorphic connections possible.

We introduce PV connections, the main characters of this work, in Chapter 2. We introduce their projective version, known as Riccati foliations. It will be very helpful in providing a geometrical interpretation of the residual spectral data. In Section 2.2, we describe the normal form, in which the parameters $(t, q, \hat{p})$ explicitly appear. Finally, in Section 2.3 we put together the information we have gathered to give a geometric interpretation also of the parameters $(t, q, \hat{p})$ and to define the Okamoto space.

Next, in Chapter 3, the construction of the moduli space begins. In Sections 3.1 and 3.2 we consider the space $\mathcal{M} \times \mathbb{C}$. In particular, we properly define irregular curves and their stable nodal version in order to precisely describe the compactification of their moduli space and the universal curve. We have now all the necessary knowledge to study in Section 3.3 the limits of connections and we compactify the moduli space of PV connections.

Finally, the compactification of the isomonodromic foliation is studied in Chapter 4. Sections 4.1.1, 4.1.2 and 4.1 are devoted to explaining the local and global monodromy of a PV connection, and to stating a generalized Riemann-Hilbert correspondence for irregular connections. In Section 4.2 we describe the fifth Painlevé equation and its parameters to deduce, in Section 4.2.1, the Hamiltonian vector field defining the isomonodromic leaves. In the last Section 4.2.2, we study the integral curves and the first integrals given by the residual spectral data at the node.

STORY OF THE PROBLEM

Riemann-Hilbert Problems

The great ancestor of this thesis is the twenty-first Hilbert problem. In 1900 David Hilbert wrote a list of twenty-three open problems and several proved to very influential for 20th-century mathematics. Some of them have been relatively fast solved, while some of them, as the famous Riemann hypothesis, still remain unsolved nowadays. Lots of literature have been written about these problem, their alternative and more precise formulations, their solutions and the further questions that raised in the twenty century. We will briefly report some parts of the long and fascinating story of the twenty-first problem. Indeed, one of the branches of the genealogy tree that see this problem as founder of the dynasty, leads, way afar from its origins, to the research of my thesis. For this historical survey we will refer in particular to the articles of N. M. Katz ([34], 1976), T. Bothner ([9], 2021) and X. Tang ([51], 2024).

The original formulation of twenty-first problem in its english translation of 1902 is the following:

Hilbert's 21st Problem. *Proof of the existence of linear differential equations having a prescribed monodromic group.*

In the theory of linear differential equations with one independent variable z , I wish to indicate an important problem one which very likely Riemann himself may have had in mind. This problem is as follows: To show that there always exists a linear differential equation of the Fuchsian class, with given singular points and monodromic group. The problem requires the production of n functions of the variable z , regular throughout the complex z -plane except at the given singular points; at these points the functions may become infinite of only finite order, and when z describes circuits about these points the functions shall undergo the prescribed linear substitutions. The existence of such differential equations has been shown to be probable by counting the constants, but the rigorous proof has been obtained up to this time only in the particular case where the fundamental equations of the given substitutions have roots all of absolute magnitude unity. L. Schlesinger (1895) has given this proof, based upon Poincaré's theory of the Fuchsian zeta-functions. The theory of linear differential equations would evidently have a more finished

appearance if the problem here sketched could be disposed of by some perfectly general method.

The statement of this problem, as written by Hilbert, can lead to different interpretations and some more precise formulations are required in order to state if the problem admits or not a positive answer.

What Hilbert calls *linear differential equation* is what we should call today a p th order scalar linear ODE of the form

$$a_0(z)y + a_1(z)y' + \cdots + a_p(z)y^{(p)} = 0$$

where $a_0, \dots, a_p$ are complex meromorphic functions. Note that such equation is equivalent to a $p \times p$ linear ODE system of the form

$$d \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{p-1} \\ y_p \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0(z) & -a_1(z) & -a_2(z) & \cdots & a_p(z) \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{p-1} \\ y_p \end{pmatrix}$$

which the vector $(y, y', \dots, y^{(p)})$ solves. Still, at the time of the congress in Paris of 1900, in which Hilbert presented some of these problems, it was known the negative answer of the twenty-first problem for scalar equations unless additional singularities are included. So, by a *linear differential equation* Hilbert likely meant a generic $p \times p$ linear ODE system. This first confusion lead to some great consequences. Indeed the notion of Fuchsian singularity is equivalent to the notion of regular singularity for scalar equations (Fuchs, 1868), but it is not the case for linear systems. In this latter case all Fuchsian singularities are always regular, while the converse statement is in general false. We recall that, for a linear ODE system, a regular singular point is characterized by the fact that the growth of solutions is bounded by an algebraic function, while a Fuchsian singularity is by definition a pole of order one. The main reason why the two definition do not coincide for systems is encoded by meromorphic gauge transformations. By a meromorphic gauge transformation, as we will precisely see in Section 1.3.1, we can increase the order of a pole without changing nor the solutions nor the monodromy. Scalar linear ODE are on the contrary invariant by the action of meromorphic gauge transformations. Therefore there are three different possible more precise

reformulation of the problem as stated in [9].

Riemann-Hilbert Problem. *Given a monodromy group with encoded singularity locations, are we asked to realize it by*

(RHP1) *a Fuchsian linear p th order differential equation?*

(RHP2) *a linear $p \times p$ system having only regular singularities?*

(RHP3) *a Fuchsian system on the whole Riemann sphere $\mathbb{P}^1$?*

But... what is the monodromy of a differential equation? In few words, the monodromy measure how the solutions of the differential system (or equation) evolves under analytic continuation along smooth path. Indeed, if we imagine a solution $Y(z)$ of a holomorphic differential equation over a simply connected complex curve C (as, for instance, $\mathbb{C}$ or $\mathbb{P}^1$, as in the Riemann-Hilbert problems) we expect that its analytic continuation along a loop $\gamma: [0, 1] \rightarrow C$ behaves in such a way that $Y(\gamma(0)) = Y(\gamma(1))$. Well, this is in general not the case when the curve C presents a non-trivial fundamental group and/or when the differential system presents a singularity encircled by the loop γ . In this latter case, the (local) monodromy manifests as follows:

$$Y(\gamma(0)) = Y(\gamma(1))M^\gamma$$

for a constant complex invertible matrix M^γ . While compute the local monodromy around a singularity is quite easy, to compute the global monodromy is way harder as the loops can encircle different singularities. The curious and impatient reader should go to see the examples in Section 1.5 in which the most elementary differential systems on the Riemann sphere are treated, and in particular we compute their global monodromy representation.

As we already mentioned, the answer to RHP1 is negative. In 1908 Plemelj solved RHP2 and believed to settle RHP3 as well. In the followings years Kohn and Arnold, Il'yashenko discovered a serious gap in the Plemelj's proof. Here follows a timeline of the attempts of solution:

- (1956) Krylov explicitly solved RHP3 for 2×2 systems and three singular points. His work made crucial use of Gauss hypergeometric functions.
- (1979) Dekkers solves RHP3 for 2×2 systems and arbitrary singularities.
- (1983) Kohn solved RHP3 for any rank and singularities supposing that at least one of the generators of the monodromy group is semisimple.

- (1989) Bolibruch find the counterexample to the general case for rank three systems
- (1992) Kostov and Bolibruch show that RHP3 has a positive solution when the monodromy representation is irreducible.

In the meanwhile, in 1957, an another generalization of this problem was settled by Röhrl. He was the first one to study the problem on the viewpoint of fiber bundles over Riemann surfaces and he showed a one-to-one correspondence between differential systems on $\mathbb{P}^1$ with prescribed monodromy representation and singular points, and the set of holomorphic section of a particular principal bundle. This inspired the famous and fundamental work [15] of Deligne in 1970 in which Riemann-Hilbert correspondence is shown. Its original formulation in French is the following:

Theorem (Théorème 2.23 of [15]). *Soit $f : X \rightarrow S$ un morphisme lisse d'espaces analytiques complexes.*

- (i) *Pour tout système local relatif V sur X , il existe sur le faisceau analytique cohérent $\mathcal{V} = \mathcal{O}_X \otimes_{f^* \mathcal{O}_S} V$ une et une seule connexion relative, dite canonique, telle qu'une section locale v de $\mathcal{V}$ soit horizontale ($\nabla v = 0$) si et seulement si v est une section de V ; cette connexion est intégrable.*
- (ii) *Etant donné un système local relatif V sur X , le complexe de De Rham défini par $\mathcal{V} = \mathcal{O}_X \otimes_{f^* \mathcal{O}_S} V$, muni de sa connexion canonique, est une résolution du faisceau V .*
- (iii) *Les foncteurs suivantes*
 - (a) V (système relatif) $\mapsto \mathcal{V} = \mathcal{O}_X \otimes_{f^* \mathcal{O}_S} V$, muni de sa connexion canonique,
 - (b) $\mathcal{V}$, faisceau analytique cohérent sur X , muni d'une connexion intégrable relative $\mapsto$ le sous faisceau de ses sections horizontales ($\nabla v = 0$)

sont des équivalences de catégories quasi-inverses l'une de l'autre entre la catégorie des systèmes locaux relatifs sur X et la catégorie des faisceaux analytiques cohérents sur X , munis d'une connexion intégrable relative.

In 1984 the Japanese mathematician M. Kashiwara showed in [33] a even more generalized Riemann-Hilbert correspondence, proving an equivalence between the category of regular holonomic D -modules on an analytic complex variety X and the category of perverse sheaves on X . In 2013, Kashiwara and A. D'Agnolo [14] were able to drop the regularity condition, enhancing the target category.

Painlevé Transcendents

A differential equation $f(x, y, y', \dots, y^{(n)}) = 0$ on the complex plane is said to have the Painlevé property (PP) if there exists a finite subset $\Sigma \subseteq \mathbb{C}$ such that, for any local solution $y(x)$ analytic in a neighborhood of $x_0 \in \mathbb{C} \setminus \Sigma$, and for any curve $\gamma \subset \mathbb{C} \setminus \Sigma$ based in x_0 , the function $y(x)$ admits a meromorphic continuation along γ . An equivalent definition is that a differential equation has the PP if its general solution has no movable critical singularities, where a point is said to be movable if its position in $\mathbb{C}$ depends on the initial conditions.

Before the PP as precisely defined, in the end of the XIX century, H. Poincaré and L. Fuchs proved that if $P(x, y, y') = 0$ is a polynomial differential equation satisfying the PP then the equation is either linear, a Riccati equation, a Weierstrass equation, or else all its solutions are algebraic.

In the beginning of the XX century, Paul Painlevé attempted to classify second order equations with the PP of the form $y'' = R(x, y, y')$, where R is rational in y' , algebraic in y and analytic in x . His goal was to discover new transcendents (new special functions that arise in the solution of differential equations). After an extensive study, unfortunately marred by some serious calculation errors, Painlevé identified five families of differential equations satisfying the PP. He forgot the famous Painlevé VI equation, which was added to the list a few years later, in 1910, by B. Gambier, a student of Painlevé, who correcting the errors in Painlevé's work. Independently of Painlevé and Gambier, equation Painleve VI was found in 1905 by R. Fuchs (son of L. Fuchs) [23] from completely different considerations: he studied isomonodromic deformations of linear differential equations with regular singularities.

Isomonodromic Deformations

The work of Fuchs inspired L. Schlesinger [50] et R. Garnier [24] whose works in 1912 deeply studied isomonodromic deformations problems. Their work was forgotten for a long time until, in 1980, Jimbo, Miwa and Ueno [31] published their famous article relating Painlevé equations and isomonodromic deformations of linear ordinary differential equations. The interested reader should look at the introduction of their first article, in which the contributions of Fuchs, Schlesinger and Garnier are detailed.

Two years later, B. Malgrange [43] published a study on the isomonodromic deformations of Fuchsian connections on the Riemann sphere. His greatest contribution was

the construction of the maximal deformation.

In the early 2000s, various contributions of P. Boalch [4, 7] provided an interesting geometrization of the problem. In particular, an analytic (not algebraic) moduli space of irregular connections was constructed, allowing one to interpret isomonodromic deformations as a foliation. A few years later, Inaba, Iwasaki and Saito [29] gave an algebraic version of the moduli space of logarithmic connections, thus providing a geometric interpretation of the Painlevé property in terms of the isomonodromic foliation. The long introduction to their first article is an excellent survey of their work.

PRELIMINARY NOTIONS

The main character of this thesis are PV connections: a class of rank 2 irregular connections on the Riemann sphere. The goal of this first introductory section is to make the reader comfortable with the elements of complex and analytic geometry needed to make friends with PV connections. The contents of this section are quite classical, but we suggest to pay a particular attention to Section 1.3.2 because residual spectral data play a crucial role in Chapter 2 and 3.

1.1 Geometry of $\mathbb{P}^1$ and Moebius Transformations

There are several equivalent way in literature to define the Riemann sphere and any of them brings its advantages. Let, for the moment, call X the Riemann sphere.

Definition (Riemann sphere, Forster, [22]). Let $X := \mathbb{C} \cup \infty$, where ∞ is a symbol not contained in $\mathbb{C}$. Introduce the following topology on X . The open sets are the usual open sets $U \subseteq \mathbb{C}$ together with sets of the form $V \cup \{\infty\}$, where $V \subseteq \mathbb{C}$ is the complement of a compact set $K \subseteq \mathbb{C}$. With this topology X is a compact Hausdorff topological space, homeomorphic to the 2-sphere S^2 .

We find sometimes in the literature also the symbols $\overline{\mathbb{C}}$ or $\hat{\mathbb{C}}$ to denote the Riemann sphere with such definition. They make reference to the Alexandroff compactification of $\mathbb{C}$ by adding the point at infinity.

For the moment we have just defined a topological space, but we would like to give him the structure of a complex curve: set

$$U_0 := X \setminus \{\infty\} = \mathbb{C} \quad \text{and} \quad U_\infty := X \setminus \{0\} = \mathbb{C}^* \cup \{\infty\}.$$

Define maps $\varphi_i: U_i \rightarrow \mathbb{C}, i = 0, \infty$, as follows. φ_0 is the identity map and

$$\varphi_\infty(x) = \begin{cases} \frac{1}{x} & \text{if } x \in \mathbb{C}^* \\ 0 & \text{if } x = \infty \end{cases}$$

Clearly these maps are homeomorphisms and thus X is a two-dimensional topological manifold. Since U_0 and U_∞ are connected and have non-empty intersection, X is also connected.

The complex structure on X is now defined by the atlas consisting of the charts $\varphi_i: U_i \rightarrow \mathbb{C}, i = 0, \infty$. We must show that the two charts are holomorphically compatible. But $\varphi_0(U_0 \cap U_\infty) = \varphi_\infty(U_0 \cap U_\infty) = \mathbb{C}^*$ and

$$\varphi_\infty \circ \varphi_0^{-1}: \mathbb{C}^* \rightarrow \mathbb{C}^* \quad x \mapsto z = \frac{1}{x}$$

is biholomorphic.

Definition. The *complex projective line* $\mathbb{P}_\mathbb{C}^1$ is

$$\mathbb{P}_\mathbb{C}^1 := (\mathbb{C}^2 \setminus \{0\})/\sim,$$

where $u \sim v$ if, and only if, there exists $\lambda \in \mathbb{C}^*$ such that $u = \lambda v$.

Lemma 1.1.1. *The Riemann sphere X is biholomorphic to the complex projective line $\mathbb{P}_\mathbb{C}^1$.*

Sketch of proof. We can use the biholomorphism $F: \mathbb{C} \cup \{\infty\} \rightarrow \mathbb{P}_\mathbb{C}^1$ given by

$$F(x) = \begin{cases} [x : 1] & \text{if } x \in \mathbb{C} \\ [1 : 0] & \text{if } x = \infty \end{cases}.$$

□

From now on we will denote the Riemann sphere by $\mathbb{P}^1$ (omitting the underscore $\mathbb{C}$). The reader should keep in mind both the interpretations as the compactification $\mathbb{C} \cup \{\infty\}$ and as the projective line.

To better understand the geometry of $\mathbb{P}^1$ we need to study its automorphism group, that we will use in the following of the text.

Proposition 1.1.2. *The automorphism group of the Riemann sphere is $\text{Aut}(\mathbb{P}^1) \cong \text{PSL}_2(\mathbb{C})$.*

Proof. Let $f \in \text{Aut}(\mathbb{P}^1)$ such that $f(\infty) \neq \infty$. The function $f_0 := \varphi_0 \circ f \circ \varphi_0^{-1}: \mathbb{C} \rightarrow \mathbb{C}$ must be meromorphic since, by surjectivity, there must be at least an element $d \in \mathbb{C}$ such that $f_0(d) = \infty$. By the injectivity of f , the pole d is unique, moreover the pole is simple since f must be invertible. The function then globally writes as $f_0 = g_0/(x - d)$ for a holomorphic function g_0 such that $g_0(d) \neq 0$. By hypothesis we asked that $f(\infty) \neq \infty$, then we have that $\lim_{x \rightarrow \infty} g_0(x)/(x - d) \neq \infty$ that imposes that the degree of g_0 is less or equal than one, that is $g_0(x) = ax + b$ is linear. In particular it holds that any such f can be chosen the form

$$f(x) = \frac{ax + b}{x - d}.$$

If, using the same notations, we have $f(\infty) = \infty$, then f_0 must be a holomorphic automorphism of $\mathbb{C}$, and then of the form $f_0 = ax + b$. In particular, also f is of the form

$$f(x) = ax + b.$$

This implies that any automorphism of the Riemann sphere is of the form

$$f(x) = \frac{ax + b}{cx + d}$$

for some complex coefficient $a, b, c, d \in \mathbb{C}$. In order to prove the isomorphism with $\text{PSL}_2(\mathbb{C})$ we remark that $ad - bc \neq 0$ because it assure the invertibility of f . Therefore we can associate to each $f \in \text{Aut}(\mathbb{P}^1)$ an invertible matrix

$$f(x) = \frac{ax + b}{cx + d} \rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbb{C}).$$

Moreover, for any $\lambda \in \mathbb{C}^*$,

$$f(x) = \frac{\lambda ax + \lambda b}{\lambda cx + \lambda d} = \frac{ax + b}{cx + d},$$

concluding the proof. □

Finally, we conclude this section by stating one property of this automorphism group.

Proposition 1.1.3. *The action $\text{Aut}(\mathbb{P}^1) \curvearrowright \mathbb{P}^1$ is 3-transitive, that means that the action is transitive, and for any choice of $a, b, c \in \mathbb{P}^1$ pairwise distinct and $a', b', c' \in \mathbb{P}^1$ pairwise*

distinct, there exists $f \in \text{Aut}(\mathbb{P}^1)$ such that $f(a) = a'$, $f(b) = b'$, $f(c) = c'$.

Proof. Comes from a direct calculation using Proposition 1.1.2. $\square$

Remark 1. We will often use the previous proposition with $\{a', b', c'\} = \{0, 1, \infty\} \subseteq \mathbb{P}^1$.

1.2 Rank 2 Holomorphic Vector Bundles on $\mathbb{P}^1$

The geometry of holomorphic vector bundles on the Riemann sphere is governed by the Birkhoff-Grothendieck Theorem 1.2.2: any holomorphic vector bundle on the Riemann sphere can be expressed as a direct sum of line bundles. We then study the Picard group of $\mathbb{P}^1$ in Proposition 1.2.1 and some basic properties of line bundles, as, for instance, the existence of holomorphic sections. Finally, we give a brief description of the Hirzebruch surfaces, arising from the projectivisation of certain rank 2 vector bundles.

Definition. The *tautological bundle* $\mathcal{O}(-1)$ is the line bundle over the Riemann sphere in which the fiber over a point $x = [l] \in \mathbb{P}^1$ is exactly the line l . Its dual is $\mathcal{O}(1)$ and is called the *hyperplane bundle*.

These two line bundles are the building blocks for any holomorphic vector bundle over the Riemann sphere as Proposition 1.2.1 and Theorem 1.2.2 show. Let us then study in details how we can compute the cocycle of the tautological bundle. Let us consider a point $p \in U_0 \cap U_\infty$. Therefore $p = [x_0 : 1] = [1 : z_0]$ for some $x_0 = 1/z_0 \in \mathbb{C}^*$. The fiber $\mathcal{O}(-1)_p$ is therefore the line in $\mathbb{C}^2$ passing through 0 and p . In particular we can express its coordinates in the trivialising atlas of $\mathcal{O}(-1)$. In U_0 the line express as $l_0 := \text{Vect}((x_0, 1))$ and in U_∞ as $l_\infty := \text{Vect}((1, z_0))$. To pass from an expression to an another we use the fact that $x_0 = 1/z_0$, and in particular

$$\mu(x_0, 1) = \lambda(1, z_0) \iff \mu = \frac{1}{x_0} \lambda,$$

showing that the cocycle $\mathcal{O}(-1)$ in the overlap $U_0 \cap U_\infty$ is x^{-1} . By duality, the cocycle of $\mathcal{O}(1)$ on the same overlap is x .

Proposition 1.2.1. *The Picard group of the Riemann sphere is isomorphic to $\mathbb{Z}$ and the hyperplane bundle $\mathcal{O}(1)$ is a generator. In particular any line bundle over the Riemann sphere*

is isomorphic to $\mathcal{O}(k) := \mathcal{O}(1)^{\otimes k}$ for some $k \in \mathbb{Z}$ (where for negative k this notation means $\mathcal{O}(-1)^{\otimes |k|}$).

Proof. Let us consider the exponential exact sequence for $\mathbb{P}^1$. Since $\mathbb{P}^1$ is simply connected we know that $H^1(\mathbb{P}^1, \mathbb{Z}) = 0$.

$$0 \longrightarrow \cancel{H^1(\mathbb{P}^1, \mathcal{O})} \longrightarrow H^1(\mathbb{P}^1, \mathcal{O}^*) \longrightarrow H^2(\mathbb{P}^1, \mathbb{Z}) \longrightarrow \cancel{H^2(\mathbb{P}^1, \mathcal{O})} \longrightarrow \dots$$

Let us recall that $\text{Pic}(\mathbb{P}^1) \cong H^1(\mathbb{P}^1, \mathcal{O}^*)$. Since $\mathbb{P}^1$ is a connected manifold, it holds that $H^2(\mathbb{P}^1, \mathbb{Z}) \cong \mathbb{Z}$. It suffices then to show that $H^1(\mathbb{P}^1, \mathcal{O}) = H^2(\mathbb{P}^1, \mathcal{O}) = 0$. The H^2 is zero since we can choose $U_0 \cup U_\infty$ as Leray cover composed by just two open set, and there are no triple intersections. The H^1 is zero since any $\{f_{0\infty}\} \in H^1(\mathbb{P}^1, \mathcal{O}^*)$ has a Laurent expression in $U_0 \cap U_\infty$ given by $f_{0\infty}(x) = \sum_{-\infty}^{\infty} a_n x^n$. We can then decompose it as $\sum_0^{\infty} a_n x^n + \sum_{-\infty}^{-1} a_n x^n = f_0 - f_\infty$. Clearly $f_0 \in \mathcal{O}(U_0)$ and $f_\infty \in \mathcal{O}(U_\infty)$, concluding the proof. $\square$

Remark 2. Since U_0 and U_∞ are contractible, any line (and, more generally, vector) bundle admits $\{U_0, U_\infty\}$ as trivialising cover.

Example 1. The line bundle of holomorphic 1-forms over $\mathbb{P}^1$, that we denote by Ω^1 , is isomorphic to $\mathcal{O}(-2)$. Indeed, if we consider the trivialising atlas $\{U_0, U_\infty\}$, we have that a holomorphic one form ω restricted to U_0 is of the form $f(x)dx$ and the change of coordinates $z = 1/x$ gives

$$\omega|_{U_\infty} = f(1/z)d(1/z) = -\frac{1}{z^2}f(1/z)dz$$

showing that the cocycle of Ω^1 is $-z^{-2}$. By the isomorphisms $\text{Pic}(\mathbb{P}^1) \cong H^1(\mathbb{P}^1, \mathcal{O}^*) \cong \mathbb{Z}$, we conclude.

We can now increase the rank by one.

Definition. A rank 2 holomorphic vector bundle over $\mathbb{P}^1$, that we often denote by $E \rightarrow \mathbb{P}^1$, is a vector bundle such that its fibers are all homeomorphic to $\mathbb{C}^2$ and the coordinate change is holomorphic.

Holomorphic vector bundles over the Riemann sphere have a very special description. It holds for any rank.

Theorem 1.2.2 (Birkhoff-Grothendieck Theorem). *Every rank r holomorphic vector bundle $E \rightarrow \mathbb{P}^1$ is holomorphically isomorphic to a direct sum of line bundles*

$$E \cong \mathcal{O}(d_1) \oplus \cdots \oplus \mathcal{O}(d_r).$$

As for line bundles, also for higher rank vector bundles the cover $\{U_0, U_\infty\}$ is trivialising. In particular, thanks to the Birkhoff-Grothendieck theorem it holds that cocycles of rank 2 holomorphic vector bundles are of the form

$$g_{0\infty} = \begin{pmatrix} x^{d_1} & 0 \\ 0 & x^{d_2} \end{pmatrix}.$$

From now on we will refer to holomorphic vector bundles as just vector bundles since we will never deal with differential or topological vector bundles.

Definition. A global holomorphic section of a vector bundle $E \rightarrow \mathbb{P}^1$ is a holomorphic function $\sigma: \mathbb{P}^1 \rightarrow E$ such that $\pi \circ \sigma = 1$. We will denote by $\Gamma(E)$ or $H^0(\mathbb{P}^1, E)$ the vector space of global holomorphic sections.

Example 2. Let $k \geq 0$. It holds that $\Gamma(\mathcal{O}(k)) = \mathbb{C}[x]_k$, that is the space of polynomials of degree k . First of all we remark that if $k = 0$, then a section is a holomorphic function $\mathbb{P}^1 \rightarrow \mathbb{C}$ that is hence constant.

If now $k > 0$ and σ is a section of $\mathcal{O}(k)$, then we can restrict it to U_0 getting $\sigma_0 := \sigma|_{U_0}: U_0 \rightarrow U_0 \times \mathbb{C}$. It admits an expression as a Taylor series $\sigma_0 = \sum_{i=0}^{\infty} a_i x^i$. Since σ is a global holomorphic section we know that $\sigma_0 = g_{0\infty} \sigma_\infty$ and we have that $g_{0\infty} = x^k$. In particular we have that, setting $z = 1/x$,

$$\sigma_\infty(z) = z^k \sum_{i=0}^{\infty} a_i z^{-i} = \sum_{i=0}^{\infty} a_i z^{k-i},$$

that is holomorphic if, and only if, $a_i = 0$ for all $i > k$.

Remark 3. 1. If $k < 0$, $\mathcal{O}(k)$ does not have global holomorphic sections.

2. If $k \geq 0$, any global holomorphic section of $\mathcal{O}(k)$ vanishes in exactly k points.

3. If $\mathcal{O}(k)$ admits a never vanishing global holomorphic section, then $k = 0$.

4. Any $\mathcal{O}(k)$ admits a unique zero section whose image is biholomorphic to $\mathbb{P}^1$.

Thanks to the Birkhoff-Grothendieck theorem we can see holomorphic sections of a rank 2 vector bundle as morphisms $\sigma: \mathbb{P}^1 \rightarrow E$ that locally express as

$$\sigma_0: U_0 \rightarrow U_0 \times \mathbb{C}^2; \quad x \mapsto \left(x, \begin{pmatrix} y_1(x) \\ y_2(x) \end{pmatrix} \right).$$

In particular the bundle $\mathcal{O}(d_1) \oplus \mathcal{O}(d_2)$ admits global holomorphic sections if and only if $d_1 \geq 0$ or $d_2 \geq 0$. We conclude this section by exploring the projectivisation of a vector bundle. This construction exists for topological vector bundles of any rank, over any manifold and over any base field, but we will specify in the holomorphic rank 2 case over the Riemann sphere, to simplify notations and introduce the Hirzebruch surfaces.

Definition. Let $\pi: E \rightarrow \mathbb{P}^1$ be a rank 2 vector bundle and let $E' := E \setminus \mathbb{P}^1$ be the complement of the zero section. The quotient space of E' with respect to the equivalence relation

$$u \sim v \iff \begin{cases} \pi(u) = \pi(v) \\ u = \lambda v \text{ for some } \lambda \in \mathbb{C}^* \end{cases}$$

is denoted by $\mathbb{P}(E)$ and together with the projection $[v] \mapsto \pi(v)$ it is called the projective bundle associated to E .

Roughly speaking, the projective bundle $\mathbb{P}(E)$ has as fibers the projectivisation of the fibers of E . Its cocycle is the same as the one of E : since the g_{ij} are linear morphisms they descend to the quotient.

Definition. Let us consider the rank 2 vector bundle $\mathcal{O} \oplus \mathcal{O}(k) \rightarrow \mathbb{P}^1$. We denote by $\mathbb{F}_k := \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(k))$ the associated projective bundle and we call it the k -th *Hirzebruch surface*.

Given a vector bundle $E \rightarrow \mathbb{P}^1$ and a line bundle $L \rightarrow \mathbb{P}^1$ it holds that $\mathbb{P}(E) \cong \mathbb{P}(L \otimes E)$. Indeed, if the cocycle of E is $\{g_{ij} \in \mathrm{GL}_2(\mathbb{C})\}$ and the cocycle of L is $\{\lambda_{ij} \in \mathbb{C}^*\}$, then the cocycle of $L \otimes E$ is $\{\lambda_{ij} \cdot g_{ij}\}$. Once we pass to the projective quotient the cocycles $\{g_{ij}\}$ and $\{\lambda_{ij} \cdot g_{ij}\}$ are the same and they define the same projective bundle. In particular it holds that if $E \cong \mathcal{O}(d_1) \oplus \mathcal{O}(d_2)$, then $\mathbb{P}(E) \cong \mathbb{F}_{d_2-d_1} \cong \mathbb{F}_{d_1-d_2}$.

Remark 4. In some texts the Hirzebruch surface $\mathbb{F}_k$ is defined as the compactification of the total space of the line bundle $\mathcal{O}(k)$ by adding a copy of $\mathbb{P}^1$ representing the infinity section.

1.3 Meromorphic Connections

We are interested in the study of a special class of linear meromorphic connections, of PV type, but let us start by introducing what a meromorphic connection is in a general setting.

Definition. Let X be a complex manifold. A linear meromorphic connection is a triple (E, D, ∇) , where

1. $E \rightarrow X$ is a holomorphic vector bundle,
2. D is an effective divisor of X ,
3. ∇ is a $\mathbb{C}$ -linear morphism of sheaves $\nabla: \mathcal{E} \rightarrow \mathcal{E} \otimes \Omega_X^1(D)$, satisfying the Leibniz rule given by the $\mathcal{O}_X$ -module structure of the sheaf $\mathcal{E}$ of holomorphic sections:

$$\nabla(f \cdot \sigma) = df \cdot \sigma + f \cdot \nabla \sigma.$$

The divisor D is called *polar divisor* and it describes the order and the position of the poles that the connection must have. We recall the notation $\Omega_X^1(D) := \Omega_X^1 \otimes \mathcal{O}_X(D)$.

Remark 5. If $D = \emptyset$, we get a holomorphic connection.

Even if some of the notions we introduce hold for any complex manifold and any vector bundle, for the following of the article we always suppose that $X = \mathbb{P}^1$ and $\text{rk} E = 2$. This significantly simplify notations.

Definition. A point $a \in \mathbb{P}^1$ is called a *singularity* (or a *pole*) for the connection (E, ∇, D) if $a \in |D|$. Its *Poincaré rank* is the integer number $\deg(D|_a) - 1$.

Definition. A singularity is said *logarithmic* if its Poincaré rank is zero. Otherwise, it is said *regular* if it can be reduced to a logarithmic singularity and *irregular* if not.

Example 3 (Trivial Connection on the Trivial Bundle). If the vector bundle is trivial we can consider the connection $(d, \mathcal{O} \oplus \mathcal{O}, \emptyset)$. We recall that global holomorphic sections of the trivial bundle are just constant (ad then in the kernel of d), but we have local sections of the form $Y: U \rightarrow \mathbb{C}^2$. The trivial connection acts in the following way:

$$\text{if } Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad \text{then } dY = \begin{pmatrix} dy_1 \\ dy_2 \end{pmatrix},$$

clearly satisfying the Leibniz rule $d(fY) = df \cdot Y + f dY$.

We can generalise a bit more this example via the following proposition.

Proposition 1.3.1. *Any meromorphic connection $(\nabla, \mathcal{O} \oplus \mathcal{O}, D)$ on the trivial bundle can be globally expressed as*

$$\nabla = d + \Omega \quad \text{for some} \quad \Omega = \begin{pmatrix} \omega_{1,1} & \omega_{1,2} \\ \omega_{2,1} & \omega_{2,2} \end{pmatrix} \in \mathfrak{gl}_2(\Omega^1(D)).$$

Sketch of proof. First of all we notice that for any $\Omega \in \mathfrak{gl}_2(\Omega^1(D))$ the operator $d + \Omega$ is a connection on the trivial bundle since it is $\mathbb{C}$ -linear and satisfies the Leibniz rule

$$(d + \Omega)(f \cdot \sigma) = d(f \cdot \sigma) + \Omega(f \cdot \sigma) = df \cdot \sigma + f \cdot d\sigma + f \cdot \Omega\sigma = df \cdot \sigma + f(d + \Omega)\sigma.$$

As a second step, a computation shows that the difference between a connection $(\nabla, \mathcal{O} \oplus \mathcal{O}, D)$ and the trivial connection $(d, \mathcal{O} \oplus \mathcal{O}, \emptyset)$ is a differential operator of order zero, that is $\nabla - d = \Omega \in \mathfrak{gl}_2(\Omega^1(D))$. The thesis then follows since we get $\nabla = d + \Omega$. $\square$

Roughly speaking, we can see any connection on the trivial bundle as a rank 2 system of ODEs over the projective line. Indeed, we can write

$$\nabla Y = (d + \Omega)Y = \begin{pmatrix} dy_1 \\ dy_2 \end{pmatrix} + \begin{pmatrix} \omega_{1,1} & \omega_{1,2} \\ \omega_{2,1} & \omega_{2,2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 0.$$

Remark 6. Note that if the vector bundle E is not trivial, then no connection (E, D, ∇) can be globally expressed as $\nabla = d + \Omega$. Of course, for any trivialising open set U , it holds that $\nabla|_U = d + \Omega_U$ for some $\Omega_U \in \mathfrak{gl}_2(\Omega^1(D \cap U))$. In Proposition 1.3.3 we will see how this constriction behaves under the coordinate change.

More generally, thanks to the previous remark, we can see a connection (E, D, ∇) as a collection of systems of ODEs, one for each trivialising open set for E . It becomes then natural to wonder what can be local and global solutions of the differential systems induced by ∇ . In Proposition 1.3.3 we will show how these local systems glue on the overlaps.

Definition. A local section Y satisfying $\nabla Y = 0$ is called *parallel* or *horizontal* section for the connection (∇, E, D) . Equivalently, we say that Y is a solution of the differential equation $\nabla = 0$

We are interested in studying the behavior of horizontal sections in a neighborhood of a singularity $x = a$.

Theorem 1.3.2. *Let X be a Riemann surface and (∇, E, D) a connection on it. Let $U \subseteq X \setminus \text{Supp}(D)$ a simply connected open set trivialising E . Then:*

1. *There exists a fundamental matrix of solutions $Y(x)$ defined everywhere over U .*
2. *Any other matrix of solutions $Y'(x)$ differs from $Y(x)$ by a constant matrix factor $Y'(x) = Y(x) \cdot C$.*
3. *For any $x_0 \in U$, the evaluation map $Y(\cdot) \mapsto Y(x_0)$ is an isomorphism between the linear space of solutions of $\nabla|_U Y = 0$ and $\mathbb{C}^2$.*

Proof. Theorem 14.1 and remark 19.2 of [28] □

We can specify the theorem via the following facts coming from Propositions 15.10, 20.20, 21.3 of [28].

Fact. If $x = a \ni U$ is a logarithmic singularity a local basis of solution can be expressed as $Y(x) = H(x)x^A$ where A is a constant matrix and H is a meromorphic invertible matrix function.

Fact. If $x = a \ni U$ is an irregular singularity, then the local solutions of $\nabla = 0$ can be expressed as $Y(x) = F(x)x^A$, where A is a constant matrix and F is a matrix function single valued and holomorphically invertible in $U \setminus \{a\}$, but eventually having an essential singularity in $x = a$.

The description of solutions around an irregular singularity is much more complicated since Stokes phenomena appear. In fact, for simplicity, assume that $U \cong \mathbb{D}$ is isomorphic to the unitary complex disk centered in the singularity. We cannot express a solution as a holomorphic function in $\mathbb{D}^* := \mathbb{D} \setminus \{0\}$, but we are only allowed to define it on angular sectors that cover $\mathbb{D}^*$, whose amplitude depends on the order of the singularity. For example, in the case of an order two singularity, we have two sectors:

$$V^k := \left\{ x \in \mathbb{D}^* \mid \left| \arg(x) - (\theta + k\pi) \right| < \pi \right\} \quad \text{for } k = 0, 1$$

for a θ that depends on the spectral data of $\Omega_{\mathbb{D}}$, and on V^k the solutions are holomorphic. Moreover, a solution $Y(x)$ will grow exponentially fast on V^0 and decreases faster than any finite power $|x|^{N>0}$ in V^1 . We will treat an example in Section 4.1.2, in which

we introduce the irregular monodromy. For more details, see for instance paragraph 21.1 of [28] and section 1.3 of [38].

Up to now, we tried to understand what solutions of $\nabla = 0$ look like locally, or on the trivial bundle. When the vector bundle E is not globally trivial we need to understand how to glue together the local systems. In particular we know that any holomorphic vector bundle is trivial when restricted to any contractible open set of $\mathbb{P}^1$, and then it is always sufficient to consider $\{U_0, U_\infty\}$ as trivialising open cover. A connection on $\mathbb{P}^1$ is then the datum of two different differential systems $\nabla_0 = d + \Omega_0$ and $\nabla_\infty = d + \Omega_\infty$ and the two connection matrices are related as follows.

Proposition 1.3.3. *Let E be a rank 2 vector bundle over the Riemann sphere with cocycle $\{g_{0\infty}\}$. The two connection matrices are related in the overlap $U_0 \cap U_\infty$ by the formula*

$$\Omega_\infty = g_{0\infty}^{-1} \cdot \Omega_0 \cdot g_{0\infty} + g_{0\infty}^{-1} \cdot dg_{0\infty}.$$

Proof. The trivialization change acts on sections as $Y_0 = g_{0\infty}Y_\infty$ on the overlap. In particular

$$(d + \Omega_0)Y_0 = (d + \Omega_0)(g_{0\infty}Y_\infty) = dg_{0\infty}Y_\infty + g_{0\infty}dY_\infty + \Omega_0g_{0\infty}Y_\infty.$$

Collecting $g_{0\infty}$ it can be rewritten as

$$(d + \Omega_0)Y_0 = g_{0\infty} \left(dY_\infty + [g_{0\infty}^{-1} \cdot \Omega_0 \cdot g_{0\infty} + g_{0\infty}^{-1} \cdot dg_{0\infty}]Y_\infty \right) = g_{0\infty}(d + \Omega_\infty)Y_\infty$$

in the chart U_0 , and as $(d + \Omega_\infty)Y_\infty$ in the chart U_∞ , as desired. $\square$

We are interested to give an explicit expression to the connection matrix in a neighborhood of a singularity: we can then shrink a trivializing open set for E in order to get a simply connected open set U that contains at most one singularity at, for instance, $x = a$. On this open set U , the connection matrix has a very simple and explicit description:

$$\text{if } E|_U \cong U \times \mathbb{C}^2 \text{ and } |D| \cap U = \{a\}, \quad \text{then } \nabla_U = d + \Omega_U,$$

for some $\Omega_U \in \text{gl}(\Omega^1(n[a]))$ that writes:

$$\Omega_U := \sum_{k=1}^n A_k^{(a)} \frac{dx}{(x-a)^k} + \text{holomorphic terms}$$

where $A_k^{(a)} \in \text{gl}_2(\mathbb{C})$ are constant matrices and n is the order of the pole in $x = a$.

Definition. We call $\Omega_{U,a}^{<0} := \sum_{k=1}^n A_k^{(a)} \frac{dx}{(x-a)^k}$ the *principal part* of the connection matrix Ω_U around the singularity $a \in \mathbb{P}^1$. Note that it depends on the chosen trivialization.

Definition. Let (E, D, ∇) be a meromorphic connection and $a \in U \subseteq \mathbb{P}^1$ be a singularity. We say that the connection is in *diagonal principal form* at the singularity $x = a$ if it has no holomorphic part and the principal part is diagonal:

$$\Omega_U := \sum_{k=1}^n D_k^{(a)} \frac{dx}{(x-a)^k}.$$

1.3.1 Gauge Transformations

Definition. Two connections (E, D, ∇) and (E', D, ∇') are said *gauge equivalent* if there exists a holomorphic isomorphism of bundles $\Phi \in \text{Iso}(E, E')$ that sends ∇ -horizontal sections into ∇' -horizontal sections. Equivalently, we have that

$$\nabla = \Phi^* \nabla'.$$

Moreover if $E = E'$ and $\nabla = \nabla'$, we call it a *symmetry* of the connection (E, D, ∇) .

We would like to deduce a formula that relates the local forms of gauge equivalent connections.

Proposition 1.3.4. *Let (E, D, ∇) and (∇', E, D) be holomorphically gauge equivalent and let $\Psi: E \rightarrow E$ be the gauge transformation. Let $U \subseteq \mathbb{P}^1$ be a trivialising open set for both connections. Then the connection matrices Ω_U and Ω'_U are related by the formula*

$$\Omega_U = \Phi^{-1} \Omega'_U \Phi + \Phi^{-1} d\Phi.$$

Proof. By the definition it follows that if Y is a local horizontal section for ∇ , then $Y' = \Psi Y$ is a local horizontal section for ∇' . The proof is then (*mutatis mutandis*) the same as Proposition 1.3.3. $\square$

Example 4. The symmetries of the trivial holomorphic connection $(\mathcal{O} \oplus \mathcal{O}, \emptyset, d)$ are given by constant matrices. Indeed, let Ψ be a symmetry, that is $\Psi^* d = d$, and let Y a solution matrix. It holds that

$$0 = dY = (\Psi^* d)(Y) = d(\Psi Y) = d(\Psi)Y + \Psi d(Y),$$

and hence $d(\Psi) = 0$, and $\Psi = C$ is a constant matrix, as desired.

Example 5. Let us consider a connection (E, D, ∇) expressed locally in U around a double pole say centered in $x = 0$. Its connection matrix has expression

$$\Omega = A_2^{(0)} \frac{dx}{x^2} + A_1^{(0)} \frac{dx}{x} + \text{holomorphic terms}.$$

A holomorphic gauge transformation can be locally written as $\Phi|_U = M_0 + M_1x + \dots$, with $M_0 \in GL_2(\mathbb{C})$. The result of the gauge action on Ω is then

$$\left(M_0^{-1} A_2^{(0)} M_0\right) \frac{dx}{x^2} + \left(M_0^{-1} A_1^{(0)} M_0 + M_0^{-1} A_2^{(0)} M_1 - M_0^{-1} M_1 M_0^{-1} A_2^{(0)} M_0\right) \frac{dx}{x} + \text{hol terms}.$$

More generally, a holomorphic gauge transformation then acts by conjugation on the highest term of the connection matrix.

Remark 7. If $A_2^{(0)}$ is semisimple, we can always diagonalize the principal part of a connection via a finite number of holomorphic gauge transformations.

Holomorphic gauge transformations allow us to understand when two connections on isomorphic bundles are equivalent. We would like to have a more general definition, allowing us to establish an equivalence of connections over different bundles of different degree.

Definition. A meromorphic gauge transformation $\Psi: (\nabla, E, D) \rightarrow (\nabla', E', D')$ is a birational morphism of bundles $\Psi: E \rightarrow E'$ such that $\Psi^* \nabla' = \nabla$.

Remark 8. Meromorphic gauge transformations are monodromy preserving, as we will see in Sections 1.3.2 (for instance, Proposition 1.3.13 in the logarithmic case).

A particular and interesting class of meromorphic gauge transformations are elementary transformations. They are meromorphic gauge transformations Ψ such that $\det \Psi$ has only one zero or one pole of order one. Any meromorphic gauge transformation is a composition of elementary transformations.

Definition. Let E be a rank 2 vector bundle and l be a line in E_{x_0} . We denote via (E, l) the parabolic vector bundle. The elementary transformations $\text{elm}_{l, x_0}^\pm$ are meromorphic gauge transformations

$$\text{elm}_{l, x_0}^+ : (E, l) \rightarrow (E^+, l^+) \quad \text{and} \quad \text{elm}_{l, x_0}^- : (E, l) \rightarrow (E^-, l^-).$$

We choose a local trivialisation $E|_U \cong U \times \mathbb{C}^2$ based in $x_0 = 0$ and such that $l = \mathbb{C} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, then the elementary transformations have local expression

$$\text{elm}_{l,x_0}^+ = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix} : \mathbb{C}^2 \rightarrow \mathbb{C}^2 \quad \text{and} \quad \text{elm}_{l,x_0}^- = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{x} \end{pmatrix} : \mathbb{C}^2 \rightarrow \mathbb{C}^2 \quad (1.1)$$

Proposition 1.3.5. *Let us consider a point $x_0 \in U \subseteq \mathbb{P}^1$ and the elementary transformation $\text{elm}_{l,x_0}^\pm$. It holds that $\det(E^\pm) = \det(E) \otimes \mathcal{O}(\pm[x_0])$ and therefore $\deg E^\pm = \deg E \pm 1$.*

Proof. Section 6 of [41]. □

In order to better understand how the elementary transformations acts on the connection matrix, let us consider the following example.

Example 6. Let us consider the following connection on the trivial bundle given by $(\mathcal{O} \oplus \mathcal{O}, D, d + \Omega)$ and a point $q \notin |D|$. Then

$$\text{elm}_q^\pm(\mathcal{O} \oplus \mathcal{O}, D, d + \Omega) = (\mathcal{O} \oplus \mathcal{O}(1), D + [q], d + \tilde{\Omega}).$$

Indeed here we can globally find a choice of coordinates on $\mathcal{O} \oplus \mathcal{O}$ to express the elementary transformation as a matrix, and it acts in the standard way on Ω :

$$\begin{aligned} \tilde{\Omega} &= \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{x-q} \end{pmatrix}^{-1} \begin{pmatrix} \omega_{1,1} & \omega_{1,2} \\ \omega_{1,2} & \omega_{1,2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{x-q} \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{x-q} \end{pmatrix}^{-1} \begin{pmatrix} 0 & 0 \\ 0 & -\frac{dx}{(x-q)^2} \end{pmatrix} = \\ &= \begin{pmatrix} \omega_{1,1} & (x-q)\omega_{1,2} \\ \frac{1}{(x-q)}\omega_{2,1} & \omega_{2,2} - \frac{dx}{x-q} \end{pmatrix}. \end{aligned}$$

Remark 9. When we refer to the isomorphism class of a connection, we will implicitly mean under the action of meromorphic gauge transformations. Otherwise we will specify that we consider only holomorphic gauge equivalence.

One can wonder what happens when we apply a meromorphic gauge transformation based in a point $p \in |D|$. We can study the problem locally, so let us consider a local connection $(E, n[0], \nabla)$ on the disk $\mathbb{D}$ and the elementary transformation $\text{elm}_{l,0}^-$. Consider then the principal part of the connection matrix and the action of the elementary

transformation:

$$\text{elm}_{l,0}^- : \begin{pmatrix} \sum_{i=1}^n \frac{a_i}{x^i} & \sum_{i=1}^n \frac{b_i}{x^i} \\ \sum_{i=1}^n \frac{c_i}{x^i} & \sum_{i=1}^n \frac{d_i}{x^i} \end{pmatrix} dx \mapsto \begin{pmatrix} \sum_{i=1}^n \frac{a_i}{x^i} & x \sum_{i=1}^n \frac{b_i}{x^i} \\ \frac{1}{x} \sum_{i=1}^n \frac{c_i}{x^i} & \sum_{i=1}^n \frac{d_i}{x^i} - \frac{1}{x} \end{pmatrix} dx.$$

In particular the degree of the pole $x = 0$ can be decreased or increased.

The polar divisor D of a meromorphic connection is then not well defined up to meromorphic gauge equivalence. We still have the following lemma.

Lemma 1.3.6. *Let (E, D, ∇) be a meromorphic connection with $D = \sum_{i=1}^k a_i[p_i]$. Then there exist integers $n_1, \dots, n_k$, with $0 \leq n_i \leq a_i$, such that after a finite number of meromorphic gauge transformations the connection (E, D, ∇) is transformed into the equivalent (E', D', ∇') where $D' = \sum_{i=1}^k n_i[p_i]$ presents minimal pole orders n_i , in the sense that they can not be decreased anymore by further action via meromorphic gauge transformations.*

Proof. A complete proof of this fact can be found in Chapter 6 of [10]. We indeed give an operative algorithm that allows one to establish whenever a pole is reduced or not. We can, of course, study the problem pole by pole, so let us consider the local connection $(E, n[0], \nabla)$ on the disk and let express the principal part of the connection as

$$\Omega_{\mathbb{D}} := \sum_{k=1}^n A_k^{(0)} \frac{dx}{x^k}.$$

Let us suppose $n \geq 2$. Up to holomorphic gauge transformation we can suppose that the matrix $A_n^{(0)}$ is upper triangular. The principal part of the connection matrix then express

$$\begin{pmatrix} a_n & b_n \\ 0 & d_n \end{pmatrix} \frac{dx}{x^n} + \begin{pmatrix} a_{n-1} & b_{n-1} \\ c_{n-1} & d_{n-1} \end{pmatrix} \frac{dx}{x^{n-1}} + \dots$$

- If $a_n \neq d_n$, then the order of the pole is minimal and it cannot be reduced.
- If $a_n = d_n$ and $b_n = 0$ (that is, the matrix $A_n^{(0)}$ is a multiple of the identity), then we can reduce the order.
- If $a_n = d_n$, but $b_n \neq 0$, then we have two sub cases. Let us apply the elementary transformation as in Formula 1.1. If $c_{n-1} = 0$, then the order of the pole can be reduced. If $c_{n-1} \neq 0$ then the order of the pole remains the same and it cannot be reduced.

□

Definition. We refer to the *minimal polar divisor* of a meromorphic connection (E, D, ∇) as the minimal polar divisor within its meromorphic gauge equivalence class, as the divisor D' in Lemma 1.3.6.

1.3.2 Residual Spectral Data

Cauchy's residue theorem shows the role played by the residue of a meromorphic function in computing integrals along closed path. We are looking for a similar invariant for meromorphic connections. We need to be careful since in Example 5 we have seen that the spectrum of the residual matrix is not invariant under gauge transformations for poles of order higher than one.

Definition. We call $A_1^{(a)} = \text{Res}_{x=a} \Omega_U$ the *residual matrix* of Ω_U at $x = a$.

Notice that the expression of $A_1^{(a)}$ strongly depends on the chosen trivialization. We need then a precise algorithm to compute the residual spectral data in such a way that they do not depend on the representative of the meromorphic gauge class of equivalence.

Definition. The *residual spectral datum* of a connection (E, D, ∇) at a pole $x = a$ is computed as follows. Let Ω be any connection matrix for (E, D, ∇) in an open trivialising neighborhood of a pole $a \in \mathbb{P}^1$, and let us denote with $\Omega^{<0}$ its principal part. Up to a holomorphic gauge transformation, we can diagonalize it (Example 5 and Remark 7). Let $D_1^{(a)}$ be the new (diagonal) residual matrix and let us denote by $\{\kappa_a^+, \kappa_a^-\}$ its spectrum. It does not depend on the holomorphic gauge transformation we have chosen. We refer to the *residual spectral data* of a connection (E, D, ∇) as the collection

$$\left(\{\kappa_a^+, \kappa_a^-\} \right)_{a \in |D|}.$$

Remark 10. Residual spectral data does not depend on the holomorphic part of a connection, and that is the reason why they do not depend on the chosen holomorphic gauge transformation used to diagonalize $\Omega^{<0}$.

We would like to have an explicit formula to directly compute the residual spectral datum of a connection in a pole of order two $x = a$. For simplicity, we will suppose that it is a pole of order two. Let then us consider any connection matrix Ω as in the

definition and let us consider its diagonalized principal part

$$\begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}, \quad \text{for } \lambda, \mu \in \mathbb{C}((x-a)).$$

Notice that $\kappa_a^+ = \text{Res}_{x=a} \lambda$ and $\kappa_a^- = \text{Res}_{x=a} \mu$. In particular, it holds that $\text{tr} \Omega = \lambda + \mu$ and $\det \Omega = \lambda \mu$. To find $\kappa_a^\pm$, we need to compute the residue of $\lambda - \mu$: indeed $\text{Res}_{x=a} \text{tr} \Omega = \kappa_a^+ + \kappa_a^-$ and $\text{Res}_{x=a}(\lambda - \mu) = \kappa_a^+ - \kappa_a^-$. We can then use the following formula

$$(\lambda - \mu)^2 = (\text{tr} \Omega)^2 - 4 \det \Omega =: R(\Omega).$$

In particular, considering the principal parts $(\lambda - \mu)^{<0} = \sum_{i=1}^2 \alpha_i (x-a)^{-i}$ and $R(\Omega)^{<0} = \sum_{i=1}^4 c_i (x-a)^{-i}$, for some coefficient $\alpha_i, c_i \in \mathbb{C}$, it holds that

$$((\lambda - \mu)^2)^{<-2} = R(\Omega)^{<-2} \implies \frac{\alpha_2^2}{(x-a)^4} + \frac{2\alpha_1\alpha_2}{(x-a)^3} = \frac{c_4}{(x-a)^4} + \frac{c_3}{(x-a)^3}$$

and we deduce the residue α_1 of $\lambda - \mu$:

$$\alpha_1 = -\frac{c_3}{2\sqrt{c_4}} = \kappa_a^+ - \kappa_a^-. \quad (1.2)$$

Remark 11. The residual spectral data must satisfy the Fuchs' relation

$$\sum_{a \in |D|} \kappa_a^+ + \kappa_a^- = -\deg E$$

Definition. The residual spectral data of a connection are called *generic* if the following conditions

$$\sum_{a \in |D|} \kappa_a^{\epsilon_i} \notin \mathbb{Z} \quad \text{and} \quad \forall a \in |D|, \kappa_a^+ - \kappa_a^- \notin \mathbb{Z} \quad (1.3)$$

hold for all choices of $\epsilon_i \in \{+, -\}$.

We always assume that the residual spectral data are generic. This implies that the connection is irreducible, that is, there does not exist a ∇ -invariant line bundle $L \subseteq E$.

Remark 12. Meromorphic gauge transformations may shift some residual data by an integer, as shown in Proposition 1.3.8 below. Since the degree of the bundle changes as well, the residual spectral data still satisfy the new Fuchs' relation 11. This implies

that the residual spectral data of an isomorphism class of a connection are well defined only up to integers. When some residual data are zero some pathological behaviors arise, it also explains why we asked the generic condition to be satisfied.

Proposition 1.3.7. *Any irreducible meromorphic connection (E, D, ∇) with $\deg D = 4$ is meromorphic gauge equivalent to a connection (∇', E', D') where $E' \cong \mathcal{O}(-1) \oplus \mathcal{O}$.*

Proof. It is a consequence of the Fuchs' relation and the irreducibility of the connection (implied by the generic choice of the residual spectral data) as shown in Proposition 2.7.4 of [35]. An explicit description of such connections is given in Section 4.1 of [46]. $\square$

We have not still introduced monodromy, but meromorphic gauge transformations play an important role since they are monodromy preserving, as shown in Proposition 1.3.12. It is the key point for the logarithmic and irregular Riemann-Hilbert correspondence (1.4.1, 4.1.3)

Proposition 1.3.8. *Elementary transformations shift by integers the residual spectral data.*

Proof. As said in Remark 10, we are not interested in the holomorphic part of the connection and up to holomorphic gauge transformations we can suppose that the principal part is diagonal. After applying the elementary transformation it becomes

$$\begin{pmatrix} \sum_{i=1}^n \frac{a_i}{x^i} & 0 \\ \star & \sum_{i=1}^n \frac{d_i}{x^i} - \frac{1}{x} \end{pmatrix} dx.$$

In particular the residual matrix has spectral data $\{a_1, d_1 - 1\}$ as desired. $\square$

1.3.3 Local Diagonal Principal Form

Let $(E, n[0], \nabla)$ be a meromorphic connection over a disk $\mathbb{D} \subseteq \mathbb{P}^1$. We can for instance see $(E, n[0], \nabla)$ as the restriction of a meromorphic connection (E, D, ∇) over $\mathbb{P}^1$ on a small contractible open set containing just one singularity.

Remark 13. We denote such connections over the disk with $(E, n[0], \nabla)$, but, since the disk is contractible, E is the trivial bundle and the connection ∇ express globally as $d + \Omega$. The equation $\nabla = 0$ is therefore just a differential system.

We wonder if we can always find a connection $(E, n[0], \tilde{\nabla})$ that is in diagonal principal form and holomorphically gauge equivalent to $(E, n[0], \nabla)$.

When $n = 1$, the answer depends on the spectrum of the residual matrix.

Theorem 1.3.9 (Poincaré-Dulac in rank 2). *Let $(E, [0], \nabla)$ be a logarithmic connection over the disk. Up to holomorphic gauge transformation, we can assume that its matrix connection writes*

$$\Omega = \begin{pmatrix} \kappa^+ & 0 \\ 0 & \kappa^- \end{pmatrix} \frac{dx}{x} \quad \text{or} \quad \begin{pmatrix} \kappa & x^n \\ 0 & \kappa + n \end{pmatrix} \frac{dx}{x}.$$

Moreover, this form is unique up to permutation of κ^+ and κ^- .

Proof. Theorem 1.2.1 of [38]. □

For generic residual spectral data, any logarithmic connection is then locally holomorphically gauge equivalent to a connection in diagonal principal form. We call the first case the non-resonant and resonant the second. In the non-resonant case the residual spectral data are indeed $\{\kappa^+, \kappa^-\}$.

When the singularity is irregular things become much more complicated. We study the case for $n = 2$. We do not have yet all the ingredients to state a more precise theorem, we will say something more about the diagonal principal form of irregular connection in Section 4.1.2, after defining properly the irregular monodromy. Consider the local connection on the disk $(E, 2[0], \nabla)$, with connection matrix Ω as in Example 5. As stated in the same Example, we see that the spectrum of the matrix $A_2^{(0)}$ is well defined up to holomorphic gauge transformations, since over this matrix the action resume to a conjugation. We choose a simpler and general case by supposing that $A_2^{(0)} = \text{diag}(\lambda^+, \lambda^-)$ with $\lambda^+ \neq \lambda^-$. This is often called in the literature the *non-ramified case*.

Theorem 1.3.10 (Turrittin-Levelt). *Let us consider $(E, 2[0], \nabla)$ as before. Up to formal gauge transformation $\hat{M} \in GL_2(\mathbb{C}[[x]])$, we can assume that its matrix connection writes*

$$\hat{\Omega} = \begin{pmatrix} \lambda^+ & 0 \\ 0 & \lambda^- \end{pmatrix} \frac{dx}{x^2} + \begin{pmatrix} \kappa^+ & 0 \\ 0 & \kappa^- \end{pmatrix} \frac{dx}{x} =: \begin{pmatrix} \kappa^+ & 0 \\ 0 & \kappa^- \end{pmatrix}.$$

Proof. Proposition 1.3.1 of [38] □

We stress the fact that $\hat{M}$ is only a formal gauge transformation, and in Section 4.1.2 we will see a necessary and sufficient condition on the monodromy of the connection to let $\hat{M}$ converge.

Remark 14. The residual spectral data $\{\kappa^+, \kappa^-\}$ are still well defined. Indeed with a holomorphic gauge transformation we can get to a connection matrix that has diagonal principal part, but with still infinite holomorphic terms.

Let us denote $\kappa^\pm = \lambda^\pm \frac{dx}{x^2} + \kappa^\pm \frac{dx}{x}$ as in Turrittin-Levelt theorem.

Proposition 1.3.11. *The formal gauge transformation $\hat{M}$ is unique up to right multiplication by a diagonal or anti-diagonal constant matrix. In particular the pair $\{\kappa^+, \kappa^-\}$ is uniquely defined, and its elements are principal part of the two solutions of*

$$\kappa^2 - \text{tr}(\Omega)\kappa + \det(\Omega) = 0.$$

Proof. Proposition 1.3.2 of [38]. □

We finally remark that we find back what we said in Section 1.3.2.

1.3.4 Local Logarithmic Monodromy

The monodromy of a connection arises by analytic continuation of a horizontal section along a closed path. If the connection is holomorphic, monodromy is only related to the manifold topology, while, when the connection is meromorphic, monodromy can also arise when we perform analytic continuation along path encircling singularities. As we will see in Section 1.4, the global monodromy of a logarithmic connection over $\mathbb{P}^1$ can be very difficult to compute. Other difficulties arise when the connection presents irregular singularities, as we see in details in Section 4.1.2. These are the reasons why we decided to start by presenting the local monodromy of a logarithmic connection, avoiding for the moment all the extra difficulties.

Let us then consider a logarithmic connection $(E, [0], \nabla)$ over the disk $\mathbb{D}$, an horizontal section Y satisfying $\nabla Y = dY + \Omega Y = 0$, and a point $p \in \mathbb{D}^*$. The monodromy of the connection measures the evolution of Y under analytic continuation along a closed path encircling the singularity $x = 0$. We can indeed consider a smooth loop $\gamma: [0, 1] \rightarrow \mathbb{D}^*$ such that $\gamma(0) = \gamma(1) = p$. Since $\gamma([0, 1])$ is compact, we can cover it with a finite number of trivialising set U_i , such that $p \in U_0 \cap U_n$. The cocycle change

states that $Y_i = g_{i,i+1}Y_{i+1}$ on $U_i \cap U_{i+1}$ for any $i = 0, \dots, n-1$. Let us set Y^γ the analytic continuation of the horizontal section Y along γ , that is:

$$Y^\gamma_{|U_0 \cap U_n} = Y_{|U_0 \cap U_n} \left(\prod_{i=0}^{n-1} g_{i,i+1} \right) = Y_{|U_0 \cap U_n} \cdot M^\gamma,$$

for a constant matrix $M^\gamma \in \mathrm{GL}_2(\mathbb{C})$ that is called the monodromy matrix. It is a classical result (Lemma 1.1.4 [38]) that M^γ only depends on the homotopy class $[\gamma] \in \pi_1(\mathbb{D}^*, p)$. Intuitively, it comes from the fact that we can deform the path γ keeping the same trivialising cover. We deduce hence that nearby a logarithmic singularity we have a local monodromy representation

$$\begin{aligned} \rho: \pi_1(\mathbb{D}^*, p) &\rightarrow \mathrm{GL}_2(\mathbb{C}) \\ [\gamma] &\mapsto M^\gamma. \end{aligned}$$

Our construction of the monodromy map strongly depends on the trivialisation we choose for the bundle $E_{|U_i}$, since we express Y in coordinates. We should investigate what happens when we change basis.

Proposition 1.3.12. *Given a local horizontal section Y of the connection $(\nabla, E, [0])$ expressed in a local chart $U \subseteq \mathbb{D}^*$, a basis change $Y' = YH$ for $H \in \mathrm{GL}_2(\mathbb{C})$ has the effect to conjugate the monodromy representations: $\rho' = H^{-1}\rho H$.*

Proof. It comes from the direct calculation

$$Y'M'^\gamma = (Y')^\gamma = (YH)^\gamma = Y^\gamma H^\gamma = (YM^\gamma)H = Y'H^{-1}M^\gamma H.$$

Remark that, being H constant, it holds that $H^\gamma = H$. □

Finally, we wonder what happens to monodromy when we apply a gauge transformation.

Proposition 1.3.13. *Let $(E, [0], \nabla)$ and $(E, [0], \nabla')$ be two (meromorphic) gauge equivalent logarithmic connections on the disk, then $\rho = \rho'$.*

Proof. Let us call $\Psi: E \rightarrow E$ the gauge transformation. As proved in Proposition 1.3.4, if Y is a horizontal section for ∇ , then $Y' = \Psi Y$ is a horizontal for ∇' . In particular, if γ

is a loop around the singularity:

$$Y'M'^\gamma = (Y')^\gamma = (\Psi Y)^\gamma = \Psi'^\gamma Y^\gamma = \Psi Y^\gamma = \Psi Y M^\gamma = Y'M^\gamma,$$

and then $M'^\gamma = M^\gamma$. Remark that, being Ψ univalued, it holds that $\Psi'^\gamma = \Psi$. $\square$

We resume these two propositions by noticing that a trivialization change induces a conjugation of the monodromy matrix, while a (meromorphic) gauge transformation keeps the monodromy untouched. In any case, the monodromy of a gauge class of meromorphic connection is only defined up to conjugation since we do not want it to depend on the choice of trivialization.

Thanks to the Poincaré-Dulac theorem in the logarithmic case 1.3.9, we have an easy expression of a solution matrix and its monodromy for the system $\nabla = 0$.

Proposition 1.3.14. *Let $(E, [0], \nabla)$ be a logarithmic connection on the disk in diagonal principal form and let $\{\kappa^+, \kappa^-\}$ be its spectral residual data satisfying $\kappa^+ - \kappa^- \notin \mathbb{Z}$. A multiform local solution matrix of the system $\nabla = 0$ is*

$$Y = \begin{pmatrix} x^{\kappa^+} & 0 \\ 0 & x^{\kappa^-} \end{pmatrix}.$$

Moreover, if we consider a simple loop $\gamma \subseteq \mathbb{D}^*$ turning one time around the singularity $x = 0$, the monodromy matrix is

$$M^\gamma = \begin{pmatrix} e^{2i\pi\kappa^+} & 0 \\ 0 & e^{2i\pi\kappa^-} \end{pmatrix}$$

Proof. It follows from a direct integration of the differential system. $\square$

Corollary 1.3.15. *Given a logarithmic connection $(E, [0], \nabla)$ on the disk with residual spectral data $\{\kappa^+, \kappa^-\}$, then its monodromy M^γ has spectral data $\{e^{2i\pi\kappa^+}, e^{2i\pi\kappa^-}\}$. Conversely, given a matrix M with these latter spectral data, there exists a unique local logarithmic connection $(E, [0], \nabla)$ having spectral data $\{\kappa^+, \kappa^-\}$ and monodromy matrix M .*

We remark that all the systems with residual spectral data $\{\kappa^+ + n, \kappa^- + m\}$, for $(m, n) \in \mathbb{Z}^2$, have the same monodromy matrix M^γ .

Example 7. Let us consider the trivial holomorphic connection $(d, \mathcal{O} \oplus \mathcal{O}, \emptyset)$. We can

apply to it the elementary transformation elm_0^- . The connection matrix becomes

$$\tilde{\Omega} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{x} \end{pmatrix}^{-1} \begin{pmatrix} 0 & 0 \\ 0 & -\frac{dx}{x^2} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \frac{dx}{x}.$$

It is in its Poincaré-Dulac form and the residual spectral data are $\{0, -1\}$. In particular the monodromy matrix around the singularity in 0 is just the identity.

1.4 Regular Riemann-Hilbert Correspondence

In Section 1.3.4 we studied the monodromy of a local logarithmic connection. The goal of this section is to improve this link, stating the global Riemann-Hilbert correspondence. Before this, we need to properly define some objects.

Definition. Let us consider a rank 2 vector bundle $E \rightarrow \mathbb{P}^1$, an effective divisor $D = [x_1] + \dots + [x_n]$ and a set of residual spectral data that are generic (and of course satisfy the Fuchs' relation) $\Theta = \{\kappa_i^+, \kappa_i^-\}_{i=1, \dots, n}$. We denote via $\text{Con}_{\Theta, E}^D$ the set of all logarithmic connections (E, D, ∇) with such residual spectral data Θ . We finally denote via $\mathfrak{Con}_{\Theta}^D := \text{Con}_{\Theta, E}^D / \sim$ the moduli space of those connection up to gauge equivalence.

We notice that Θ is now well defined just up to integer shifts due to the action of meromorphic gauge transformations, as shown in Proposition 1.3.8. This anyway does not affect the monodromy as pointed out in Corollary 1.3.15. Moreover, the bundle E does not appear anymore, that is because meromorphic gauge transformations can also change the bundle E . Of course the Fuchs' relation is still satisfied as noticed in Remark 12.

Definition. Let us set $\pi_1^D := \pi_1(\mathbb{P}^1 \setminus \{x_1, \dots, x_n\})$. We denote

$$\text{Rep}_{\Theta}^D := \text{Hom}(\pi_1^D, \text{GL}_2(\mathbb{C}))$$

the set of representation of the fundamental group of $\mathbb{P}^1 \setminus |D|$ having spectral data $\exp(\Theta)$. We finally denote via

$$\chi_{\Theta}^D := \text{Rep}_{\Theta}^D / \text{GL}_2(\mathbb{C}) \cong \text{spec} \left(\mathbb{C} \left[\text{Hom}(\pi_1^D, \text{GL}_2(\mathbb{C})) \right]^{\text{GL}_2(\mathbb{C})} \right)$$

the GIT quotient by the conjugation action. The set χ_{Θ}^D is called the *character variety* associated to Rep_{Θ}^D .

Proposition 1.4.1 (Logarithmic RH Correspondence). *Considering the polar divisor $D = \sum_{i=1}^n [x_i]$ and a generic (1.3) set of residual spectral data Θ , the monodromy map*

$$\text{Mon}: \mathfrak{Con}_{\Theta}^D \rightarrow \chi_{\Theta}^D$$

is a bijection.

Proof. First of all we notice that Propositions 1.3.12 and 1.3.13 holds for the whole Riemann sphere, assuring that the monodromy map is well defined. In fact it does not depend on the basis chosen for E , and equivalent connections induce the same monodromy representation.

Injectivity: Let us now consider $[\rho] \in \chi_{\Theta}^D$. We suppose that both (E, D, ∇) and (E', D, ∇') have $[\rho]$ as (conjugacy class of) monodromy representation. Up to a basis change for ∇ , we can suppose that both connections have exactly the same monodromy representation ρ . We want to show that they are gauge equivalent. Let us consider Y and Y' two local solution matrices respectively for ∇ and ∇' in a open set U not containing any singularities. The transformation $\Psi_U := Y' \circ Y^{-1}$ is a local gauge transformation conjugating the connections. We can extend Ψ_U to $\mathbb{P}^1 \setminus |D|$ since it has trivial monodromy indeed

$$\Psi_U^{\gamma} = (Y')^{\gamma} \circ (Y^{-1})^{\gamma} = (Y' \circ \rho(\gamma)) \circ (Y \circ \rho(\gamma))^{-1} = Y' \circ Y^{-1} = \Psi_U.$$

Let us call Ψ this holomorphic extension. We have to show that it extends holomorphically to the poles. Let us suppose that one pole is zero and let us consider a small disk $\mathbb{D}$ containing just this singularity. Solutions matrices Y_0 and Y'_0 induce trivializations such that ∇ and ∇' are in their Poincaré-Dulac form. We get to the following commutative diagram:

$$\begin{array}{ccc} (E, D, \nabla)_{\mathbb{D}} & \xrightarrow{\Psi} & (E', D, \nabla')_{\mathbb{D}} \\ Y \uparrow & & Y' \uparrow \\ (\mathbb{D}^*, [0], d) & \xrightarrow{\hat{\Psi}} & (\mathbb{D}^*, [0], d) \end{array}$$

Since ∇ and ∇' share the same monodromy, the normal form is the same and $\hat{\Psi} = (Y')^{-1} \circ \Psi \circ Y$ is then a symmetry of the normal form over $\mathbb{D}^*$ and, by Proposition 1.2.4

of [38], it extends to the whole $\mathbb{D}$.

Surjectivity: Let $[\rho] \in \chi_{\Theta}^D$. We want to build a connection (E, D, ∇) that realizes ρ as monodromy representation. Let us set $X := \mathbb{P}^1 \setminus \{x_1, \dots, x_n\}$ and $\pi: \tilde{X} \rightarrow X$ its universal cover. Consider now the trivial bundle $\tilde{X} \times \mathbb{C}^2$. We can act on it via deck transformations in the following way:

$$\begin{aligned} \pi_1(X) \times (\tilde{X} \times \mathbb{C}^2) &\rightarrow (\tilde{X} \times \mathbb{C}^2) \\ (\gamma, (\tilde{x}, Y)) &\mapsto (\gamma \cdot \tilde{x}, \rho(\gamma)Y). \end{aligned}$$

This action is discrete and free and the quotient $E := (\tilde{X} \times \mathbb{C}^2)/\pi_1(X)$ inherit the structure of a holomorphic vector bundle on X . The trivial holomorphic connection $(\tilde{X} \times \mathbb{C}^2, \emptyset, d)$ is invariant under the action because $d(\rho(\gamma)Y) = \rho(\gamma)dY$. It induces then a well defined holomorphic connection ∇ on E . By construction the monodromy representation is ρ . We have done half of the job, since we have realized $[\rho] \in \chi_{\Theta}^D$ as a holomorphic connection on X , while we would like to realize it as a meromorphic connection (E', D, ∇') on $\mathbb{P}^1$. Consider then the punctured disk $\mathbb{D}_i \subseteq \mathbb{P}^1$ containing only the point x_i and denote $\mathbb{D}_i^* := \mathbb{D}_i \setminus \{x_i\} \subseteq X$. The local monodromy of $\nabla_{\mathbb{D}_i^*}$ can be realized by the monodromy of a logarithmic connection $(E'_i, [x_i], \nabla'_i)$ on $\mathbb{D}_i$ having a simple pole at x_i and spectrum as prescribed by Θ (Corollary 1.3.15). By injectivity, $\nabla_{\mathbb{D}_i^*}$ and $\nabla'_{\mathbb{D}_i^*}$ are gauge equivalent, and we can patch $(E, \emptyset, \nabla)$ with the meromorphic local connections $(E'_i, [x_i], \nabla'_i)$ over any of the small disks $\mathbb{D}_i$ getting finally the logarithmic connection (E', D, ∇') . $\square$

1.5 An Example

We have seen that a meromorphic connection on $\mathbb{P}^1$ is the datum of a differential operator ∇ , a (rank 2) holomorphic vector bundle E and a polar divisor D . In some sense, we can say that the "complexity" of a rank 2 meromorphic connection over $\mathbb{P}^1$ is given by the degree of the polar divisor D . Indeed, if the connection is logarithmic, it counts the number of singularities.

Since there exists no connection on the Riemann sphere with polar divisor of degree 1 (Poincaré-Hopf theorem), the easiest case is $\deg D = 2$ called Euler systems. Since we are mainly interested to monodromy, we study in details this cases up to gauge equivalence. We remark that PV connections, defined in Section 4.2, are meromorphically

gauge equivalent to a Heun equation, characterized by $\deg D = 4$.

Euler Systems

Let us consider the trivial bundle $E = \mathcal{O} \oplus \mathcal{O}$ and a connection $(E, [p] + [q], \nabla)$. Up to Moebius transformations, we can suppose that $p = 0$ and $q = \infty$. We know that the connection globally writes as $\nabla = d + \Omega$, and in the local chart U_0 it has principal form (not necessarily diagonal)

$$\Omega = A \frac{dx}{x}, \quad \text{with } A \in \mathfrak{gl}_2(\mathbb{C}). \quad (1.4)$$

It is consistent because to find the principal form in U_∞ we can apply the coordinate change $z = 1/x$. Note that on the trivial bundle the cocycle is just the identity.

$$A \frac{dx}{x} = Az d\left(\frac{1}{z}\right) = -Az \frac{dz}{z^2} = -A \frac{dz}{z}$$

We find hence back the simple pole at infinity as desired. Let us call $\{\kappa_0^+, \kappa_0^-\}$ the eigenvalues of A .

Remark 15. The Euler systems satisfy the Fuchs' relation, indeed:

$$\sum_{a \in |D|} \kappa_a^+ + \kappa_a^- = \kappa_0^+ + \kappa_0^- + \kappa_\infty^+ + \kappa_\infty^- = 0 = -\deg E$$

since the residual matrix for the singularity $x = \infty$ is $-A$ and hence its spectrum is the opposite to the one of A .

In the chart U_0 the solutions take the form

$$Y = x^A := \exp(A \log x)$$

that is a multi-valued function (because of the nature of the complex logarithm). We notice also that any other solution of the system take the form $Y' = YB$ for a matrix $B \in \text{GL}_2(\mathbb{C})$. Indeed, if Y is a solution then Y' solves the system too: $dY' = d(YB) = dYB = -\Omega YB = -\Omega Y'$.

We wonder how gauge transformations acts on Euler systems.

Proposition 1.5.1. *Up to holomorphic gauge transformation, there exists exactly one Euler system for any generic residual spectral datum $\Theta := \{\kappa_0^+, \kappa_0^-\}$.*

Sketch of proof. First of all we notice that, up to a Moebius transformation, we can always suppose the singularities to be $x = 0, \infty$. Then, thanks to Theorem 1.3.9, we now that in a neighborhood of $x = 0$ we can suppose the system to be in its Poincaré-Dulac form, and the same holds for the singularity at $x = \infty$. The Fuchs' relation and the coordinate change tells us that $\{\kappa_0^+, \kappa_0^-\} = \{-\kappa_\infty^+, -\kappa_\infty^-\}$. $\square$

In particular, the moduli space of Euler system with prescribed spectral data Θ , that we denote $\mathfrak{Eul}_\Theta$, is just a point. In this case the Riemann-Hilbert correspondence is still true, but way less interesting since it associate to the point $\mathfrak{Eul}_\Theta$ its monodromy representation class. What is interesting is that the simplicity of Euler systems allows us to directly compute such representation. Let us consider the solution $Y(x) = x^A := \exp(A \ln x)$ and the loop $\gamma(t) := x e^{2i\pi t}$ turning around 0. We can study the solution restricted to this path:

$$\begin{aligned} Y(x \exp(2i\pi t)) &= \exp\left(A \ln(x \exp(2i\pi t))\right) = \\ &= \exp\left(A \ln x + A \ln(\exp(2i\pi t))\right) = Y(x) \cdot \exp(2i\pi t A). \end{aligned}$$

We obviously have that $\gamma(0) = \gamma(1) = x$, but $Y(\gamma(0)) = Y(x)$ and $Y(\gamma(1)) = Y(x) \cdot \exp(2i\pi A)$. The monodromy matrix is then $M_0 = \exp(2i\pi A) \in \mathrm{GL}_2(\mathbb{C})$. In the other chart we get the monodromy matrix $M_\infty = \exp(-2i\pi A) = M_0^{-1}$. Since $\pi_1(\mathbb{P}^1 \setminus \{0, \infty\}) \cong \pi_1(\mathbb{C}^*) \cong \mathbb{Z}$, we have the following monodromy representation

$$\begin{aligned} \rho_\Theta: \mathbb{Z} &\rightarrow \mathrm{GL}_2(\mathbb{C}) \\ n &\mapsto \exp(2i\pi n A). \end{aligned}$$

PAINLEVÉ V TYPE CONNECTIONS

We introduce now the main characters of this work: connections of PV type. They are the simplest examples of meromorphic connections with polar divisor of degree four, which present an irregular singularity. They are the first degeneracy of the very famous and studied Painlevé VI type connections, see for instance [35].

Definition. A meromorphic connection (∇, E, D) over $\mathbb{P}^1$ is of *Painlevé type* if: $\text{rank} E = 2$, and the minimal polar divisor within its meromorphic gauge equivalence class has degree four.

Remark 16. A complete classification of Painlevé type connections can be found in Section 1.4 of [46].

Definition. A Painlevé type connection is of *PV type* if the minimal divisor within its meromorphic gauge equivalence class is Moebius equivalent to $D^V := [0] + 2[1] + [\infty]$.

Painlevé type connections with fixed reduced polar divisor and residual spectral data are the simplest examples of meromorphic connections with a positive dimensional moduli space (w.r.t. meromorphic gauge equivalence). Indeed, the dimension of the moduli space depends on the degree of the polar divisor and on the rank of the vector bundle and we have seen in Section 1.5 that the moduli space of rank 2 meromorphic connection with reduced polar divisor of degree less than four is just a point. For rank 2 connections, we then need a polar divisor of degree at least four to have a moduli space of positive dimension.

2.1 Induced Riccati Foliation

Any rank 2 meromorphic connection (E, D, ∇) over $\mathbb{P}^1$ induces a singular Riccati foliation on the projective bundle $\mathbb{P}(E)$.

Remark 17. Given the vector bundle E , we can define its projective bundle $\mathbb{P}(E)$. The fibers of $\mathbb{P}(E)$ are just the projectivisation of the fibers of E . The cocycle of E descends to the quotient, since the action is linear. Finally, given a fiber E_a for $a \in \mathbb{P}^1$, we will denote by $(a, p) = (a, [1 : p]) \in F_a := \mathbb{P}(E_a)$ the point representing the line $\langle (1, p) \rangle \in E_a$.

Since ∇ is a $\mathbb{C}$ -linear morphism, it descends to the projectivisation. Indeed, in any open set U trivialising the connection, we have an expression in coordinates $\nabla = d + \Omega_U$ and let $Y = (y_1, y_2)$ an horizontal section. We know that $\mathbb{P}(E|_U) \cong U \times \mathbb{P}^1$, and let $[1 : y] = [y_1 : y_2]$ be the coordinate on the fibers. Then:

$$d \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} \omega_{1,1} & \omega_{1,2} \\ \omega_{2,1} & \omega_{2,2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 0 \quad \implies \quad dy = d \left(\frac{y_2}{y_1} \right) = \frac{dy_2 \cdot y_1 - dy_1 \cdot y_2}{y_1^2} = \omega_{1,2} y^2 + (\omega_{1,1} - \omega_{2,2}) y - \omega_{2,1},$$

for some $\omega_{i,j} \in \Omega^1(D|_U)$. Solutions of this Riccati differential equation define a Riccati foliation on $\mathbb{P}(E|_U)$. As one can expect, the following result holds.

Proposition 2.1.1. *Holomorphic gauge equivalent connections induce biholomorphic Riccati foliations.*

Proof. Section 1.4.3 of [35]. □

A fundamental consequence of this result and Proposition 1.3.7 is that any meromorphic gauge class of Painlevé V connection can be represented as a Riccati foliation over $\mathbb{F}_1 := \mathbb{P}(\mathcal{O}(-1) \oplus \mathcal{O})$, that is the so called first Hirzebruch surface. We recall that Hirzebruch surfaces $\mathbb{F}_k$ are smooth algebraic ruled surfaces birational to $\mathbb{P}^1 \times \mathbb{P}^1$. Additionally, they can be interpreted as the compactification of the total space of the line bundle $\mathcal{O}(k)$, obtained by adding a so-called "section at infinity". Hirzebruch surfaces have a very rich birational geometry, the interested reader will find more details in [3]. A geometrical property that we will use in the following is that in each Hirzebruch surface $\mathbb{F}_k$ there is a unique curve of self intersection $-k$: it corresponds to the line $\mathbb{P}(\mathcal{O}(k)) \subseteq \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(k))$, which is exactly the "section at infinity" we mentioned before to compactify the total space of $\mathcal{O}(k)$.

For the following, it is useful to understand how elementary transformations act on the Riccati foliation (see also Section 1.5 of [27]). Consider a point $x_0 \in U \subseteq \mathbb{P}^1$ and $\{e_1, e_2\}$ a basis of $E|_U \cong U \times \mathbb{C}$ such that the elementary transformations have the coordinate

expression as in the formula 1.1. If we consider a section $Y = (y_1, y_2)$, then

$$\text{elm}_{x_0}^+(Y) = \begin{pmatrix} x - x_0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} (x - x_0)y_1 \\ y_2 \end{pmatrix}.$$

In the projectivisation, we have hence that

$$\text{elm}_{x_0}^+[y_1 : y_2] = [(x - x_0)y_1 : y_2],$$

and we see that on the fiber above x_0 , any local section is forced to pass through the point $[0 : 1]$. In fact, an elementary transformation is nothing more than a flip.

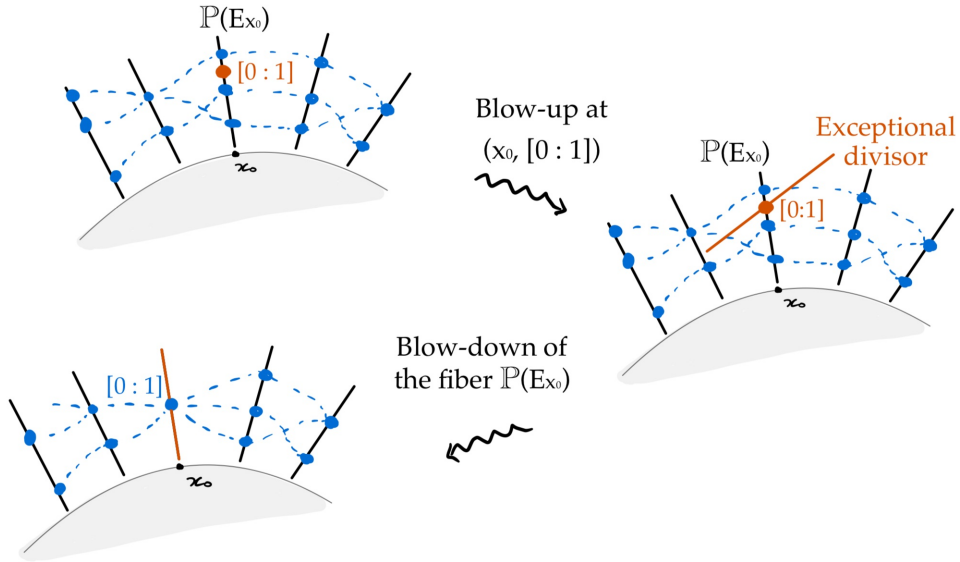


Figure 2.1 – Titre plus long avec des explications.

Finally, as a consequence of Proposition 1.3.5, we deduce that an elementary transformation is a birational morphism between $\mathbb{F}_k$ and $\mathbb{F}_{k\pm 1}$.

Let us study in detail the case of PV type connections. Let $(\mathcal{O}(-1) \oplus \mathcal{O}, D, \nabla)$ be a PV type connection. The same reasoning used in [37], Section 5, for a Painlevé VI type connection, shows that the Riccati foliation induced on $\mathbb{F}_1$ has exactly one tangency point with the section at infinity. Let us call q the projection over $\mathbb{P}^1$ of that point. We can then apply an elementary transformation based in the tangency point, landing then in $\mathbb{F}_2$. All the leaves of the new Riccati foliation need then to pass through a certain point $(q, [1 : \hat{p}]) \in \mathbb{F}_2$, rising a radial singularity. The value of $\hat{p}$ is determined by the

curvature of the leaf in $\mathbb{F}_1$ at its tangency point with s_∞ .

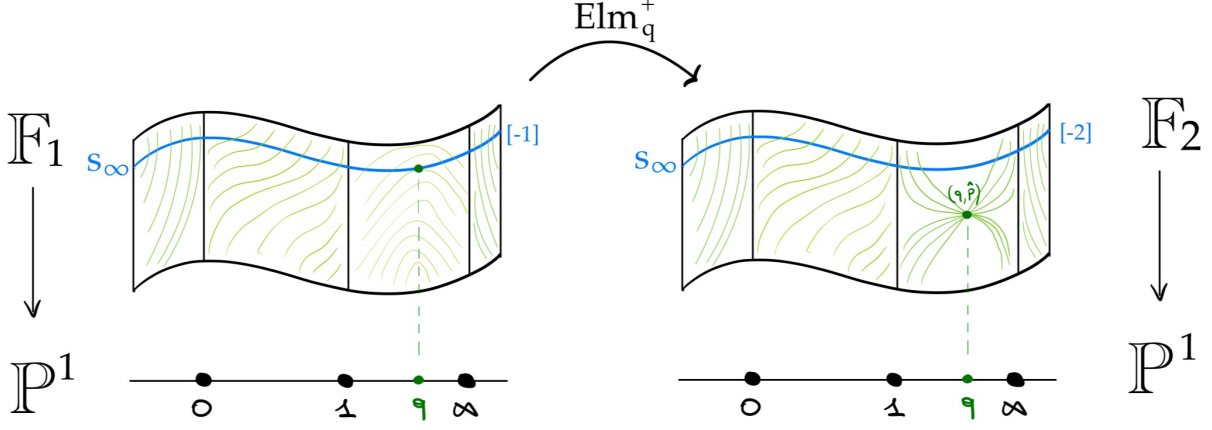


Figure 2.2 – The action of an elementary transformation on the Riccati foliation.

2.2 Normal Form

The goal of this section is to use the action of meromorphic gauge transformations to find a suitable representative in each isomorphism class of connections of PV type. This points out the link with the Riccati foliation, becoming the cornerstone for the construction of the moduli space.

In a recent paper, Diarra and Loray [19] proved that we can choose a special representative by considering a connection in $E = \mathcal{O} \oplus \mathcal{O}(2) \rightarrow \mathbb{P}^1$ in companion form:

$$\nabla|_0 = d + \Omega_0 = d + \begin{pmatrix} 0 & 1 \\ 0 & t \end{pmatrix} \frac{dx}{(x-1)^2} + \begin{pmatrix} 0 & -1 \\ 0 & -\kappa_1 \end{pmatrix} \frac{dx}{x-1} + \begin{pmatrix} 0 & 1 \\ 0 & -\kappa_0 \end{pmatrix} \frac{dx}{x} + \quad (2.1)$$

$$+ \begin{pmatrix} 0 & 0 \\ -\rho^V & 0 \end{pmatrix} x dx + \begin{pmatrix} 0 & 0 \\ \hat{p} & -1 \end{pmatrix} \frac{dx}{x-q} + \begin{pmatrix} 0 & 0 \\ \hat{K} & 0 \end{pmatrix} dx, \quad (2.2)$$

where

- $\kappa_0, \kappa_1, \kappa_\infty$ are generic residual spectral data, as explained in Section 1.3.2;
- $\rho^V = \frac{(\kappa_0 + \kappa_1 - 1)^2}{4} - \frac{\kappa_\infty^2}{4}$;

- $\hat{K} = \frac{\hat{p}^2}{q(q-1)^2} + \frac{(\kappa_0(q-1)^2 + (\kappa_1-1)q(q-1) - tq)\hat{p}}{q(q-1)^2} + \rho^V q + \frac{\hat{p}}{q-1};$
- $t, \hat{p}$ and q are three free parameters uniquely defining the normal form.

Remark 18. In the literature, as in [46], the parameter κ_1 can be found under the form as $\theta + 1$. The advantage of using θ is its more direct link to the Painlevé fifth equation.

Remark 19. This normal form can also be obtained from the connection on $\mathcal{O}(-1) \oplus \mathcal{O}$ presented in [46] by changing the variables (x, y, z) into $(x, q, -\frac{\hat{p}}{q(q-1)^2})$ and by applying the elementary transformation

$$\begin{pmatrix} 1 & 0 \\ 0 & \frac{-1}{x(x-1)^2} \end{pmatrix},$$

landing on $\mathcal{O} \oplus \mathcal{O}(2)$.

Remark 20. The choice of the line bundle $\mathcal{O} \oplus \mathcal{O}(2) \cong \mathcal{O} \oplus \Omega^1(D^V)$ to represent a normal form seems to be quite arbitrary. We state here some motivations.

First of all let us notice that on the bundle $\mathcal{O} \oplus \mathcal{O}(1)$ we can choose a connection matrix of the form

$$\begin{pmatrix} A(x) & B(x) \\ C(x) & D(x) \end{pmatrix} \frac{dx}{P(x)}$$

with $P(x) = x(x-1)^2$ and A, B, C polynomials with some restrictions on the degrees, in particular $\deg(A) \leq 2$ (Section 3 of [19]). This implies that $\mathcal{O} \oplus \mathcal{O}(2)$ is precisely the line bundle with enough automorphisms to kill $A(x)$ and get a connection matrix in companion form.

Second, to pass from $\mathcal{O} \oplus \mathcal{O}(1)$ to $\mathcal{O} \oplus \mathcal{O}(2)$ we can choose precisely the elementary transformation shown in Figure 2.2, making the parameters (q, p) appear in the expression of the connection matrix.

Finally, the projectivisation of $\mathcal{O} \oplus \mathcal{O}(2)$ is the second Hirzebruch surface $\mathbb{F}_2$ that will play a central role in the following as an Okamoto space.

An apparent singularity in $x = q$ arises. It corresponds to the radial singularity of the Riccati foliation in $\mathbb{F}_2$ shown in Figure 2.2.

Definition. A singularity for a meromorphic connection is said *apparent* if it can be removed via a (meromorphic) gauge transformation.

Remark 21. Apparent singularities have nilpotent or diagonalizable residual matrix with integer eigenvalues and trivial monodromy.

We have now more material for "decorating" a little bit more the picture of the Riccati foliation induced by the connection in normal form. Let us consider for instance the residual matrix of the logarithmic pole 0. Its diagonal form is

$$\begin{pmatrix} 0 & 1 \\ 0 & -\kappa_0 \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{\kappa_0} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & -\kappa_0 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{\kappa_0} \\ 0 & 1 \end{pmatrix},$$

pointing out the two eigenvalues 0 and $-\kappa_0$, with the respective eigenvectors. We can then point out the two eigenvectors in the fiber over 0, represented by two points in $\mathbb{F}_2$, respectively $(0, [1 : 0])$ and $(0, [1 : -\kappa_0])$, in red in Figure 2.3. The same (with much more involved computations) can be applied for the other logarithmic pole at ∞ . In the following of this section, the local coordinates on a trivialising set $U \subseteq \mathbb{F}_2$ are then called $(x, y) \in \mathbb{P}^1 \times \mathbb{P}^1$.

Remark 22. There is a geometrical interpretation of these points, making the link with the Riccati foliation more explicit. Indeed, considering the induced Riccati differential equation in a small neighborhood near $x = 0$ yields

$$\frac{dy}{dx} = -\frac{1}{x}y^2 + \frac{\kappa_0}{x}y \quad \text{and, for } x = 0, \quad 0 = -y^2 + \kappa_0 y,$$

whose solutions are precisely 0 and $-\kappa_0$. They correspond to the first terms of the Taylor expansions of the invariant curves of the foliation in a neighborhood of $x = 0$.

For the double pole, the residual matrix alone is not sufficient: one has to consider the order two matrix as well. It becomes crucial to better understand the fiber over $x = 1$. The local Riccati differential equation is in this case

$$\left[\frac{dy}{dx} = \frac{x-2}{(x-1)^2}y^2 + \frac{-\kappa_1(x-1)+t}{(x-1)^2}y \quad \text{and for } x = 1, \quad 0 = (x-2)y^2 + (-\kappa_1 x + t + \kappa_1)y \right]$$

giving as solutions $y = 0$ and $y = -\frac{t-\kappa_1(x-1)}{x-2}$. Their Taylor expansion of order 2 in $x = 1$ are 0 and

$$t + (t - \kappa_1)(x - 1) + O((x - 1)^2).$$

The special red points in the fiber over $x = 1$ are then $(1, 0)$ and $(1, t) \in \mathbb{F}_2$ (corresponding indeed to the eigenvectors of the second order matrix at $x = 1$), and to get to the other terms of the Taylor expansion we have to blow up both those points getting to

the following picture (in which we omitted the Riccati foliation) that perfectly represents the Okamoto divisor relative to the PV type connections, that is composed by the section at infinity, the singular fibers, and the exceptional divisors $E_{1,0}^\pm$, as shown in the picture below.

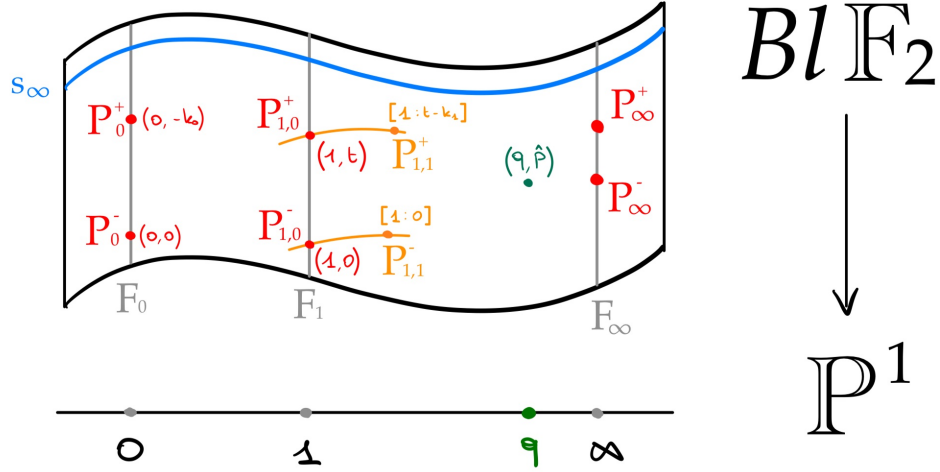


Figure 2.3 – The Okamoto space Ok_t .

Remark 23. We refer to the Okamoto divisor as defined in [49]. We denote by F_a the fiber above $x = a$ and with $E_{1,0}^\pm$ the two exceptional divisors respectively arising from the blowing up of $(1, 0)$ and $(1, t) \in \mathbb{F}_2$, then the Okamoto divisor is

$$Ok^V := 2[s_\infty] + [F_0] + [F_1] + [F_\infty] + [E_{1,0}^+] + [E_{1,0}^-].$$

Note that in [49], each irreducible components of the Okamoto divisor has self intersection -2. This can be achieved if we blow-up the six points $P_0^\pm$, $P_{1,1}^\pm$ and $P_\infty^\pm$. The natural ambient space of the Okamoto divisor is then more likely the one appearing in Figure 2.4.

We finally remark that also the point $(q, \hat{p})$ has, as all the other special red points, the interpretation as a point $q \in \mathbb{P}^1$ and an eigenvector $(1, \hat{p})$ of the residual matrix at $x = q$. In particular we stress that for each PV type connection we have a picture as the one above depending on the values of t, q and $\hat{p}$.

2.3 Confluence of Singularities and Okamoto Space

Let us conclude the present section with a preliminary study about confluence of singularities, that we will use for building the compactification of the moduli space of PV type connections. We wonder what happens when the apparent singularity q converges into a pole. In other words: we wonder under which conditions, for $a = 0, 1, \infty$, the $\lim_{q \rightarrow a} \Omega_0$ does exist, giving a well defined connection matrix and how it looks like.

From the expression of the normal form, we see that for the coefficient

$$\hat{K} := \frac{\hat{p}^2}{q(q-1)^2} + \frac{(\kappa_0(q-1)^2 + (\kappa_1-1)q(q-1) - tq)\hat{p}}{q(q-1)^2} + \rho^V q + \frac{\hat{p}}{q-1},$$

the $\lim_{q \rightarrow a} \hat{K}$ is not well defined for any value of $a = 0, 1, \infty$.

Theorem 2.3.1. *Given a PV type connection in normal form, the confluence of the apparent singularity into a pole $x = a$ produces a new well defined connection if, and only if, the point $(q, \hat{p}) \in \mathbb{F}_2$ follows the trajectory of the invariant curve at $x = a$ of the induced Riccati foliation up to the order r of its Taylor expansion, where $r + 1$ is the order of the pole $x = a$. Moreover, the connection depends on the $(r + 1)$ -th coefficient of the Taylor expansion of the trajectory.*

Proof. We should explicitly compute $\lim_{q \rightarrow a} \Omega_0$, and we already noticed that only the coefficient $\hat{K}$ is divergent.

Let us firstly study the case for $a = 0$, that, we recall, is a logarithmic singularity. In Section 2.2 we proved that the Taylor expansion up to the order 0 of the invariant curves of the Riccati foliation at $x = 0$ are

$$\hat{p} = 0 \quad \text{and} \quad \hat{p} = -\kappa_0.$$

We recall that, since $x = 0$ is a logarithmic pole, the Taylor expansion of order 0 should be sufficient.

We can hence make a substitution into $\hat{K}$ by replacing $\hat{p}$ with $\alpha q + \beta$. Our goal is then to show that the limit $q \rightarrow 0$ is well defined if, and only if, $\beta = 0, -\kappa_0$. It then holds that

$$\text{Res}_{q=0} \hat{K} = \beta(\beta + \kappa_0)$$

that is indeed zero if, and only if, $\beta = 0, -\kappa_0$, as desired. Moreover a straightforward

computation shows that the resulting connection depends on α .

The computations for $a = \infty$ (that is the other logarithmic pole) are the same.

For the pole of order two the procedure is similar, but we have to take in account two terms of the Taylor expansion of the invariant curve:

$$\hat{p} = 0 \quad \text{and} \quad \hat{p} = -\kappa_1(q - 1) + t$$

We can hence make a substitution into $\hat{K}$ by replacing $\hat{p}$ with $\alpha(q - 1)^2 + \beta(q - 1) + \gamma$ and, as before, one can show that the limit $q \rightarrow 1$ is well defined if, and only if, $(\beta, \gamma) = (0, 0), (-\kappa_1, t)$. It then holds that

$$\text{Res}_{q=1} \hat{K} = \gamma^2 + (t + 2\beta - \theta - 1) \gamma + \beta t \quad \text{and} \quad \text{Res}_{q=1} (q - 1) \hat{K} = \gamma(\gamma - t).$$

Moreover a straightforward computation shows that the resulting connection depends on α , concluding the proof. $\square$

We are now ready to state the following result, which will be useful in Section 3.3.3 to understand the whole 3-dimensional moduli space and its compactification.

Corollary 2.3.2. *The moduli space of PV type connections for a fixed value of $t \in \mathbb{C} \setminus \{0\}$ (called the Okamoto moduli space) is given by the complement of the Okamoto divisor inside $\text{Bl}\mathbb{F}_2$, that is the blow-up of $\mathbb{F}_2$ in the eight points $P_i^\pm$ and $P_{1,j}^\pm$ of Figure 2.3.*

Sketch of proof. The normal form tells us that for $q \neq 0, 1, \infty$ and $\hat{p} \neq \infty$, the point $(q, \hat{p})$ uniquely define a PV type connection. The last part of Theorem 2.3.1 shows that for $q = 0, 1, \infty$, there exist PV type connections lying exactly on those exceptional divisors, in green in the following picture. For a complete proof in the Painlevé VI type case see Theorem 2.7.6 of [35]. $\square$

We stress the fact that the coordinates of the points $P_i^\pm$ and $P_{1,j}^\pm$ encodes perfectly the geometry of the invariant curves of Theorem 2.3.1. For instance $P_0^+ = (0, -\kappa_0)$ and $P_0^- = (0, 0)$.

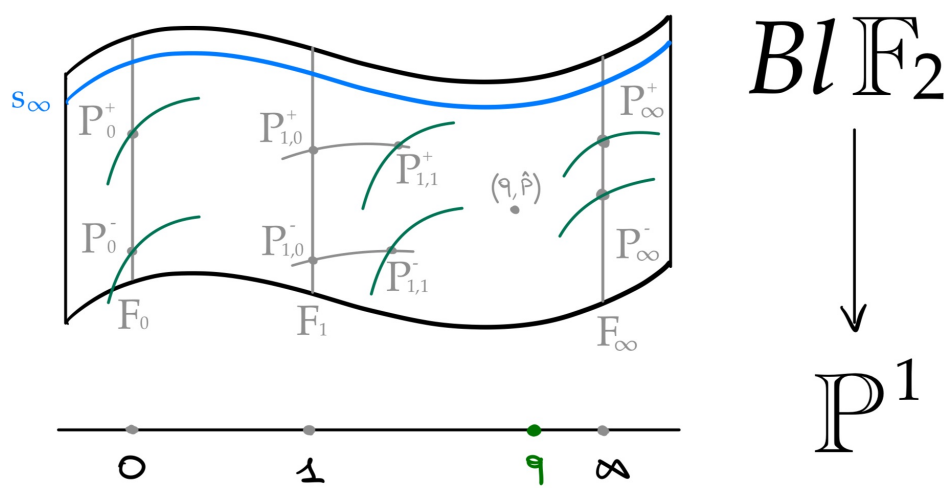


Figure 2.4 – Okamoto space.

MODULI SPACE OF PV CONNECTIONS

3.1 Generic PV Connections

The goal of this section is to study $\mathcal{C}on_{\Theta}^V$: the moduli space of connections of PV type up to gauge equivalence and with fixed generic residual spectral data Θ . As mentioned in Section 2.2, Diarra and Loray [19] proved that, for a generic choice of t and q , we can uniquely choose a particular representative of the gauge equivalence class by considering a connection in normal companion form 2.1 on the vector bundle $\mathcal{O} \oplus \mathcal{O}(2)$. The three free parameters $t, q, \hat{p}$ are then good candidates for being the coordinates in a dense open set of the moduli space of connections of PV type. We can not hope that they parametrize the entire moduli space, since we have seen in the previous section that inside $\mathcal{C}on_{\Theta}^V$ we should allow connections with q equal to 0, 1 or ∞ , and they are not taken into account in the normal form.

In Corollary 2.3.2 we have seen that the moduli space of connection $\mathcal{C}on_{\Theta}^V$ contains the union $\cup_{t \in \mathbb{C}^*} Ok_t$ of Okamoto spaces. The goal is to understand what happens for $t = 0, \infty$.

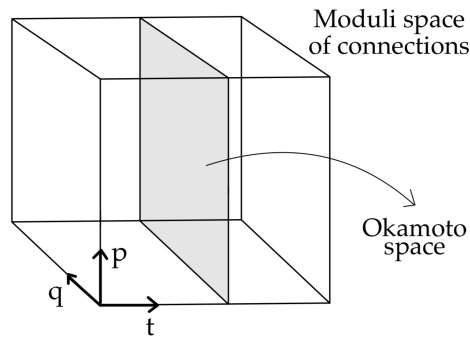


Figure 3.1 – The moduli space of connection as a family of Okamoto spaces.

We already explored the geometric interpretation and the role of q and $\hat{p}$ in Section

2.3. The geometric interpretation of the eigenvalue t seems more mysterious, and it can be understood by studying how it behaves under the action of a change of coordinates in the base $\mathbb{P}^1$.

Lemma 3.1.1. *Let $f \in \text{Aut}(\mathbb{P}^1)$ be a Moebius transformation and (E, D, ∇) a connection of PV type in normal form. Then $f^*\nabla$ is holomorphically gauge equivalent to (E, D, ∇) , with the same residual spectral data and the eigenvalue t is modified by $t \mapsto (Df)(1) \cdot t$.*

Proof. First of all note that the pullback acts on the connection as $(f^*E, f(D), f^*\nabla)$. Since f is an automorphism of the basis $\mathbb{P}^1$, the bundles E and f^*E are equivalent and horizontal sections of ∇ are sent to horizontal sections of $f^*\nabla$ as desired.

The variable change $X = f(x)$ transforms the connection matrix Ω into $f^*\Omega$, and we are interested to study the behavior of the parameter t and of the residual spectral data under this pullback. A straightforward calculation shows that

$$\text{Res}_{X=f(1)}((X - f(1)) \cdot f^*\Omega) = (Df)(1) \cdot \begin{pmatrix} 0 & 1 \\ 0 & t \end{pmatrix}.$$

It implies that the second order matrix relative to the pole $f(1)$ is transformed by the multiplication of the non zero complex number $(Df)(1)$, as desired.

We want now to study the residual matrices at the poles. For any $i \in \{0, 1, \infty\}$, the pole has been moved in $f(i)$. The same computation as before shows that the residues

$$\text{Res}_{X=f(i)}(f^*\Omega) = \text{Res}_{x=i}(\Omega),$$

stay untouched. □

In particular we deduce that t transforms like a vector tangent to $1 \in \mathbb{P}^1$, that is $t \in T_1\mathbb{P}^1 \setminus \{0\}$. To summarise: we consider the rank 2 vector bundle $E = \mathcal{O} \oplus \mathcal{O}(2) \rightarrow \mathbb{P}^1$ with a PV type connection in normal form.

- $q \in \mathbb{P}^1$, the base of the vector bundle,
- $\hat{p} \in \mathbb{P}(E_1)$, the projectivisation of the fiber above 1,
- $t \in T_1\mathbb{P}^1$, the tangent space to $\mathbb{P}^1$ in 1.

In conclusion, we see t, q as elements of the quasi projective variety

$$\mathcal{M} := \left\{ q \in \mathbb{P}^1 \setminus \{0, 1, \infty\}; \quad t \in T_1\mathbb{P}^1 \setminus \{0\} \right\} \cong \mathbb{P}^1 \times \mathbb{P}^1 \setminus (Q^0 \cup Q^1 \cup Q^\infty \cup A_0 \cup A_\infty),$$

Definition. We refer to a *generic PV connection* as a Pv connection belonging to the Zariski open set $\mathcal{M} \times \mathbb{C}$.

Okamoto [47] proposed a compactification of $\mathcal{M} \times \mathbb{C}$ for the confluence of the apparent singularity q into a pole. The goal of this chapter is to build an algebraic compactification $\mathbb{C}on_{\Theta}^V$ that extends the Okamoto's one for $t = 0, \infty$.

3.2 Moduli Space of Rational Irregular Curves

When discussing moduli spaces of configurations of points on the Riemann sphere, the spaces $\mathcal{M}_{0,n}$ naturally come to mind. The space $\mathcal{M}$ considered here may be viewed as a natural generalization of the moduli space $\mathcal{M}_{0,5}$, which parametrizes configurations of five distinct points on $\mathbb{P}^1$. In our setting, however, the fifth marked point is replaced by a tangent vector at one of the points.

Before giving the main definition, let us recall the definition of a jet, as we will use it in the following.

Definition. A r -jet of coordinates over $\mathbb{P}^1$ is a point of $J^r(\mathbb{R}_0, \mathbb{P}^1)$. In particular, given a smooth curve $\gamma: \mathbb{R}_0 \rightarrow \mathbb{P}^1$ and a coordinate of $\mathbb{P}^1$ around $\gamma(0)$, we set $j^r \gamma = (\gamma(0), \gamma^{(1)}(0), \dots, \gamma^{(r)}(0))$, where $\gamma^{(i)}$ is the i -th derivative of γ .

This lead us to give the following definition, that has been introduced in [18] and [6].

Definition. An irregular curve (C, D, J) is the datum of a complex curve C , an effective divisor $D = \sum_{i \in I} a_i [p_i] \in \text{Div}(C)$ and a collection of jets $J = \{j^{a_i-1} \gamma_i\}_{i \in I}$, where $\gamma_i: \mathbb{R}_0 \rightarrow \mathbb{P}^1$ is a smooth curve such that $\gamma_i(0) = p_i$. Sometimes we will omit D or J in the notation.

Remark 25. The quasi projective variety $\mathcal{M}$ is the moduli space of irregular curves $(\mathbb{P}^1, [a] + 2[b] + [c] + [d], J)$, for distinct $a, b, c, d \in \mathbb{P}^1$ and with $J = \{j^0(a) = a, j^1(b) = (b, t), j^0(c) = c, j^0(d) = d\}$.

Remark 26. An automorphisms $\varphi \in \text{Aut}(\mathbb{P}^1)$ acts on $(\mathbb{P}^1, D, J)$ as $(\varphi(\mathbb{P}^1), \varphi_* D, \varphi_* J)$. If we call $p = \gamma(0)$, we have that $\varphi_*(j^k \gamma) = (\varphi(p), D\varphi(p)(\gamma^{(1)}(0)), \dots, D^k \varphi(p)(\gamma^{(k)}(0)))$.

We can now rephrase Proposition 3.1.2 in the following terms.

Proposition 3.2.1. *The moduli space of connection of PV type $\mathfrak{Con}_\Theta^V$ contains as a dense open set the trivial line bundle $\mathcal{M} \times \mathbb{C}$, where $\mathcal{M}$ is the moduli space of rational irregular curves.*

$$\begin{array}{c} [\nabla] \\ \text{generic} \end{array} \iff \left\{ \begin{array}{l} (\mathbb{P}^1, [0] + 2[1] + [\infty] + [q], J) \\ J = \{0, (1, t), \infty, q\} \end{array} \right\} \text{ and } \hat{p} \in \mathbb{C} \iff (t, q, \hat{p}) \in \mathcal{M} \times \mathbb{C}.$$

3.2.1 Deligne-Mumford Compactification of $\mathcal{M}$

If we want to compactify $\mathfrak{Con}_\Theta^V$, and study its birational geometry, Proposition 3.2.1 suggests to understand before how to compactify $\mathcal{M}$. We proceed by the following steps:

- First, we are inspired by Deligne-Mumford's article to compactify $\mathcal{M}$ studying the irregular stable nodal curves lying on Q^i and A_j .
- Next, we describe the connections on such curves. That will be crucial to establish a suitable set of coordinates in $\mathfrak{Con}_\Theta^V$.
- Finally, we understand how the trivial bundle $\mathcal{M} \times \mathbb{C}$ extends over the compactified basis $\overline{\mathcal{M}}$.

We take inspiration by the compactification that M. Kapranov [32] described for $\mathcal{M}_{0,5}$, the moduli space of configurations of five points in $\mathbb{P}^1$.

His main idea is to establish an isomorphism between $\mathcal{M}_{0,5}$ and the moduli space of smooth marked conics in $\mathbb{P}^2$ passing through four given points in general position (with the marking that is outside the four given points). This isomorphism associates to any such marked conic an element in $\mathcal{M}_{0,5}$, that is indeed a $\mathbb{P}^1$ with five punctures. The advantage of this isomorphism is that a compactification of the moduli space of smooth marked conics is much easier to get, and it is then straightforward to deduce the desired compactification $\overline{\mathcal{M}_{0,5}}$.

Remark 27. We can see $\mathbb{P}^1$ with five punctures as a rational irregular curve $(\mathbb{P}^1, [a] + [b] + [c] + [d] + [e])$. Note that the divisor has degree five.

We adapt this reasoning to $\mathcal{M}$, that is the moduli space of rational irregular curves of the form $(\mathbb{P}^1, [0] + 2[1] + [\infty] + [q], J)$ up to Moebius transformations.

We then consider smooth marked conics in $\mathbb{P}^2$ passing through three given points $A, B, C \in \mathbb{P}^2$ in general position and with a prescribed tangent vector $l \in T_A \mathbb{P}^2 \setminus \{0\}$.

Since we want all these data to be "in general position", we ask the tangent direction l not to point to B or C . Up to automorphisms of $\mathbb{P}^2$, we can suppose these data to be

$$A = [1 : 0 : 0], \quad B = [0 : 1 : 0], \quad C = [0 : 0 : 1], \quad l = \overrightarrow{AD} \text{ for } D = [1 : 1 : 1].$$

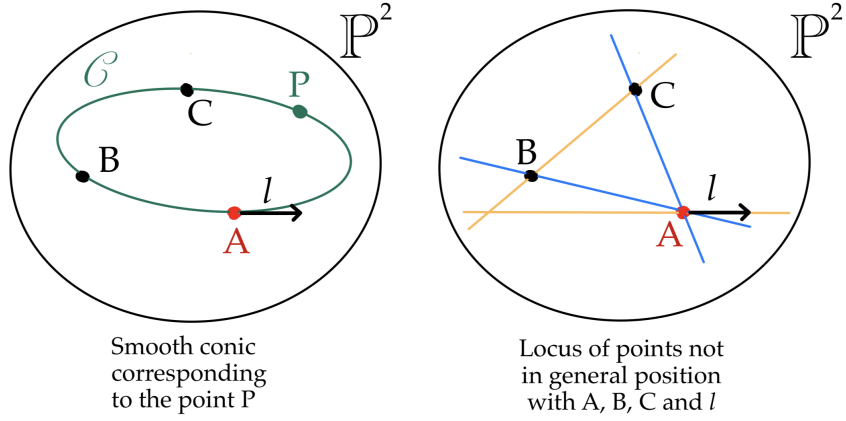
From now on, we denote by $\mathfrak{C} := \mathbb{P}^2 \setminus (\overline{AB} \cup \overline{AC} \cup \overline{BC} \cup \overline{AD})$, where $\overline{AB}$ denotes the unique line in $\mathbb{P}^2$ passing through A and B .

Lemma 3.2.2. *The quasi projective variety $\mathfrak{C}$ is the moduli space of smooth marked conics passing through three fixed point $A, B, C \in \mathbb{P}^2$ in general position, with prescribed tangent direction $l \in T_A\mathbb{P}^2$ not pointing B or C .*

Proof. It is well known that the moduli space of conics in $\mathbb{P}^2$ is isomorphic to $\mathbb{P}^5$ by taking the projectivisation of $\mathbb{C}^6$, the space of the coefficients of the polynomial equation. In particular, a conic has five degrees of freedom. Imposing the four requested conditions, the requirement that the conic pass through an additional point $P \in \mathbb{P}^2 \setminus \{A, B, C\}$ uniquely determines the conic. Moreover, the conic is smooth if, and only if, P is chosen in $\mathfrak{C}$. Finally, let P and P' be two different points defining the same conic C , then the marked conics $(C; A, B, C, P, l)$ and $(C; A, B, C, P', l)$ are not isomorphic since $\dim \text{Aut}(\mathbb{P}^2) = 8$ and hence the only automorphism of $\mathbb{P}^2$ fixing pointwise A, B, C and a $D \in l \setminus \{A\}$ is the identity. $\square$

In the picture below, on the left, we show a conic C represented by a point $P \in \mathfrak{C}$, while, on the right, it is shown that picking a $P \in \mathbb{P}^2 \setminus \mathfrak{C}$ gives rise to a singular conics, represented in yellow and blue. Now, let us construct the explicit isomorphism $\mathcal{M} \cong \mathfrak{C}$. Once we have it, we can build the compactification $\overline{\mathfrak{C}}$ and deduce $\overline{\mathcal{M}}$. We recall that a point $(t, q) \in \mathcal{M}$ represents a class of rational irregular curves, that we would like to identify with a conic in $\mathfrak{C}$. To do so, we have to find a suitable way to associate with each point $P \in \mathfrak{C}$ the parameters (t, q) .

Let us call $[X_0 : X_1 : X_2]$ the coordinates on $\mathbb{P}^2$. Imposing that the conics pass through A, B, C and be tangent to l , we find that points of $\mathfrak{C}$ represent conics of the form $X_1X_2 + a(X_0X_1 - X_0X_2) = 0$ for $a \in \mathbb{C}$. We want to consider these curves with the


 Figure 3.3 – Pencil of conics in $\mathbb{P}^2$.

parametrisation $\mathbb{P}^1 \rightarrow \mathbb{P}^2$ that identifies $0, 1, \infty \in \mathbb{P}^1$ with respectively $B, A, C \in \mathbb{P}^2$:

$$\begin{array}{ccc}
 \mathbb{P}^1 & \dashrightarrow & \mathbb{P}^2 \\
 [s_0 : s_1] & \mapsto & [s_0 s_1 : a s_1 (s_0 - s_1) : a s_0 (s_0 - s_1)] \\
 0 = [0 : 1] & \mapsto & [0 : 1 : 0] = B \\
 1 = [1 : 1] & \mapsto & [1 : 0 : 0] = A \\
 \infty = [1 : 0] & \mapsto & [0 : 0 : 1] = C
 \end{array}$$

Imposing that the conic passes through the point $P = [X_0^0 : X_1^0 : X_2^0]$, we get

$$a = \frac{X_2^0 X_1^0}{X_2^0 X_0^0 - X_1^0 X_0^0}.$$

In fact, the point P plays the role of q in the rational irregular curve, and in order to get its expression, we have to find the preimage of P under the parametrisation.

Since $\mathcal{C} \cap \{X_0 \neq 0\} = \mathcal{C}$, we can work in the affine charts $V_1 := \{s_1 \neq 0\} \subseteq \mathbb{P}^1$ and $U_0 := \{X_0 \neq 0\}$, with coordinates $x_i = X_i/X_0$. The parametrisation becomes

$$\begin{array}{ccc}
 \gamma : & V_1 & \rightarrow U_0 \\
 \frac{s_0}{s_1} = s & \mapsto & \left(\frac{a(s-1)}{s}, a(s-1) \right).
 \end{array}$$

The preimage of P is then

$$q = \gamma^{-1}(P) = \frac{x_2^0}{x_1^0}.$$

The role of t is, of course, played by the tangent vector of the conics in A . We already know that it has the same direction as $\overline{AD}$. A computation shows that

$$\gamma'(1) = (a, a) \quad \text{and then} \quad t = \frac{1}{a} = \frac{x_2^0 - x_1^0}{x_2^0 x_1^0}$$

We get then the following birational map

$$\begin{aligned} \varphi: \quad \mathbb{P}^2 & \dashrightarrow \mathbb{P}^1 \times \mathbb{P}^1 \\ [X_0 : X_1 : X_2] & \mapsto ([X_2 X_0 - X_1 X_0 : X_1 X_2], [X_2 : X_1]) \\ (x_1, x_2) \in U_0 & \mapsto (t, q) = \left(\frac{x_2 - x_1}{x_2 x_1}, \frac{x_2}{x_1} \right) \in V_1 \times V_1 \end{aligned}$$

that is not defined in A, B, C . We prove that

Lemma 3.2.3. *The restricted map $\varphi|_{\mathfrak{C}}: \mathfrak{C} \rightarrow \mathcal{M}$ is an isomorphism.*

Proof. We refer to Figure 3.2 for the notations, recalling that

$$\mathfrak{C} \cong \mathbb{P}^2 \setminus (\overline{AB} \cup \overline{AC} \cup \overline{BC} \cup \overline{AD}) \quad \text{and} \quad \mathcal{M} \cong \mathbb{P}^1 \times \mathbb{P}^1 \setminus (Q^0 \cup Q^1 \cup Q^\infty \cup A_0 \cup A_\infty).$$

We have that $\varphi|_{\mathfrak{C}}$ is everywhere well defined and injective, and hence it induce an isomorphism into its image. It suffice to show $\varphi(\mathfrak{C}) = \mathcal{M}$. First of all we remark that Q^0, Q^1, Q^∞ and A_∞ are not in the image of φ . Moreover $A_0 = \varphi(\overline{BC})$ and $\overline{BC} \not\subset \mathfrak{C}$. The other lines not appearing in $\mathfrak{C}$ are such that

$$\varphi(\overline{AB}) = ([1 : 0], [0 : 1]) = (\infty, 0) = A_\infty \cap Q^0,$$

$$\varphi(\overline{AC}) = ([1 : 0], [1 : 0]) = (\infty, \infty) = A_\infty \cap Q^\infty, \quad \varphi(\overline{AD}) = ([0 : 1], [1 : 1]) = (0, 1) = A_0 \cap Q^1.$$

And then $\varphi(\mathfrak{C}) = \mathcal{M}$ as desired. $\square$

We construct $\overline{\mathfrak{C}}$ by identifying the limit conics that lie on the boundary components.

First Step:

We can easily add to $\mathfrak{C}$ the points that lie on $(\overline{AB} \cup \overline{AC} \cup \overline{BC} \cup \overline{AD}) \setminus \{A, B, C\}$ by taking into account singular conics that represent the locus on points that are not in general

position with the data, as shown on the right part of Figure 3.3.

Second Step:

Since we are less comfortable working with tangent vectors, we can blow-up the point $A \in \mathbb{P}^2$, and call L the point in the exceptional divisor E_A corresponding to the direction l , as shown in the following picture. Notice that, now, the condition of tangency has been translated by the condition of "passing through L ".

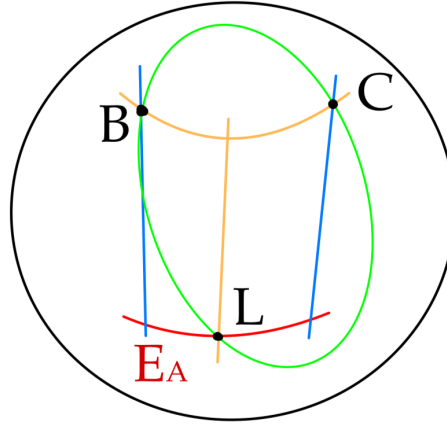


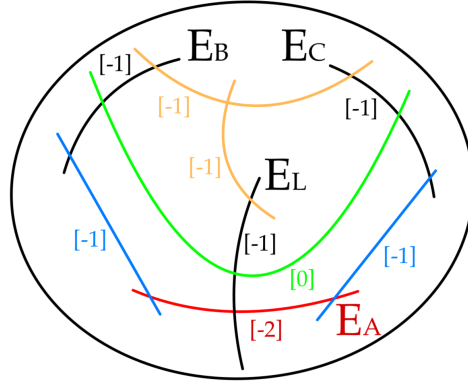
Figure 3.4 – Blow up of the point A .

Third Step:

In order to distinguish conics that pass through B , C and L by the slope of incidence, let us blow-up these three points, denoting respectively by E_B , E_C and E_L the exceptional divisors. In the following picture we denote into brackets the self-intersection number of any shown curve.

Lemma 3.2.4. *Let us denote by X the blowup of $\mathbb{P}^2$ shown in Picture 3.5 and by P_1, P_2 the intersection points between E_A and the blue lines. There is a bijection between $(X \setminus E_A) \cup \{P_1, P_2\}$ and the set of marked conics in $\mathbb{P}^2$ passing through A, B, C and with prescribed tangent l , with the exception that P_1 and P_2 define isomorphic marked conics. In particular $\overline{\mathcal{C}}$ is isomorphic to the (singular) space obtained by the contraction of the (-2) -curve E_A inside X .*

Proof. Recalling that the exceptional divisor E_A represents the incidence slopes into A , the first part follows by construction since we only consider conics incident in A with the prescribed slope l . Finally, the points P_1 and P_2 induce isomorphic marked conics, since we can exchange the two branches via an automorphism. As a consequence, we can contract E_A to P_1 giving the thesis. $\square$

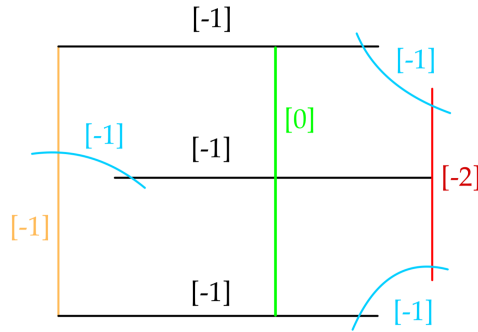

 Figure 3.5 – Blow-up of B, C and L and self intersection numbers.

We postpone to the next section the study of the stable nodal curve corresponding to points lying in the boundary components of $\overline{\mathcal{C}}$.

We are now ready to build the compactification $\overline{\mathcal{M}}$.

Lemma 3.2.5. *The blowup of $\mathbb{P}^1 \times \mathbb{P}^1$ at the points $(0, 1), (\infty, 0)$ and (∞, ∞) give rise to a smooth variety $\widetilde{\mathcal{M}}$ that is isomorphic to the variety X appearing in Lemma 3.2.4. Moreover, the isomorphism φ of Lemma 3.2.3 lifts to an isomorphism $\widetilde{\varphi}: \widetilde{X} \rightarrow \widetilde{\mathcal{M}}$.*

Proof. The following picture shows the intersection numbers of the curves arising in $\widetilde{\mathcal{M}}$. Both $\widetilde{\mathcal{M}}$ and X are then isomorphic to a weak Del Pezzo surface of degree 5 and


 Figure 3.6 – The space $\widetilde{\mathcal{M}}$.

since they have the same number of (-1) and (-2) -curves, they are then isomorphic, since the moduli space of such weak Del Pezzo surfaces is just a point as shown in [25]. We have now to show that φ lifts to an isomorphism between X and $\widetilde{\mathcal{M}}$ as shown in Figures 3.5 and 3.6. We already noticed in Lemma 3.2.3 that $\varphi(\overline{BC}) = A_0$. Let us call

E_A, E_B and E_C the exceptional divisors of the points A, B and C respectively. It holds that

$$\begin{aligned}\varphi(E_A) &= \lim_{X_1 \rightarrow 0} \varphi([1 : X_1 : \alpha X_1]) = \lim_{X_1 \rightarrow 0} ([\alpha X_1 - X_1 : \alpha X_1^2], [\alpha X_1 : X_1]) = \\ &= \lim_{X_1 \rightarrow 0} ([\alpha - 1 : \alpha X_1], [\alpha : 1]) = ([1 : 0], [\alpha : 1]) = A_\infty.\end{aligned}$$

With a similar computation we find that $\varphi(E_B) = Q^0$ and $\varphi(E_C) = Q^\infty$.

In Lemma 3.2.3, we have also shown that $\varphi(\overline{AB}) = (\infty, 0)$, $\varphi(\overline{AC}) = (\infty, \infty)$ and $\varphi(\overline{AD}) = (0, 1)$. We now blow-up these points in $\mathbb{P}^1 \times \mathbb{P}^1$ and we show that the images of the lines in $\mathbb{P}^2$ fit perfectly the exceptional divisors. We have that

$$\text{Bl}_{(\infty, \infty)}(\mathbb{P}^1 \times \mathbb{P}^1) = \{([t_1 : t_2], [q_1 : q_2], [R_1 : R_2]) \in (\mathbb{P}^1)^3 \mid t_2 R_1 = q_2 R_2\}.$$

Therefore $\varphi(\overline{AC}) = \varphi([1 : 0 : R]) = ([1 : 0], [1 : 0], [R : 1]) = B_\infty^\infty$. And the same holds for the two other points. This concludes the proof. $\square$

This implies the following result.

Theorem 3.2.6. *The compactification of the moduli space of stable nodal curves $\overline{\mathcal{M}}$ is given by the contraction of the (-2) -curve A_∞ inside $\widetilde{\mathcal{M}}$.*

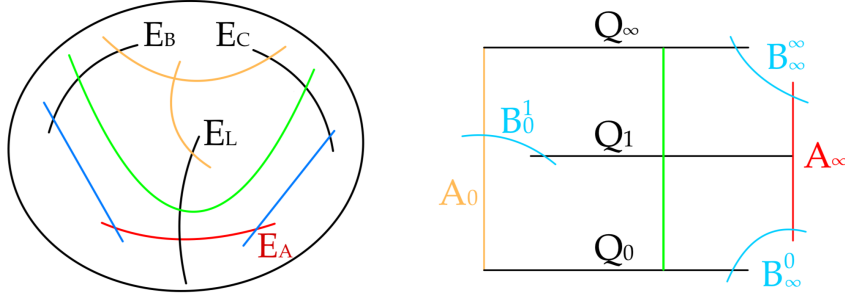
Proof. We proved in Lemma 3.2.3 that $\mathcal{M} \cong \mathbb{C}$. Then we deduced in Lemma 3.2.4 the compactification $\overline{\mathbb{C}}$. Following the same path for $\mathcal{M}$, we get to the variety $\widetilde{\mathcal{M}}$ that we showed being isomorphic to X in Lemma 3.2.5, identifying A_∞ with E_A , that is the curve we contracted in X to get to $\overline{\mathbb{C}}$. $\square$

We point out the analogy of the curves in $\widetilde{\mathcal{M}}$ and X . In order to make the comparison clear, let us give a name to some of the curves in $\widetilde{\mathcal{M}}$ that are needed later on in this article.

$$\begin{array}{llll} E_B & \rightsquigarrow & Q^\infty & \{\text{Blue singular conic}\} \rightsquigarrow B_\infty^0 \cup B_\infty^\infty \\ E_L & \rightsquigarrow & Q^1 & \text{and } E_A \rightsquigarrow A_\infty \\ E_C & \rightsquigarrow & Q^0 & \{\text{Yellow singular conic}\} \rightsquigarrow A_0 \cup B_0^1 \end{array}$$

Where B_j^i is the exceptional divisor relative to the point $(t, q) = (j, i)$. We end this section with the following result that describes the singularities arising in $\overline{\mathcal{M}}$.

Proposition 3.2.7. *The space $\overline{\mathcal{M}}$ presents strictly canonical singularities of A_1 -type.*


 Figure 3.7 – Boundary components of $\widetilde{\mathcal{M}}$.

Proof. First of all we recall that $\overline{\mathcal{M}}$ is given by the contraction of the (-2) -curve A_∞ of $\widetilde{\mathcal{M}}$.

If we set (t, q) the local coordinates of $\widetilde{\mathcal{M}}$ centred at $A_0 \cap Q^0$, the volume form $\frac{dt}{t^2} \wedge \frac{dq}{q^2}$ gives the canonical divisor

$$K_{\widetilde{\mathcal{M}}} \equiv -2(Q^0 + A_0) - B_0^1 - B_0^0 + B_\infty^\infty.$$

In particular it holds that

$$(K_{\widetilde{\mathcal{M}}} \cdot A_\infty) = 0,$$

and the singularity arising in $\overline{\mathcal{M}}$ is hence canonical. To show that are strictly canonical, we can compute the discrepancy: let $\pi: \widetilde{\mathcal{M}} \rightarrow \overline{\mathcal{M}}$ the contraction. Then

$$K_{\widetilde{\mathcal{M}}} \equiv \pi^* K_{\overline{\mathcal{M}}} + \alpha A_\infty,$$

and computing the intersection product with A_∞ we find that

$$0 = (K_{\widetilde{\mathcal{M}}} \cdot A_\infty) = (\pi^* K_{\overline{\mathcal{M}}} \cdot A_\infty) - 2\alpha,$$

and $(\pi^* K_{\overline{\mathcal{M}}} \cdot A_\infty) = 0$ since A_∞ has been contracted to a point. This implies that $\alpha = 0$, as desired. $\square$

3.2.2 Irregular Stable Nodal Curves

We would like to extend the universal curve over $\mathcal{M}$ (Figure 3.2) to the boundary components of $\overline{\mathcal{M}}$. To do so, we introduce the Deligne and Mumford's notion of stable

nodal curve used in the compactification of the moduli spaces $\mathcal{M}_{g,n}$. This study will be relevant to understand how the connections on the boundary components of $\overline{\mathfrak{Con}}_{\Theta}^V$ are made, since they are connections over stable nodal curves, as we will see in the next section.

Definition. An irregular curve (C, D, J) , as defined at the beginning of Section 3.2, is *stable* if $\text{Aut}(C, D, J)$ is finite, where each element of $\text{Aut}(C, D, J)$ must fix pointwise J .

Definition. A complex curve C is called a *nodal curve* if it has only nodes as singularities.

Fact. Let C be a nodal curve with a nodal singularity at the point p and let C_1, C_2 be the two smooth components. If we do not allow the branches to be exchanged, then $\text{Aut}(C) \cong \text{Aut}(C_1; p) \times \text{Aut}(C_2; p)$. Notice that the action $\text{Aut}(C) \curvearrowright C$ decomposes into the two separated actions $\text{Aut}(C_1) \curvearrowright C_1$ and $\text{Aut}(C_2) \curvearrowright C_2$.

Definition. An *irregular nodal curve* is a triple (C, D, J) , where C is a nodal curve with smooth components $C_1, \dots, C_n$, D is a collection of divisors $D_1, \dots, D_n$ and J a collection $J_1, \dots, J_n$ of jets such that, for each $i = 1, \dots, n$, the pair (C_i, D_i, J_i) is a smooth irregular curve.

Definition. The automorphism group $\text{Aut}(C, D, J)$ must fix pointwise the jets in each J_i and the nodes. The irregular nodal curve (C, D, J) is said *stable* if its automorphism group is finite.

Of course, this is a generalisation of the stable nodal curves described in [17] and [1].

As shown in Figure 3.7, the boundary components of $\overline{\mathcal{M}}$ have an interpretation as confluence of points. For example Q^0 is the boundary components that allows curves in which q has converged into 0. Anyway, we do not want the curves over Q^0 just being a $\mathbb{P}^1$ with four markings instead of five: we want to keep the information of the confluence. We recall moreover that we are considering $\overline{\mathcal{M}}$ as a moduli space of punctured curves up to Moebius transformations. This helps us since confluences are not invariant under the action of $\text{Aut}(\mathbb{P}^1)$.

Example 8. Let us consider a Moebius transformation f that sends q to ∞ , and that fixes 0 and t . We remark that such transformation does exist since $\text{Aut}(\mathbb{P}^1)$ is 3-transitive. Some of our markings moved, but $\mathbb{P}^1$ and $f(\mathbb{P}^1)$ are of course in the same equivalence

class. We should compare the confluence $q \rightarrow 0$ in both cases. In the original $\mathbb{P}^1$ the point q is just collapsing into 0, but in $f(\mathbb{P}^1)$ the points $f(1)$ and $f(\infty)$ will collapse together into 1. Roughly speaking, the confluence $q \rightarrow 0$ is Moebius equivalent to the confluence $1 \rightarrow \infty$, and, since $\mathbb{P}^1$ and $f(\mathbb{P}^1)$ are equivalent, we want to keep trace of both configurations using stable nodal curves.

In those kind of curves we will always have two smooth components that will encode the two different behaviours of a confluence, as we will see just below.

3.2.3 Universal Curve

In this section, we study the geometry of stable nodal irregular curves lying on the boundary components of $\overline{\mathcal{M}}$, providing a detailed explanation of what was outlined in Example 8. For each boundary component, we describe the Moebius transformation we choose in order to exhibit the two distinct types of confluence behavior. These Moebius transformation will also play a central role in Section 3.3.1, where we use them to compute the connections that arise on the smooth components of these stable nodal irregular curves.

Curves over A_0 . The boundary A_0 represents those curves with $t = 0$, but it turns out that setting $t = 0$ is not the only possible representation. We recall indeed that we are considering curves up to isomorphism and, in particular, we can apply a Moebius transformation f_{A_0} that fixes 1 and such that $Df_{A_0}(1) \cdot t = 1$. We can then study the limit for $t \rightarrow 0$ and look at what the resulting curve looks like. One such Moebius transformation is

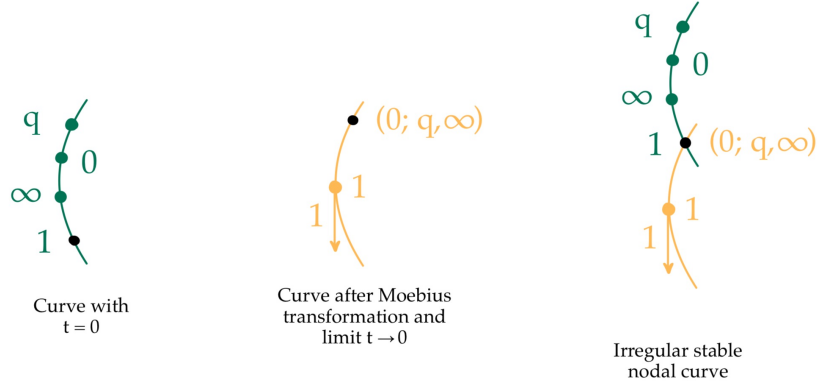
$$f_{A_0}(x) = \frac{(x+1)t}{(t-2)x+t+2}. \quad (3.1)$$

It holds that

$$\lim_{t \rightarrow 0} f_{A_0}(0) = \lim_{t \rightarrow 0} f_{A_0}(q) = \lim_{t \rightarrow 0} f_{A_0}(\infty) = 0.$$

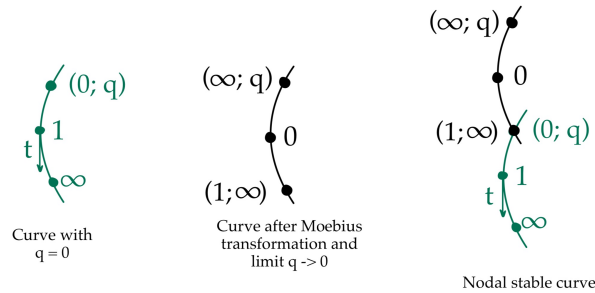
Curves over Q^0 . The boundary Q^0 represents those curves with $q = 0$. We can apply a Moebius transformation that fixes 0 and sends $q \rightarrow \infty$. We can then study the limit for $q \rightarrow 0$ and look at what the resulting curve looks like. One such Moebius transformation is

$$f_{Q^0}(x) = \frac{x}{x-q}. \quad (3.2)$$


 Figure 3.8 – Stable curve in A_0 .

By a straightforward computation it holds that

$$\lim_{q \rightarrow 0} f_{Q^0}(1) = \lim_{q \rightarrow 0} f_{Q^0}(\infty) = 1, \quad \text{and} \quad \lim_{q \rightarrow 0} Df_{Q^0}(1) \cdot t = 1.$$


 Figure 3.9 – Stable curve in Q^0 .

Curves over Q^1 . The boundary Q^1 represents those curves with $q = 1$. We can apply a Moebius transformation that fixes 1 and sends $q \rightarrow 0$. We can then study the limit for $q \rightarrow 1$ and look at what the resulting curve looks like. One such Moebius transformation is

$$f_{Q^1}(x) = \frac{t(x - q)}{(q - 1 + t)x + (-qt - q + 1)}. \quad (3.3)$$

By a straightforward computation it holds that

$$\lim_{q \rightarrow 1} f_{Q^1}(0) = \lim_{q \rightarrow 1} f_{Q^1}(\infty) = 1, \quad \text{and} \quad \lim_{q \rightarrow 1} Df_{Q^1}(1) \cdot t = 1.$$

Curves over Q^∞ . The boundary Q^∞ represents those curves with $q = \infty$. We can apply a Moebius transformation that fixes ∞ and sends $q \rightarrow 0$. We can then study the limit for $q \rightarrow \infty$ and look at what the resulting curve looks like. One such Moebius transformation is

$$f_{Q^\infty}(x) = -\frac{x}{q} + 1.$$

By a straightforward computation it holds that

$$\lim_{q \rightarrow \infty} f_{Q^\infty}(0) = \lim_{q \rightarrow \infty} f_{Q^\infty}(1) = 1, \quad \text{and} \quad \lim_{q \rightarrow \infty} Df_{Q^\infty}(1) \cdot t = 0.$$

The following pictures show how the nodal curves described so far look like. The black and yellow smooth components represent the limit curves obtained after the application of the Moebius transformation.

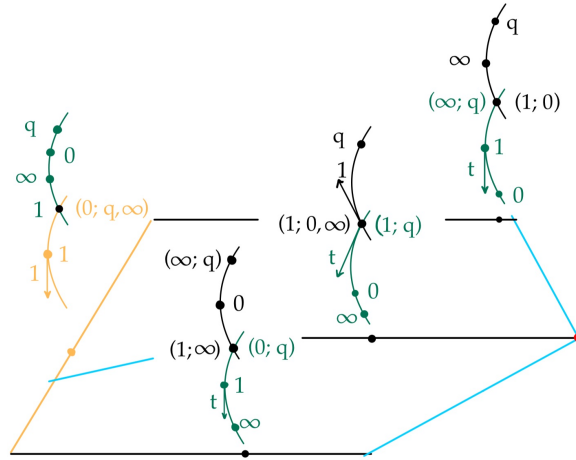


Figure 3.10 – Universal curve.

Remark 28. We see that, as expected, each curve shows only one parameter determining its position on the boundary component.

Curves over B_0^1 . The boundary B_0^1 represents those curves with $t = 0$ and $q = 1$. We denote by $B = \frac{t}{q-1}$ the coordinate on this exceptional divisor. As for the other boundary

components, one can either directly consider the curve with $t = 0$ and $q = 1$, either one can make the substitution $t = B(q - 1)$ and apply a Moebius transformation that fixes 1 and sends $q \rightarrow \infty$. We can then study the limit for $q \rightarrow 1$ and look at what the resulting curve looks like. One such Moebius transformation is

$$f_{B_0^1}(x) = \frac{(1 - q)x}{x - q}.$$

By a straightforward computation it holds that

$$\lim_{q \rightarrow 1} f_{B_0^1}(0) = 0, \quad \text{and} \quad \lim_{q \rightarrow 1} \left(D f_{B_0^1}(1) \cdot B(q - 1) \right) = B.$$

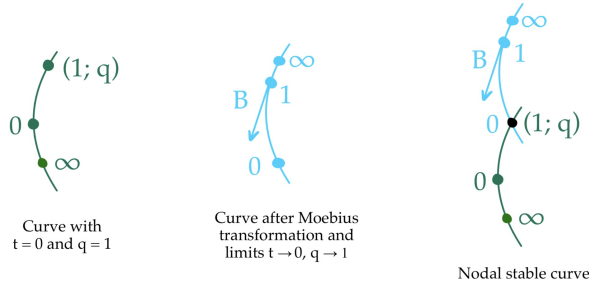


Figure 3.11 – Stable curve in B_0^1 .

Curves over B_∞^0 . The boundary B_∞^0 represents those curves with $t = \infty$ and $q = 0$. We denote by $R = qt$ the coordinate on this exceptional divisor. Unlike the other times, here one can not directly consider the curve with $t = \infty$ and $q = 0$, since we do not know what an infinite tangent vector is. We still have two possible options: either, via a Moebius transformation $f_{B_\infty^0}$, we fix 1, we normalise $t = 1$ and we fix 0, either, with $g_{B_\infty^0}$, we fix 1, we normalise $t = 1$, but we fix ∞ . These two Moebius transformations are

$$f_{B_\infty^0}(x) = \frac{(t + 1)x}{tx + 1} \quad \text{and} \quad g_{B_\infty^0}(x) = \frac{x}{t} + \frac{t - 1}{t}.$$

By a straightforward computation, after, of course, substituting $q = \frac{R}{t}$, it holds that

$$\lim_{t \rightarrow \infty} f_{B_\infty^0} \left(\frac{R}{t} \right) = \frac{R}{R+1} \quad \text{and} \quad \lim_{t \rightarrow \infty} D f_{B_\infty^0}(1) \cdot t = 1$$

$$\lim_{t \rightarrow \infty} g_{B_\infty^0}(0) = \lim_{t \rightarrow \infty} g_{B_\infty^0} \left(\frac{R}{t} \right) = 1 \quad \text{and} \quad \lim_{t \rightarrow \infty} D g_{B_\infty^0}(1) \cdot t = 1$$

Curves over B_∞^∞ . The boundary B_∞^∞ represents those curves with $t = \infty$ and $q = \infty$. We denote by $S = \frac{t}{q}$ the coordinate on this exceptional divisor. As before, we have the same two possible options, and we can use the same functions

$$f_{B_\infty^0}(x) = \frac{(t+1)x}{tx+1} \quad \text{and} \quad g_{B_\infty^0}(x) = \frac{x}{t} + \frac{t-1}{t}.$$

By a straightforward computation, after, of course, substituting $q = \frac{t}{S}$, it holds that

$$\lim_{t \rightarrow \infty} f_{B_\infty^0}(\infty) = \lim_{t \rightarrow \infty} f_{B_\infty^0} \left(\frac{t}{S} \right) = 1 \quad \text{and} \quad \lim_{t \rightarrow \infty} D f_{B_\infty^0}(1) \cdot t = 1$$

$$\lim_{t \rightarrow \infty} g_{B_\infty^0} \left(\frac{t}{S} \right) = \frac{S+1}{S} \quad \text{and} \quad \lim_{t \rightarrow \infty} D g_{B_\infty^0}(1) \cdot t = 1$$

Curve over the point $\overline{A_\infty}$. The point $\overline{A_\infty}$ represents those curves with $t = \infty$ and any value of q . As in the other cases we have two possible options: we can fix 0, 1 and normalize $t = 1$ or fix $\infty, 1$ and normalize $t = 1$. We have then the two following Moebius transformations

$$f_{A_\infty}(x) = \frac{tx}{(t-1)x+1} \quad \text{and} \quad g_{A_\infty}(x) = \frac{x+t-1}{t}.$$

In both cases

$$\lim_{t \rightarrow \infty} f_{A_\infty}(x) = \lim_{t \rightarrow \infty} g_{A_\infty}(x) = 1,$$

and then all the non fixed points converge to 1.

We can resume it with the following picture.

Let us conclude by resuming in the following picture the coordinates we introduced on $\overline{\mathcal{M}}$.

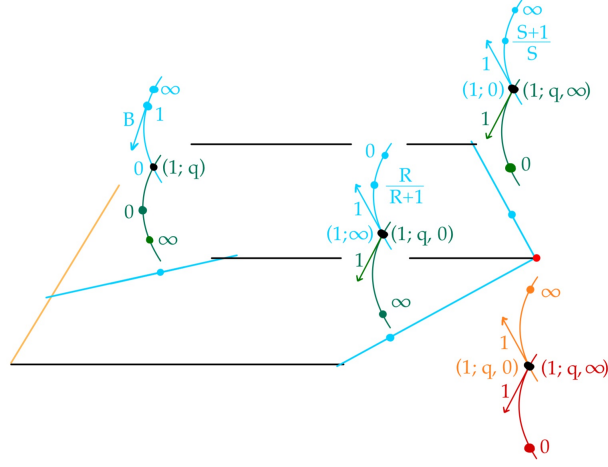


Figure 3.12 – Universal curve.

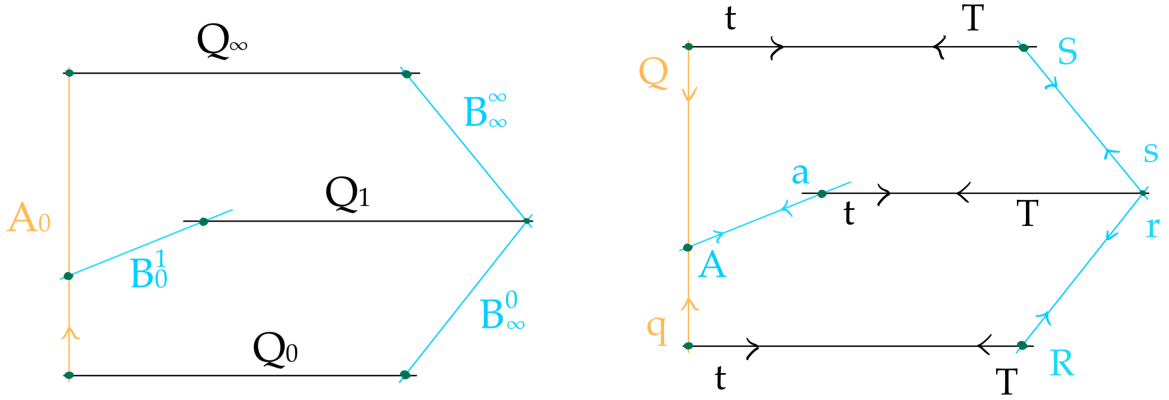


Figure 3.13 – Boundary components and their coordinate systems.

3.3 Construction of the Moduli Space of PV Connections

We already proved in Proposition 3.1.2 that the moduli space of PV type connection $\mathfrak{Conn}_{\Theta}^V$ contains $\mathcal{M} \times \mathbb{C}$ as a dense open set, and we studied in Theorem 3.2.6 the compactification of the basis $\mathcal{M}$. In order to build the compactification $\overline{\mathfrak{Conn}_{\Theta}^V}$, we have to make a choice about the "vertical" coordinate, the coordinate along the fibers, both inside $\mathfrak{Conn}_{\Theta}^V$ as along the new boundary components. Note that since we want extend a line bundle, we will mostly work on the smooth surface $\widetilde{\mathcal{M}}$ instead of the singular $\overline{\mathcal{M}}$.

3.3.1 Connections on the Boundary

Notice that a connection on a stable nodal curve is a collection of connections on any smooth component. One of the smooth components is always rigid, in the sense that it is a rational irregular curve endowed with a divisor of degree three (the node becomes a pole of the connection). There we get a (sometimes confluent) hypergeometric connection, that is rigid since there is no parameter that we can deform.

Example 9. A stable curve lying on A_0 is composed of an lower (in yellow in Figure 3.8) smooth component that is rigid since no free parameters appear on it, and an upper component (green) in which the parameter q appears. A connection on this stable nodal curve is the data of two distinct connections: one on the upper component (Heun connection) and one on the lower (confluent hypergeometric connection).

On $\mathcal{M} \times \mathbb{C}$ we use the coordinate $\hat{p}$, that appears in the normal form and that we have already seen in $\mathbb{F}_2$, as in Figure 2.3. On each boundary component we choose a suitable coordinate around a generic point that allows us to compute the resulting confluent connection on the stable nodal curve. For the following, we consider a connection ∇ of PV type, in the normal form as described in section 2.2. We call Ω_0 the connection matrix relative to the trivialising open set $U_0 = \mathbb{P}^1 \setminus \{\infty\}$.

Connections on A_0 . We have a family of stable nodal curves with two smooth components depending on the parameter q . We have hence to describe the limit connections on both smooth components.

On the first component, relative to setting $t = 0$, in green in Figure 3.8, we can easily define the limit connection by directly computing

$$\Omega_0^{t=0} := \lim_{t \rightarrow 0} \Omega_0.$$

After applying an elementary transformation based in $x = 1$, the double pole disappears. Therefore $\lim_{t \rightarrow 0} \nabla$ corresponds to an hypergeometric equation with polar divisor $[0] + [1] + [\infty]$.

We can study what happens on the other smooth components, that is the one occurring when the Moebius transformation f_{A_0} (3.1) is applied. In order to do so, we consider

the following limit:

$$\Omega_0^{f_{A_0}} := \lim_{t \rightarrow 0} f_{A_0}^* \Omega_0 = \begin{pmatrix} 0 & \frac{1}{(X-1)^2} \\ \frac{\rho(q-1)}{X^2} - \frac{\hat{p}(\hat{p}-\kappa_1+1)}{(q-1)X^2} + \frac{\hat{p}^2}{(q-1)^2 X^2} + \frac{\hat{p}(\hat{p}+\kappa_0)}{q X^2} & \frac{1}{(X-1)^2} + \frac{\kappa_1}{X} - \frac{\kappa_1}{X-1} \end{pmatrix}.$$

Remark 29. We make some observation that we will not repeat for the others computations that follow.

- The pole of order two in $X = 0$ becomes **simple** after applying an elementary transformation in $X = 0$:
- The leading coefficient relative to the pole $X = 1$ has **now 1 as coefficient** instead of t .

$$\begin{pmatrix} 0 & \frac{1}{X(X-1)^2} \\ \frac{\rho(q-1)}{X} - \frac{\hat{p}(\hat{p}-\kappa_1+1)}{(q-1)X} + \frac{\hat{p}^2}{(q-1)^2 X} + \frac{\hat{p}(\hat{p}+\kappa_0)}{q X} & \frac{1}{(X-1)^2} + \frac{\kappa_1-1}{X} - \frac{\kappa_1}{X-1} \end{pmatrix}.$$

Finally, we show that the spectral residual datum at the nodal singularity agrees on both connections, as shown in [2].

Proposition 3.3.1. *The two connections have the same residual spectral datum in the node, that is*

$$4\rho(q-1) + \kappa_1^2 - \frac{4\hat{p}(\hat{p}-\kappa_1+1)}{q-1} + \frac{4\hat{p}^2}{(q-1)^2} + \frac{4\hat{p}(\hat{p}+\kappa_0)}{q}.$$

Proof. It follows from a direct computation via the formula 1.2, in Section 1.3.2. $\square$

Connections on Q^0 . Here we have a family of stable nodal curves depending on the parameter t . In the first component, relative to the direct setting $q = 0$, we can try to compute the limit connection

$$\Omega_0^{q=0} := \lim_{q \rightarrow 0} \Omega_0(x; t, q, \hat{p}).$$

Lemma 3.3.2. *The limit*

$$\lim_{q \rightarrow 0} \Omega_0(x; t, q, \hat{p})$$

exists if, and only if, for $q \rightarrow 0$, the point $(q, \hat{p}) \in \mathbb{F}_2$ follows the trajectory $(q, \alpha q + \beta)$, for $\beta = 0, -\kappa_0$. Moreover, the resulting connection depends on α , the asymptotic slope of the convergence.

Proof. The first part of the Lemma follows from Theorem 2.3.1. To distinguish the different limits, we call α_-^0 the slope of the invariant curve for $\beta = 0$, and α_+^0 the one for $\beta = -\kappa_0$. Computing the limit for $\beta = 0$, we get:

$$\lim_{q \rightarrow 0} \Omega_0(x; t, q, \alpha_-^0 q) = \begin{pmatrix} 0 & \frac{1}{x(x-1)^2} \\ \alpha_-^0 \kappa_0 - x\rho & \frac{-\kappa_0-1}{x} - \frac{\kappa_1}{x-1} + \frac{t}{(x-1)^2} \end{pmatrix}$$

and, for $\beta = -\kappa_0$:

$$\lim_{q \rightarrow 0} \Omega_0(x; t, q, \alpha_+^0 q - \kappa_0) = \begin{pmatrix} 0 & \frac{1}{x(x-1)^2} \\ 2\kappa_0^2 + (-\alpha_+^0 + t + \kappa_1 - \frac{1}{x}) \kappa_0 - x\rho & \frac{-\kappa_0-1}{x} + \frac{-\kappa_1}{x-1} + \frac{t}{(x-1)^2} \end{pmatrix}.$$

In both cases the matrices depend, as desired, on the parameter t defining the stable nodal curve and on the parameter $\alpha_\pm^0$ defining the connection. $\square$

Remark 30. We can say that $\alpha_\pm^0$ are the coordinates of the exceptional divisors arising from blowing up the red points in $\mathbb{F}_2$ (represented in green in Figure 2.4).

We still have to study what happens on the other smooth component of the stable curve, with coordinate $X = f_{Q^0}(x)$ (3.2). To do so, we consider the matrix $f_{Q^0}^* \Omega_0$ with the parameter $\hat{p}$.

Lemma 3.3.3. *The limit $\Omega_0^{f_{Q^0}} := \lim_{q \rightarrow 0} f_{Q^0}^* \Omega_0$ produces a well defined connection.*

Proof. Computing the limit we get:

$$\lim_{q \rightarrow 0} f_{Q^0}^* \Omega_0(X; t, q, \hat{p}) = \begin{pmatrix} 0 & -\frac{1}{X(X-1)} \\ -\frac{\hat{p}}{X-1} + \frac{-\hat{p}(\hat{p}+\kappa_0)}{(X-1)^2} & -\frac{\kappa_0}{X} + \frac{\kappa_0+1}{X-1} \end{pmatrix}.$$

$\square$

A further computation shows that we get two different types of connections. For a generic value of $\hat{p}$ we get a connection with a ramified pole in $X = 1$, while for $\hat{p} = 0, -\kappa_0$ (that are the values appearing in Lemma 3.3.2) we get the hypergeometric equations we are expecting.

Proposition 3.3.4. *Connections with $\hat{p} = 0$ have the same residual spectral data in the node that corresponds to $(\kappa_0 + 1)^2$. Connections with $\hat{p} = -\kappa_0$ have also the same residual spectral data in the node that corresponds to $(\kappa_0 - 1)^2$.*

Proof. It follows from a direct computation via the formula 1.2, in Section 1.3.2. $\square$

Connections on Q^∞ . The same procedure as for Q^0 can be applied (see Section 3.3.4). The vertical coordinate that we should use is $P^\infty = -\frac{\hat{p}}{(\frac{1}{Q}-1)^2}$, where $Q = 1/q$.

Connections on B_0^1 . We can directly compute the limit for $t \rightarrow 0$ and $q \rightarrow 1$. To do that, we introduced in the last section the coordinate $B = \frac{t}{q-1}$.

Lemma 3.3.5. *The limit $\lim_{q \rightarrow 1} \Omega_0(x; B, q, P_0^1)$ produces a well defined connection if we set $P_0^1 = -\frac{\hat{p}}{(q-1)}$.*

Proof. Computing the limit we get:

$$\lim_{q \rightarrow 1} \Omega_0(x; B, q, P_0^1) = \begin{pmatrix} 0 & \frac{1}{x(x-1)^2} \\ (P_0^1)^2 + P_0^1(B - \kappa_1) - \rho(x-1) & -\frac{\kappa_1+1}{x-1} - \frac{\kappa_0}{x} \end{pmatrix}. \quad (3.4)$$

$\square$

On the other smooth component we have to apply the Moebius transformation $f_{B_0^1}$.

Lemma 3.3.6. *The limit $\lim_{q \rightarrow 1} f_{B_0^1}^* \Omega_0$ produces a well defined connection if we set $P_0^1 = -\frac{\hat{p}}{(q-1)}$.*

Proof. Computing the limit we get:

$$\lim_{q \rightarrow 1} f_{B_0^1}^* \Omega_0(x; B, q, P_0^1) = \begin{pmatrix} 0 & \frac{1}{(X-1)^2} \\ \frac{P_0^1}{X} + \frac{P_0^1(B+P_0^1-\kappa_1)}{X^2} & -\frac{\kappa_1}{X-1} + \frac{\kappa_1+1}{X} + \frac{A}{(X-1)^2} \end{pmatrix}. \quad (3.5)$$

$\square$

Proposition 3.3.7. *Connections 3.4 and 3.5 have the same residual spectral datum in the node, that is*

$$(\kappa_1 + 1)^2 + 4P_0^1(B + P_0^1 - \kappa_1).$$

Proof. It follows from a direct computation via the formula 1.2, in Section 1.3.2. $\square$

Let us now study connections on those stable curves that present the double pole in the node for the two smooth components. We will see that there will not be equality of the residual spectral data.

Connections on Q^1 . We expect over Q^1 a similar behaviour as over Q^0 , but the double pole at $x = 1$ makes everything more tricky. This time we will start by computing the limit for the pull-back connection $f_{Q^1}^* \Omega_0$. We will introduce a new variable for Q^1 , and we want to compute the direct limit using that variable.

Lemma 3.3.8. *The limit $\lim_{q \rightarrow 1} f_{Q^1}^* \Omega_0$ produces a well defined connection if we set $P^1 = -\frac{\hat{p}}{qt}$.*

Proof. Computing the limit we get:

$$\lim_{q \rightarrow 1} f_{Q^1}^* \Omega_0(x; t, q, P^1) = \begin{pmatrix} 0 & \frac{1}{(X-1)^2} \\ \frac{P^1((P^1+2)X-1)}{(X-1)^2 X} & \frac{1}{(X-1)^2} - \frac{1}{X} + \frac{1}{X-1} \end{pmatrix}.$$

□

We remark that ρ, κ_i disappear. It is due to the fact that the limit curve $f_{Q^1}(\mathbb{P}^1)$ for $q \rightarrow 1$ shows the apparent singularity at infinity and the double pole at one. It means that the relative connection is confluent hypergeometric with trivial monodromy around the poles, since $M_\infty M_1 = Id$ and $M_\infty = Id$.

We now have to rewrite the invariant curves of Theorem 2.3.1 in the new coordinate P^1 .

Lemma 3.3.9. *The invariant curves introduced in Theorem 2.3.1 become*

$$P^1 = 0 \quad \text{and} \quad P^1 = \frac{\kappa_1}{t}(q-1) - 1.$$

Proof. We apply the same procedure of Theorem 2.3.1 after the changing of variable $P^1 = -\frac{\hat{p}}{qt}$. □

We can now prove the following.

Lemma 3.3.10. *The limit*

$$\lim_{q \rightarrow 1} \Omega_0(x; t, q, P^1)$$

exists if, and only if, during the limit $q \rightarrow 1$, the point $(q, P^1) \in \mathbb{F}_2$ follows a trajectory that coincides up to the order two with the curve $(q, \frac{\alpha_\pm^1}{t^2}(q-1)^2 + \beta(q-1) + \gamma)$, for (β, γ) as in Lemma 3.3.9. Moreover the resulting connection depends on $\alpha_\pm^1$.

Proof. The first part of the Lemma follows from Theorem 2.3.1 and Lemma 3.3.9. Computing the limit for $(\beta, \gamma) = (0, 0)$, we get:

$$\lim_{q \rightarrow 1} \Omega_0(x; t, q, \frac{\alpha_-^1}{t^2}(q-1)^2) = \begin{pmatrix} 0 & \frac{1}{x(x-1)^2} \\ -\rho(x-1) + \alpha_-^1 & \frac{-\kappa_1-1}{x-1} - \frac{\kappa_0}{x} + \frac{t}{(x-1)^2} \end{pmatrix}.$$

and, for $(\beta, \gamma) = (\frac{\kappa_1}{t}, -1)$:

$$\lim_{q \rightarrow 0} \Omega_0(x; t, q, \frac{\alpha_+^1}{t^2}(q-1)^2 + \beta(q-1) + \gamma) = \begin{pmatrix} 0 & \frac{1}{x(x-1)^2} \\ (\kappa_0 + \frac{1}{x-1})t - (x-1)\rho - \alpha_+^1 & \frac{t}{(x-1)^2} + \frac{-\kappa_1-1}{x-1} - \frac{\kappa_0}{x} \end{pmatrix}.$$

In both cases the matrices depend, as expected, on the parameter t defining the stable nodal curve and on the parameter $\alpha_{\pm}^1$ defining the connection, as desired. $\square$

Remark 31. As mentioned before, we were expecting equality of residual spectral data in the node also in this situation, but it does not happen. For the pull-back connection we get 1, while for the direct limits we get respectively $\kappa_1 - 1$ and $\kappa_1 + 1$.

Connections on B_{∞}^0 . Here, both the smooth components of the nodal curve are obtained by a Moebius transformation. We recall that we defined $R = q/T$ the coordinate on B_{∞}^0 , where $T = 1/t$.

Lemma 3.3.11. *The limit $\lim_{T \rightarrow 0} f_{B_{\infty}^0}^* \Omega_0$ produces a well defined connection if we set $P_{\infty}^0 = -\frac{\hat{p}}{(RT-1)^2}$.*

Proof. Computing the limit we get:

$$\lim_{T \rightarrow 0} f_{B_{\infty}^0}^* \Omega_0(x; T, R, P_{\infty}^0) = \begin{pmatrix} 0 & -\frac{1}{(X-1)X} \\ -\frac{P_{\infty}^0(R+1)}{RX-R+X} + \frac{P_{\infty}^0}{X-1} + \frac{P_{\infty}^0(P_{\infty}^0+R-\kappa_0)}{(X-1)^2R} & \frac{-R-1}{RX-R+X} + \frac{\kappa_0+1}{X-1} + \frac{1}{(X-1)^2} - \frac{\kappa_0}{X} \end{pmatrix}$$

$\square$

And the second one.

Lemma 3.3.12. *The limit $\lim_{T \rightarrow 0} g_{B_{\infty}^0}^* \Omega_0$ produces a well defined connection if we set $P_{\infty}^0 = -\frac{\hat{p}}{(RT-1)^2}$.*

Proof. Computing the limit we get:

$$\lim_{T \rightarrow 0} g_{B_\infty^0}^* \Omega_0(x; T, R, P_\infty^0) = \begin{pmatrix} 0 & \frac{1}{(X-1)^3} \\ -\rho(X-1) + P_\infty^0 + \frac{P_\infty^0(P_\infty^0 - \kappa_0)}{R} & \frac{1}{(X-1)^2} + \frac{-\kappa_0 - \kappa_1 - 1}{X-1} \end{pmatrix}$$

□

Remark 32. As mentioned before, we were expecting equality of residual spectral data in the node also in this situation, but it does not happen. We get respectively

$$2P_\infty^0 - \kappa_0 - 1 + \frac{2P_\infty^0(P_\infty^0 - \kappa_0)}{R} \quad \text{and} \quad -2P_\infty^0 + \kappa_0 + \kappa_1 + 1 - \frac{2P_\infty^0(P_\infty^0 - \kappa_0)}{R}$$

Connections on B_∞^∞ . The same procedure as for B_∞^0 can be applied (see Section 3.3.4). The vertical coordinate that we should use is $P_\infty^\infty = -\frac{\hat{p}}{(\frac{1}{ST}-1)^2}$, where $S = \frac{Q}{T}$ is the coordinate on B_∞^∞ , $Q = \frac{1}{q}$, $T = \frac{1}{t}$.

Connections on A_∞ . The limit for $t \rightarrow \infty$ is more tricky than the other limits we studied before. We have indeed two possible approaches to compute it. In Section 3.3.3 we will give a geometrical interpretation to both of them.

The first approach is to consider the two Moebius transformations used to derive the stable nodal curve.

Lemma 3.3.13. *The limit $\lim_{T \rightarrow 0} f_{A_\infty}^* \Omega_0$ produces a well defined connection if we set $P_\infty^+ = -\frac{\hat{p}}{(q-1)^2}$ or $P_\infty^- = -\frac{\hat{p}}{(q-1)^2} + \frac{T\kappa_1(q-1)-q}{T(q-1)^2}$.*

Proof. Computing the limit we get:

$$\lim_{T \rightarrow 0} f_{A_\infty}^* \Omega_0(x; T, q, P_\infty^+) = \begin{pmatrix} 0 & -\frac{1}{X(X-1)} \\ \frac{P_\infty^+}{(X-1)^2} & -\frac{\kappa_0}{X} + \frac{\kappa_0}{X-1} + \frac{1}{(X-1)^2} \end{pmatrix}$$

$$\lim_{T \rightarrow 0} f_{A_\infty}^* \Omega_0(x; T, q, P_\infty^-) = \begin{pmatrix} 0 & -\frac{1}{X(X-1)} \\ \frac{-P_\infty^- + \kappa_0 + \kappa_1 - 1}{(X-1)^2} & -\frac{\kappa_0}{X} + \frac{\kappa_0}{X-1} + \frac{1}{(X-1)^2} \end{pmatrix}$$

□

Lemma 3.3.14. *The limit $\lim_{T \rightarrow 0} g_{A_\infty}^* \Omega_0$ produces a well defined connection if we set $P_\infty^+ = -\frac{\hat{p}}{(q-1)^2}$ or $P_\infty^- = -\frac{\hat{p}T-q}{T(q-1)^2}$.*

Proof. Computing the limit and applying an elementary transformation we get to, respectively

$$\lim_{T \rightarrow 0} g_{A_\infty}^* \Omega_0(x; T, q, P_\infty^+) = \begin{pmatrix} 0 & \frac{1}{(X-1)^3} \\ -\rho(X-1) + P_\infty^+ & \frac{-\kappa_0 - \kappa_1 - 1}{X-1} + \frac{1}{(X-1)^2} \end{pmatrix}$$

$$\lim_{T \rightarrow 0} g_{A_\infty}^* \Omega_0(x; T, q, P_\infty^-) = \begin{pmatrix} 0 & \frac{1}{(X-1)^3} \\ -\rho(X-1) - P_\infty^- + \kappa_0 + \kappa_1 & \frac{-\kappa_0 - \kappa_1 - 1}{X-1} + \frac{1}{(X-1)^2} \end{pmatrix}$$

□

Remark 33. As mentioned before, we were expecting equality of residual spectral data in the node also in this situation, but it does not happen. We get respectively for the "+" curve

$$2P_\infty^+ - \kappa_0 \quad \text{and} \quad -2P_\infty^+ + \kappa_0 + \kappa_1 + 1,$$

while for the "-" curve we get

$$2P_\infty^- - \kappa_0 - \kappa_1 + 1 - 2\frac{\kappa_1}{q-1} \quad \text{and} \quad -2P_\infty^- + \kappa_0 + 2\left(\kappa_1 - 1\frac{\kappa_1}{q-1}\right).$$

We can also have a different approach, in which we use the coordinate P^1 . It is the coordinate used for computing limits in Q^1 . Let us consider a Moebius transformation that fixes 0 and sets $t = 1$. We have a family depending on one parameter of such transformation and we can for instance choose to fix -1 (it suffices not to fix 0, ∞ or q). The curve we get after the limit $t = \infty$ is no more stable, because only the double pole in 1 survived. Anyway the limit we get is still interesting.

Proposition 3.3.15. *Let us consider the Moebius transformation*

$$h_{A_\infty}(x) := \frac{(t+1)x + t - 1}{(t-1)x + t + 1}.$$

The limit $\lim_{T \rightarrow 0} h_{A_\infty}^* \Omega_0(X; T, q, P^1)$ produces a well defined connection if we use $P^1 = -\frac{\hat{p}T}{q}$, the same coordinate as in Q^1 .

Proof. Computing the limit we get to:

$$\Omega_0^{h_{A_\infty}} := \begin{pmatrix} 0 & -\frac{1}{(X-1)^2} \\ \frac{4qP^1(P^1+1)}{(q-1)^2(X-1)^2} & \frac{1}{(X-1)^2} \end{pmatrix}$$

□

Remark 34. The relation between the coordinates we used is the following

$$P_\infty^+ = -\frac{\hat{p}}{(q-1)^2} = \frac{P^1 q}{T(q-1)^2} \quad \text{and} \quad P_\infty^- = -\frac{\hat{p}}{(q-1)^2} + \frac{T\kappa_1(q-1) - q}{T(q-1)^2} = \frac{q(P^1 + 1)}{T(q-1)^2} - \frac{\kappa_1}{q-1}.$$

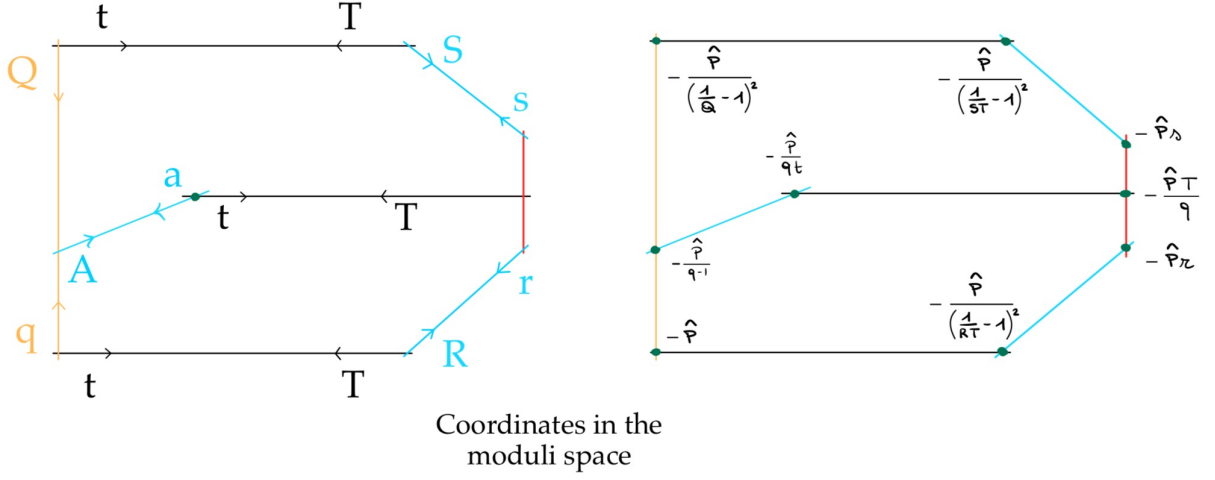
We will give a geometric interpretation of these coordinates in Section 4.2.2.

3.3.2 Extension of the Trivial Line Bundle

We recall that our final goal is to compactify the space $\mathfrak{Con}_\Theta^V$. We proceed in several steps. First, in this section, we study how the line bundle structure extends to the boundary components of $\widetilde{\mathcal{M}}$, defining a line bundle $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)$ associated to a divisor $D \in \text{Div}(\widetilde{\mathcal{M}})$. We then compactify this space and analyse its birational geometry. This analysis is necessary because the space we ultimately want to work with is the singular variety where the surface $\mathcal{A}_\infty := \overline{\mathcal{O}_{A_\infty}(D)}$ has been contracted producing only canonical singularities.

The strategy is to deduce the divisor D by studying the poles and zeros of the extension of the global section $\hat{p} = 1$ defined in $\mathfrak{Con}_\Theta^V$. In the previous section we have studied the vertical coordinate in the generic point of any boundary component. To compute the divisor, it is more suitable to define vertical coordinates based on each intersection point of the boundary components. Some computations show that one possible choice is as in Figure 3.14.

Before going into the proof, let us recall that in Proposition 3.2.7 we have shown that $\overline{\mathcal{M}}$ is a singular space, due to the fact that the (-2) -curve A_∞ has been contracted. We will hence in a first moment extend the line bundle on the smooth surface $\widetilde{\mathcal{M}}$, and then we show that it is possible to contract the surface lying over A_∞ . In what follows we denote by $\mathcal{Q}^i$ and $\mathcal{A}_j$ the total spaces of the restriction of the line bundle over, respectively, $\mathcal{Q}^i$ and A_j .


 Figure 3.14 – Expressions for the local coordinates in $\widetilde{\mathcal{M}}$.

Proposition 3.3.16. *The trivial line bundle $\mathcal{M} \times \mathbb{C}$ extends to $\widetilde{\mathcal{M}}$ as $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)$, where $D \in \text{Div}(\widetilde{\mathcal{M}})$ is the divisor*

$$D = \text{div}(\hat{p}) \equiv A_0 + 2Q^1 + 2B_0^1 - B_\infty^0 - B_\infty^\infty.$$

Proof. Since it is easier for us to work with a smooth variety, we study in a first moment the extension of the trivial vector bundle $\mathcal{M} \times \mathbb{C}$ on $\widetilde{\mathcal{M}}$.

By studying the poles and zeros of the section $\hat{p} = 1$ in all the coordinates, we get to the divisor

$$-B_0^1 + 2Q^\infty + 2B_\infty^\infty + A_\infty.$$

To get the desired expression, we can explicitly study the Picard group of $\widetilde{\mathcal{M}}$.

We now it is the blowup of $\mathbb{P}^1 \times \mathbb{P}^1$ in three points. Therefore, since $\text{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z} \times \mathbb{Z}$, we have that $\text{Pic}(\widetilde{\mathcal{M}}) \cong \mathbb{Z}^5$ and is for instance generated by

$$\text{Pic}(\widetilde{\mathcal{M}}) \cong [A_0] \cdot \mathbb{Z} + [Q^0] \cdot \mathbb{Z} + [B_0^1] \cdot \mathbb{Z} + [B_\infty^0] \cdot \mathbb{Z} + [B_\infty^\infty] \cdot \mathbb{Z}.$$

Moreover we know that in $\mathbb{P}^1 \times \mathbb{P}^1$ the divisors A_0 and A_∞ are equivalent, and the same holds for Q^0, Q^1 and Q^∞ . Since we know that a holomorphic function $f: \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{C}$ vanishing at a point $p \in \mathbb{P}^1 \times \mathbb{P}^1$, vanishes also on the exceptional divisor $E_p \subseteq \text{Bl}_p(\mathbb{P}^1 \times \mathbb{P}^1)$, if we remark that $A_i = \{t - i = 0\}$ and $Q^j = \{q - j = 0\}$, we get the following

equivalences

$$\begin{cases} A_0 + B_0^1 = A_\infty + B_\infty^0 + B_\infty^\infty \\ Q^0 + B_\infty^0 = Q^1 + B_0^1 = Q^\infty + B_\infty^\infty \end{cases} \implies \begin{cases} A_\infty = A_0 + B_0^0 - B_\infty^0 - B_\infty^\infty \\ Q^0 = Q^1 + B_0^1 - B_\infty^0 \\ Q^\infty = Q^1 + B_0^1 - B_\infty^\infty \end{cases}$$

And if we replace them in the expression of D we get

$$\begin{aligned} D &= -B_0^1 + 2(Q^1 + B_0^1 - B_\infty^\infty) + 2B_\infty^\infty + (A_0 + B_0^1 - B_\infty^0 - B_\infty^\infty) = \\ &= A_0 + 2B_0^1 + 2Q^1 - B_\infty^\infty - B_\infty^0 \end{aligned}$$

as desired. $\square$

Since the base space we would like to work with is $\overline{\mathcal{M}}$, that is the contraction of A_∞ inside $\widetilde{\mathcal{M}}$, we wonder if we can contract the surface $\mathcal{A}_\infty := \overline{\mathcal{O}_{\widetilde{\mathcal{M}}}(D)|_{A_\infty}} := \mathbb{P}(\mathcal{O} \oplus \mathcal{O}_{\widetilde{\mathcal{M}}}(D)|_{A_\infty})$.

Before doing that, we consider the compactification of the line bundle, in order to work with a compact algebraic 3-manifold.

Proposition 3.3.17. *The line bundle $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)$ is trivial over Q^0, Q^1, A_∞ and Q^∞ .*

Proof. We recall that for any pair of divisor D and D' it holds that

$$\deg \mathcal{O}_{D'}(D) = (D \cdot D').$$

In particular:

$$\begin{aligned} (D \cdot Q^0) &= ((-B_0^1 + 2Q^\infty + 2B_\infty^\infty + A_\infty) \cdot Q^0) = 0 + 0 + 0 + 0 = 0, \\ (D \cdot Q^1) &= ((-B_0^1 + 2Q^\infty + 2B_\infty^\infty + A_\infty) \cdot Q^1) = -2 + 0 + 0 + 2 = 0, \\ (D \cdot Q^\infty) &= ((-B_0^1 + 2Q^\infty + 2B_\infty^\infty + A_\infty) \cdot Q^\infty) = 0 - 2 + 2 + 0 = 0, \\ (D \cdot A_\infty) &= ((-B_0^1 + 2Q^\infty + 2B_\infty^\infty + A_\infty) \cdot A_\infty) = 0 + 0 + 2 - 2 = 0. \end{aligned}$$

As desired. $\square$

Remark 35. We already expected this result the for $Q^{i'}$'s: indeed, Lemmas 3.3.2 and 3.3.10 give us the expression of some non vanishing global sections. In particular, it

also shows that $J^1 \mathcal{O}_{\widetilde{\mathcal{M}}}(D)$ is trivial over Q^1 .

We can hence choose the following vertical coordinates that trivialise the bundle on those boundary components:

$$P^0 = -\frac{\hat{p}}{(q-1)^2} = -\frac{\hat{p}}{(RT-1)^2}, \quad P^1 = -\frac{\hat{p}}{qt} = -\frac{\hat{p}T}{q} = -\hat{p}s = -\hat{p}r, \quad P^\infty = -\frac{\hat{p}}{(\frac{1}{Q}-1)^2} = -\frac{\hat{p}}{(\frac{1}{ST}-1)^2}$$

It suffices then to choose a coordinate at the intersection $A_0 \cap B_0^1$, that is $P_0^1 = -\frac{\hat{p}}{q-1}$. With these coordinates, the sections in Q^0 are given by the equations $P^0 = 0, P^0 = -\kappa_0$

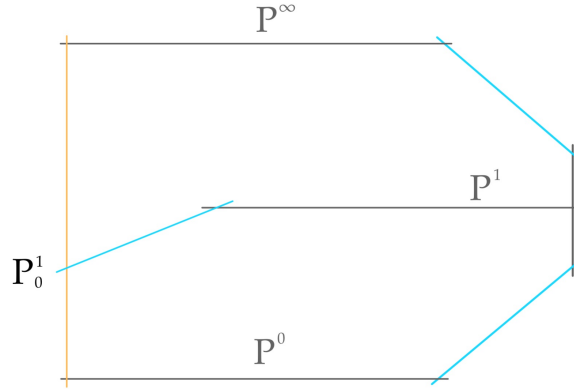


Figure 3.15 – Titre plus long avec des explications.

and in Q^1 are given by the equations $P^1 = 0, P^1 = -1$.

The following result shows that we can contract $\mathcal{A}_\infty$ producing mild singularities. It has the advantage to describe the line bundle over $\overline{\mathcal{M}}$, getting rid of $\widetilde{\mathcal{M}}$. However, we have already seen that we need to blow up some sections in Q^1 and $\mathcal{A}_\infty$ to build the moduli space $\overline{\mathcal{C}on}_\Theta^V$, making the contraction of $\mathcal{A}_\infty$ way more complicated.

Proposition 3.3.18. *Let us denote $X := \overline{\mathcal{O}_{\widetilde{\mathcal{M}}}(D)} = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}_{\widetilde{\mathcal{M}}}(D))$. The contraction of $\mathcal{A}_\infty$ inside X produces canonical singularities.*

Proof. First of all let us notice that, since $\mathcal{O}_{A_\infty}(D) \cong \mathcal{O}$, then $\mathcal{A}_\infty := \overline{\mathcal{O}_{A_\infty}(D)} \cong \mathbb{P}(\mathcal{O} \oplus \mathcal{O}) \cong \mathbb{P}^1 \times \mathbb{P}^1$.

We want then to contract the "horizontal" fibers of $\mathcal{A}_\infty$, and then we have to show that they are K_X -trivial, where K_X is the canonical bundle of X . Let us call $C_a = \mathbb{P}^1 \times \{a\}, C_b = \mathbb{P}^1 \times \{b\}$ two of those fibers. They are linearly equivalent in X since they

are in $\mathcal{A}_\infty$ and $\dim \mathcal{A}_\infty = 2$. We identify A_∞ with C_0 . It is then sufficient to show that $(K_X \cdot A_\infty) = 0$. Since $A_\infty \subseteq \widetilde{M}$, it holds that $(K_X \cdot A_\infty) = (K_X|_{\widetilde{M}} \cdot A_\infty)$.

By the adjunction formula we have that

$$K_X|_{\widetilde{M}} = K_{\widetilde{M}} - \widetilde{M},$$

and therefore

$$(K_X \cdot A_\infty) = (K_{\widetilde{M}} \cdot A_\infty) - (\widetilde{M} \cdot A_\infty).$$

We can compute the first product since we know the canonical bundle of $\widetilde{M}$:

$$(K_{\widetilde{M}} \cdot A_\infty) = ((2(Q^0 + A_0) - B_0^1 - B_0^\infty + B_\infty^\infty) \cdot A_\infty) = 0$$

and the second product is zero since $A_\infty \subseteq \widetilde{M}$.

We have then shown that all the horizontal fibers of $\mathcal{A}_\infty$ are K_X -trivial. We have now to compute their discrepancies in order to conclude the proof. Let $\hat{\pi}: X \rightarrow \hat{X}$ the contraction map. Then,

$$K_X = \hat{\pi}^* K_{\hat{X}} + \alpha \mathcal{A}_\infty$$

and the equality must hold also when intersecting with A_∞ :

$$0 = (K_X \cdot A_\infty) = (\hat{\pi}^* K_{\hat{X}} \cdot A_\infty) + \alpha (\mathcal{A}_\infty \cdot A_\infty).$$

The first intersection product is zero, since A_∞ has been contracted, while we can compute the second via the formula $(\mathcal{A}_\infty \cdot A_\infty) = -2 - (A_\infty)_{\mathcal{A}_\infty}^2 = -2 + 0$, since $\mathcal{A}_\infty \cong \mathbb{P}^1 \times \mathbb{P}^1$ and A_∞ is just a fiber. We conclude hence that $\alpha = 0$ and then that the contraction process produces only canonical singularities. $\square$

3.3.3 Compactification

We immediately remark that $\overline{\mathcal{O}_{\widetilde{M}}(D)}$ is not the moduli space we are looking for. Indeed, there are points inside it that do not represent any PV type connection (as points on the Q^i 's outside the two special sections, or most of the points in $\mathcal{A}_\infty$) and there are some points that represent infinite connections (as points in the special sections in the $Q^{i'}$'s).

As shown in Lemma 3.3.2, for Q^0 (and the same holds for Q^∞) the connections appear

only on the special sections $P^0 = 0$ and $P^0 = \kappa_0$. We recall that the resulting limits depends on the slope of incidence, and therefore each point of these sections encodes a one parameter family of connections. We can then blow-up these sections and ignore the other part of Q^0 . For Q^1 the process is similar, but, following the Lemma 3.3.10, we need to blow up two times each section to make the connections appear. We can then ignore Q^1 and the first exceptional divisor of each section. To have a picture in mind, we have exactly a family of Okamoto spaces as in Figure 2.4 for each choice of $t \in \mathbb{C}^*$. In formulae:

In $t = 0$ we have the hypersurfaces $\mathcal{A}_0$ and $\mathcal{B}_0^1$, while in $t = \infty$ the situation is more complicated.

Remark 36. We get a new sort of Okamoto divisor given by $[\mathcal{S}_\infty] + [Q^0] + [Q^1] + [Q^\infty] + [\mathcal{E}^+] + [\mathcal{E}^-]$ that we can visualize by considering in families the surfaces of Figure 2.4, with the differences given by the use of a different vertical coordinate over Q^1 .

Over $\mathcal{A}_\infty$ the coordinate P^1 has been used, which is the one we find in Proposition 3.3.15. This is a good coordinate to study the geometry of $\overline{\mathcal{O}_{\overline{\mathcal{M}}}(D)}$, but the connections defined via this coordinate are not what we are looking for, since they are not connections on the stable nodal curve we found in Section 3.2.2. In order to make them appear we have to use the coordinates $P_\infty^\pm$ introduced in Lemma 3.3.14. Since they are related by the formula

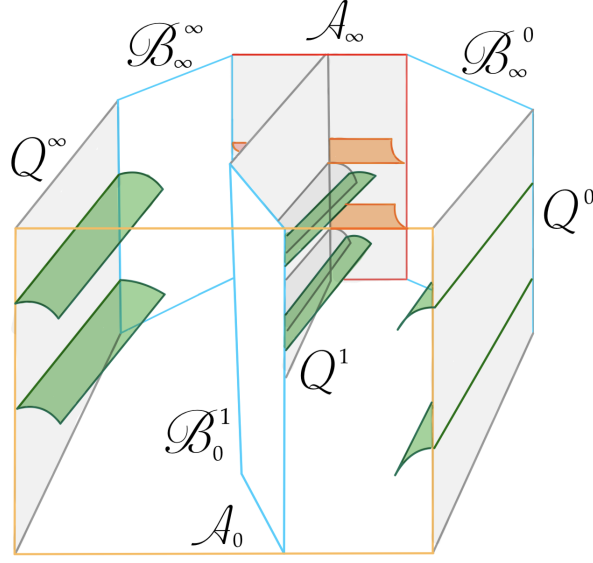
$$P_\infty^+ = \frac{P^1 q}{T(q-1)^2} \quad \text{and} \quad P_\infty^- = \frac{q(P^1 + 1)}{T(q-1)^2} - \frac{\kappa_1}{q-1}.$$

they corresponds geometrically to blow up the sections $P^1 = 0$ and $P^1 = -1$ in $\mathcal{A}_\infty$.

We can resume this section in the following result.

Theorem 3.3.19. *The compactified moduli space $\overline{\mathfrak{Con}}_\Theta^V$ comes with a fibration $\pi: \overline{\mathfrak{Con}}_\Theta^V \xrightarrow{t} \mathbb{P}^1$ such that*

- i) *For any $t \in \mathbb{C}^*$, the fiber $Ok_t := \pi^{-1}(t)$ is an Okamoto space as in Figure 2.4, that is a 8-blow-up of the second Hirzebruch space $\mathbb{F}_2$.*
- ii) *The surface $\pi^{-1}(0)$ is given by $\mathcal{A}_0 \cup \mathcal{B}_0^1$. The component $\mathcal{A}_0$ is isomorphic to $\mathbb{F}_1$, while $\mathcal{B}_0^1$ is isomorphic to a double blow up of $\mathbb{F}_1$.*
- iii) *Let us denote by $\mathcal{F}_\infty^+$ and $\mathcal{F}_\infty^-$ the blow up of the two sections $P^1 = 0$ and $P^1 = -1$ in $\mathcal{A}_\infty$. The surface $\pi^{-1}(\infty)$ is given by $\mathcal{B}_\infty^0 \cup \mathcal{B}_\infty^\infty \cup \mathcal{F}_\infty^+ \cup \mathcal{F}_\infty^-$. The surfaces $\mathcal{B}_\infty^0$ and $\mathcal{B}_\infty^\infty$ are isomorphic to a double blow up of $\mathbb{F}_1$.*


 Figure 3.16 – The moduli space $\overline{\text{Con}}_{\Theta}^V$.

Proof. Assertion (i) has been proved in the previous sections. We need to show that surfaces $\mathcal{B}_j^i$ s are isomorphic to a double blow up of $\mathbb{F}_1$, and that $\mathcal{A}_0 \cong \mathbb{F}_1$. It is equivalent to show that the line bundles $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)|_{\mathcal{B}_j^i}$ and $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)|_{\mathcal{A}_0}$ have degree equal to one.

$$\begin{aligned} (D \cdot B_0^1) &= \left((A_0 + 2Q^1 + 2B_0^1 - B_{\infty}^0 - B_{\infty}^{\infty}) \cdot B_0^1 \right) = 1 + 2 - 2 + 0 + 0 = 1, \\ (D \cdot B_{\infty}^0) &= \left((A_0 + 2Q^1 + 2B_0^1 - B_{\infty}^0 - B_{\infty}^{\infty}) \cdot B_{\infty}^0 \right) = 0 + 0 + 0 + 1 + 0 = 1, \\ (D \cdot B_{\infty}^{\infty}) &= \left((A_0 + 2Q^1 + 2B_0^1 - B_{\infty}^0 - B_{\infty}^{\infty}) \cdot B_{\infty}^{\infty} \right) = 0 + 0 + 0 + 0 + 1 = 1, \\ (D \cdot A_0) &= \left((A_0 + 2Q^1 + 2B_0^1 - B_{\infty}^0 - B_{\infty}^{\infty}) \cdot A_0 \right) = -1 + 0 + 2 + 0 + 0 = 1, \end{aligned}$$

as desired. □

Let us denote by $\mathcal{E}_{\pm}^i$ the exceptional divisor arising from blowing up the special sections in the Q^i s and being part of the moduli space $\overline{\text{Con}}_{\Theta}^V$. In particular it means that $\mathcal{E}_{\pm}^1$ correspond to the second exceptional divisors.

3.3.4 Symmetries

The construction we made presents some symmetries. The most visible one is the involution $\Phi_{0,\infty}$ exchanging the role of 0 and ∞ . There are three other involutions Ψ_a , for $a = 0, 1, \infty$, exchanging the residual spectral data $\kappa_a^+ \leftrightarrow \kappa_a^-$ of each singularity. We can express them in coordinates, on the Zariski open set $\mathcal{M} \times \mathbb{C} \subseteq \mathbb{C}on_{\Theta}^V$. They are well defined on the whole compactification of the moduli space.

$$\Phi_{0,\infty} = \begin{cases} \kappa_{\infty} = \kappa_0 \\ \kappa_0 = \kappa_{\infty} \\ q = \frac{1}{q} \\ \kappa_1 = -\kappa_1 \\ \hat{p} = \frac{q}{2} \frac{-2\hat{p}q^3 + \kappa_0q^2 + \kappa_{\infty}q^2 + 4\hat{p}q^2 + (\kappa_1 - 1)q^2 - 2\kappa_0q - 2\kappa_{\infty}q - 2\hat{p}q - 2qt + \kappa_0 + \kappa_{\infty} + 2q - \kappa_1 - 1}{(q-1)^2} \end{cases}$$

and

$$\Psi_0 = \begin{cases} t = t \\ \kappa_0 = -\kappa_0 \\ \hat{p} = \hat{p} - \frac{\kappa_0}{q} \end{cases} \quad \Psi_1 = \begin{cases} t = -t \\ \kappa_1 = -\kappa_1 \\ \hat{p} = \hat{p} + \frac{t}{(q-1)^2} - \frac{\kappa_1}{q-1} \end{cases} \quad \Psi_{\infty} = \begin{cases} t = t \\ \kappa_{\infty} = -\kappa_{\infty} \\ \hat{p} = \hat{p} \end{cases}$$

Proposition 3.3.20. *The involutions $\Phi_{0,\infty}, \Psi_0, \Psi_1$ and Ψ_{∞} generate a transformation group of order 16 acting on $\overline{\mathbb{C}on}_{\Theta}^V$. Moreover,*

$$\Phi_{0,\infty} \circ \Psi_0 = \Psi_{\infty} \circ \Phi_{0,\infty}.$$

Proof. Explicit computations show that $(\Phi_{0,\infty})^2 = (\Psi_0)^2 = (\Psi_1)^2 = (\Psi_{\infty})^2 = Id$, and that $\Phi_{0,\infty} \circ \Psi_0 = \Psi_{\infty} \circ \Phi_{0,\infty}$, for all $a = 0, 1, \infty$. $\square$

Geometrically, the symmetries Ψ_i correspond to exchange the two exceptional divisors in Q^i . For instance, in a neighbour of Q^0 , the two invariant curves $\hat{p} = 0$ and $\hat{p} = \alpha q - \kappa_0$ are switched, and the same happens for the others Q^i . The symmetry $\Phi_{0,\infty}$ exchanges Q^0 and Q^{∞} . This is the reason why, in the previous sections, we often skipped the computations for Q^{∞} .

ISOMONODROMIC DEFORMATIONS

4.1 Irregular Riemann-Hilbert Correspondence

In Section 1.4 we have discussed the Riemann-Hilbert correspondence for logarithmic connections. As studied for instance in [38], [8] or [6], an irregular version of the Riemann-Hilbert correspondence also exists. The goal of this section is to introduce the main tools we need to properly state this correspondence Theorem 4.1.3. Since a large literature can be found on irregular monodromy in a very general setting, we will specialize to the case of PV connections. We will then define the fundamental groupoid of a rational irregular curve and the local monodromy around a pole of order two.

4.1.1 Fundamental Groupoid of a Rational Irregular Curve

In Section 1.4 we have defined the logarithmic monodromy as a representation of the fundamental group of $\mathbb{P}^1 \setminus |D|$ onto $\mathrm{GL}_2(\mathbb{C})$. We would like to express the monodromy of an irregular connection also as a representation of a certain fundamental group, but we must take into account that we are dealing with higher order poles, that carry a more complicated monodromy involving Stokes phenomena, as we will see in Section 4.1.2.

For this section, we follow in particular the recent articles of F. Loray [38] and of E. Paul and J.P. Ramis [21] in which they properly define this fundamental groupoid following the ideas developed in the correspondence between P. Deligne, B. Malgrange and J.P. Ramis [16]. The main idea is to replace the arrow $t \in T_1\mathbb{P}^1$ of the rational irregular curve with a point infinitesimally close to 1. We need to be careful: since the fundamental group is a topological invariant, we cannot just add a punctured point, because we would just get the fundamental group of a 4-punctured sphere. To fix this, first we perform a real blow up at $x = 1$ and then we puncture two points close to the exceptional divisor. We will finally consider a groupoid with five basepoints. Note that

the fundamental groupoid will not depend on t , since we can continuously deform it. This construction may seem very artificial, but it will make more sense when the irregular monodromy will be defined in Section 4.1.2.

First of all, let us recall that a rational irregular curve, as defined in Section 3.2, is $(\mathbb{P}^1, D^V, J^V)$ with $D^V = [0] + 2[1] + [\infty]$ and $J = \{0, (1, t), \infty\}$.

To study the behavior around the double pole, let us for the moment work locally in an open irregular disk $(\mathbb{D}, 2[1], \{(1, t)\})$ centered in $x = 1$. Consider then the real blow up $E: \hat{\mathbb{D}} \rightarrow \mathbb{D}$, that corresponds to replace its center with a circle $\hat{C} \cong S^1$, called circle of directions. Consider now a closed annulus $A \subseteq \mathbb{D}$ whose interior boundary is $\hat{C}$ and call C its exterior boundary. Fix two antipodal points s_0, s_1 in the interior of A and consider the punctured annulus $A \setminus \{s_0, s_1\}$. Choose an another direction different from the one identified by the punctures and consider $\hat{p}_0, \hat{p}_1 \in \hat{C}$ and $p_0, p_1 \in C$ as in Figure 4.1.

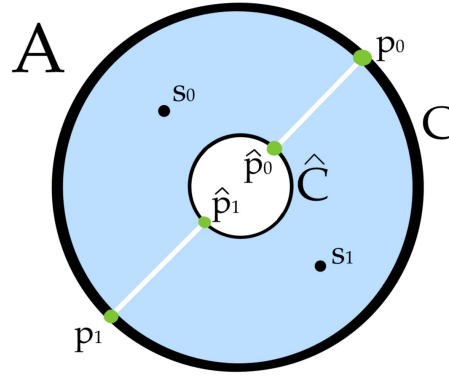


Figure 4.1 – Base points and punctures on the annulus.

Definition. The wild fundamental groupoid of the irregular disk $(\mathbb{D}, 2[1], \{(1, t)\})$ is isomorphic to

$$\pi_1(\hat{\mathbb{D}} \setminus \{s_0, s_1\}, \{p_0, p_1, \hat{p}_0, \hat{p}_1\})$$

We can then deduce the fundamental groupoid of a rational irregular curve.

4.1.2 Monodromy of a PV Connection

As in the previous section, let us start by analyze the local irregular monodromy. Let $U \cong \mathbb{D}$ be a trivialising open set for the PV connection (E, D^V, ∇) containing only

the irregular singularity at $x = 1$. Compared to the logarithmic case, parallel sections in $\mathbb{D}$ have only a formal description on the entire open punctured set $\mathbb{D}^* := \mathbb{D} \setminus \{1\}$, and these formal series converges into holomorphic sections only on the two sectors

$$V^k := \left\{ x \in \mathbb{D}^* \mid \arg(x) \neq \arg(t) + (-1)^k \frac{\pi}{2} \right\} \quad \text{for } k = 0, 1,$$

where t , as it appears in the normal form 2.1, is the non zero eigenvalue of the second order matrix at the irregular pole. We recall, to make the analogy with the previous section clear, that it can be seen also as the complex number defining the jet $j^1\gamma_1 = (1, t)$ in the associated rational irregular curve.

Remark 37. In order to understand what is happening, let us look at the one dimensional case. Consider the equation

$$dy = a(x) \frac{dx}{x^2} y, \quad \text{with } a(x) = t + a_1x + a_2x^2 + \dots$$

A solution is given by the explicit formula

$$y(x) = \exp \int \frac{a(x)}{x^2} dx = \exp \left(-\frac{t + o(1)}{x} \right).$$

The condition $\operatorname{Re}(t/x) = 0$ consists of 2 rays, that we will call *anti-Stokes directions*, dividing the disk into two equal sectors, that are precisely the V^k as before. An easy computation indeed shows that

$$\operatorname{Re} \left(\frac{t}{x} \right) = \frac{\rho_t}{\rho_x} \operatorname{Re} \left(\frac{e^{i\theta_t}}{e^{i\theta_x}} \right) = \frac{\rho_t}{\rho_x} \operatorname{Re} \left(e^{i(\theta_t - \theta_x)} \right) = 0 \iff \theta_t - \theta_x = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

Note that such rays corresponds to the sign change for $\operatorname{Re}(t/x)$. Consider now the bisectors rays of sectors $V^0 \cap V^1$ and let us call them *Stokes directions*. They corresponds to the solutions of the equation $\theta_t - \theta_x = k\pi$, and hence $\operatorname{Re}(t/x) = \pm \rho_t / \rho_x$. In particular on one Stokes direction the solution $y(x)$ tends to infinity, while, on the other, to zero.

The previous remark explains why solutions cannot converge on the whole punctured disk.

Irregular monodromy is also more complicated: passing to a sector from another also generates monodromy. Details on irregular monodromy and Stokes phenomena

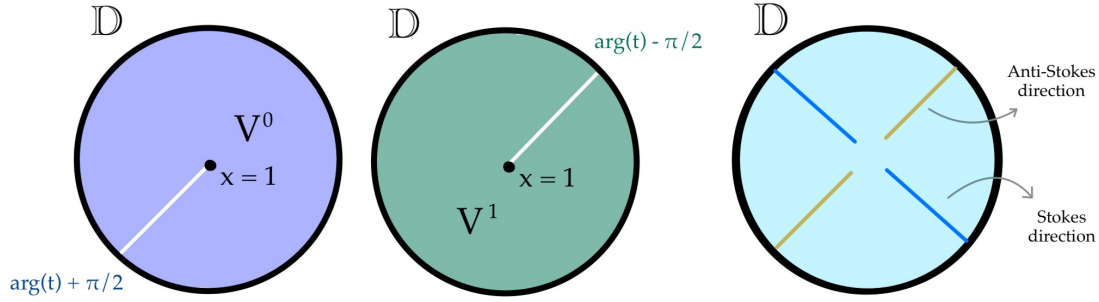


Figure 4.2 – Stokes sectors and directions.

are beyond this thesis end we refer to paragraph 21.1 of [28] and section 1.3 of [38] for a detailed description.

Let us then jump to two fundamental results describing the local irregular monodromy. We recall that in Section 1.3.3 we discussed the Turrutin-Levelt form for an irregular local connection. With a holomorphic gauge transformation, it is always possible to achieve a diagonal principal part, but it is in general not possible to kill the holomorphic part.

Lemma 4.1.1. *Let $(E, 2[0], \nabla)$ an irregular connection on the disk. Let the connection matrix Ω be such that its principal part is already diagonal. Consider $\{\kappa^+, \kappa^-\}$ the residual spectral data. The monodromy relative to a loop γ encircling the singularity $x = 0$ is*

$$M^\gamma := S_0 \cdot S_1 \cdot D = \begin{pmatrix} 1 & c_0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ c_1 & 1 \end{pmatrix} \begin{pmatrix} e^{2i\pi\kappa^+} & 0 \\ 0 & e^{2i\pi\kappa^-} \end{pmatrix}$$

for some complex numbers c_0, c_1 . Matrices S_0 and S_1 are called Stokes matrices.

Proof. Proposition 1.3.8 of [38]. □

As the following result shows, Stokes coefficients c_0, c_1 are the obstructions to analytically get to the Turrutin-Levelt form defined in Section 1.3.3.

Proposition 4.1.2. *An irregular connection $(E, 2[0], \nabla)$ on the disk is holomorphically gauge equivalent to its diagonal principal form (meaning that the formal transformation in Turrutin-Levelt Theorem 1.3.10 is convergent) if, and only if, $c_0 = c_1 = 0$.*

Proof. Theorem 1.3.6 of [38]. □

We are now ready to build the local irregular monodromy representation. In particular we associate to each path in the local fundamental groupoid of Section 4.1.1 its monodromy matrix. Let us give in Figure 4.3 a presentation to the groupoid, as in [38].

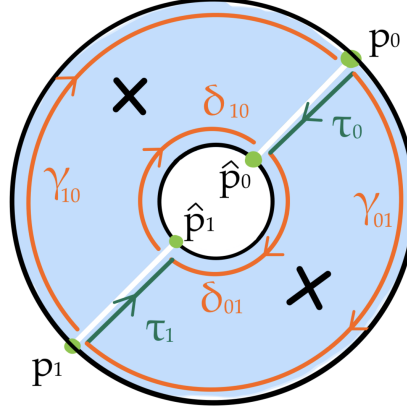


Figure 4.3 – Local fundamental groupoid.

In particular, we can associate a monodromy matrix to each path. We denote by $\star$ the concatenation of loops, with the convention that $\gamma \star \delta$ means “first follow γ , then δ ”.

$$\gamma = \gamma_{0,1} \star \tau_1 \star \delta_{0,1}^{-1} \star \tau_0^{-1} \quad \rightsquigarrow \quad M^\gamma = \begin{pmatrix} 1 & c_0 \\ 0 & 1 \end{pmatrix} =: S_0$$

$$\gamma = \gamma_{1,0} \star \tau_0 \star \delta_{1,0}^{-1} \star \tau_1^{-1} \quad \rightsquigarrow \quad M^\gamma = \begin{pmatrix} 1 & 0 \\ c_1 & 1 \end{pmatrix} =: S_1$$

$$\gamma = \delta_{1,0} \star \delta_{0,1} \quad \rightsquigarrow \quad M^\gamma = D$$

$$\gamma = \gamma_{0,1} \star \gamma_{1,0} \quad \rightsquigarrow \quad M^\gamma =: S_0 \cdot S_1 \cdot D$$

giving the local monodromy representation

$$\rho_{\mathbb{D}}: \pi_1(\hat{\mathbb{D}} \setminus \{s_0, s_1\}, \{p_0, p_1, \hat{p}_0, \hat{p}_1\}) \rightarrow \mathrm{GL}_2(\mathbb{C}).$$

We would like to describe the global monodromy representation for a PV connection. To do so, we first need to define the global wild fundamental groupoid, that is constructed as follows in [21]. Consider $\mathbb{P}^1$ and perform a real blowing up at the points

0, 1 and ∞ . Let us denote by $\hat{\mathbb{P}}^1$ this new curve, and by D_0, D_1 and D_∞ the exceptional divisors. We recall that, since we are performing a real blow-up, they are all isomorphic to a circle. Remove then two points s_1^+ and s_1^- close to D^1 , lying on two opposite Stokes lines. Choose then four base points for the fundamental groupoid, $p_0 \in D_0, p_\infty \in D_\infty$ and two points $p_1, p_2 \in D_1$, lying on two opposite anti-Stokes lines. Finally, consider the groupoid whose morphisms are the paths between these four base points up to homotopy:

$$\pi_1^V := \pi_1(\hat{\mathbb{P}}^1 \setminus \{s_1^+, s_1^-\}, \{p_0, p_\infty, p_1, p_2\}).$$

It is the *wild fundamental groupoid* associated to any PV connection. In Figure 4.4 we show a set of generators for π_1^V .

The relations between the generators of this presentation are generated by the follow-

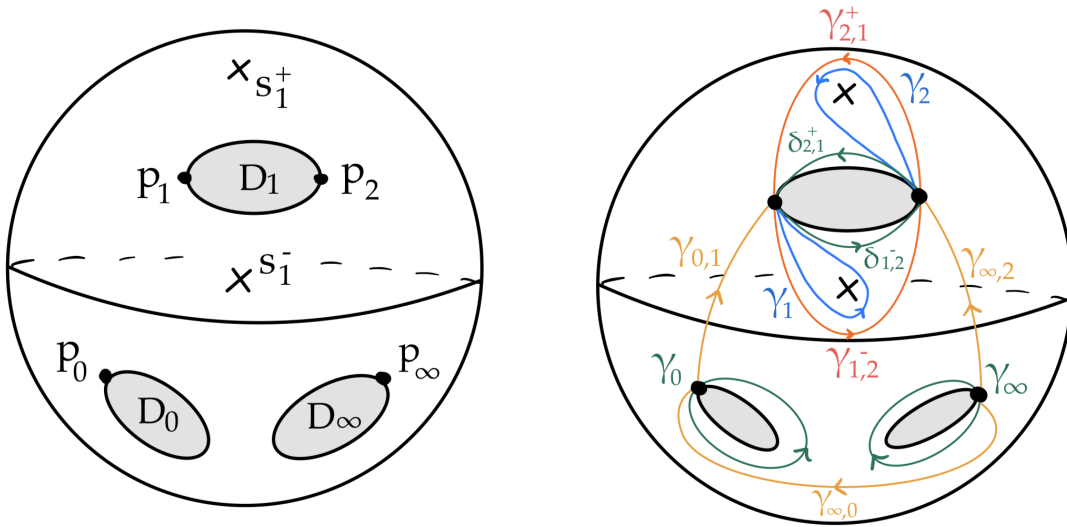


Figure 4.4 – Wild fundamental groupoid.

ing:

- $(\gamma_{2,1}^+)^{-1} \star (\gamma_{\infty,2})^{-1} \star \gamma_{\infty,0} \star \gamma_{0,1} = 1,$
- $\gamma_{1,2}^- \star (\gamma_{\infty,2})^{-1} \star \gamma_\infty \star \gamma_{\infty,0} \star \gamma_0 \star \gamma_{0,1} = 1$
- $\gamma_1 \star \delta_{1,2}^- \star \gamma_2 \star \delta_{2,1}^+ = \gamma_{1,2}^- \star \gamma_{2,1}^+$

We have now enough data to study the monodromy representation of a PV connection:

$$\begin{aligned}\rho_{\nabla}: \pi_1^V &\rightarrow \mathrm{GL}_2(\mathbb{C}) \\ [\gamma] &\mapsto M^\gamma.\end{aligned}$$

We recall that a basis change for E produces another monodromy representation conjugated to the initial one. In particular, up to basis changes, we can always achieve that

$$\rho_{\nabla}(\gamma_{\infty,2}) = \rho_{\nabla}(\gamma_{\infty,0}) = \rho_{\nabla}(\gamma_{0,1}) = \mathrm{Id}.$$

Note that this implies also that $\rho_{\nabla}(\gamma_{2,1}^+) = \mathrm{Id}$.

Definition. A representation of $\rho: \pi_1^V \rightarrow \mathrm{GL}_2(\mathbb{C})$ is said *normalized* if $\rho(\gamma_{\infty,2}) = \rho(\gamma_{\infty,0}) = \rho(\gamma_{0,1}) = \rho(\gamma_{2,1}^+) = \mathrm{Id}$.

We would like to keep the same structure that we found in the local fundamental groupoid.

Definition. A normalized representation of $\rho: \pi_1^V \rightarrow \mathrm{GL}_2(\mathbb{C})$ is said *admissible* if

- $\rho(\gamma_1) := S_1$ is upper triangular,
- $\rho(\delta_{1,2}^- \star \gamma_2 \star (\delta_{1,2}^-)^{-1}) := S_2$ is lower triangular,
- $\rho(\delta_{1,2}^- \star \delta_{2,1}^+) := D$ is diagonal.

Remark 38. We find back what we expected from the local fundamental groupoid. Indeed

$$\begin{aligned}\rho(\gamma_{1,2}^- \star \gamma_{2,1}^+) &= \rho\left(\gamma_1 \star \delta_{1,2}^- \star \gamma_2 \star \delta_{2,1}^+\right) = \\ &= \rho\left(\gamma_1 \star \delta_{1,2}^- \star \gamma_2 \star (\delta_{1,2}^-)^{-1} \star \delta_{1,2}^- \star \delta_{2,1}^+\right) = \\ &= \rho(\gamma_1) \cdot \rho(\delta_{1,2}^- \star \gamma_2 \star (\delta_{1,2}^-)^{-1}) \cdot \rho(\delta_{1,2}^- \star \delta_{2,1}^+) = \\ &= S_1 \cdot S_2 \cdot D.\end{aligned}$$

We denote by $\mathrm{Rep}_{\Theta}^V(\pi_1^V, \mathrm{GL}_2(\mathbb{C}))$ the set of normalized and admissible representations. Let us finally recall that we consider monodromy up to equivalence, that is up to global conjugation by a diagonal matrix, in order to preserve the admissibility.

Definition. The *wild character variety* associated to the fundamental groupoid of a rational irregular curve $(\mathbb{P}^1, D^V, J^V)$ is the categorical quotient

$$\chi_{\Theta}^V := \text{Rep}_{\Theta}^V \left(\pi_1^V, \text{GL}_2(\mathbb{C}) \right) // \mathcal{D},$$

where $\mathcal{D}$ is the subgroup of $\text{GL}_2(\mathbb{C})$ composed by diagonal matrices.

In Section 3.2 of [48], the interested reader can find a geometric description of χ_{Θ}^V .

4.1.3 The Correspondence

The Riemann-Hilbert correspondence for regular singular connections has been proved by Deligne [15] in 1970. The irregular case has been deeply studied by P.P. Boalch [5] in the framework of meromorphic connections. In 2024, A. D’Agnolo and M. Kashiwara proved the irregular Riemann-Hilbert correspondence in the more general setting of D -modules. We state it for PV connections in the following way:

Theorem 4.1.3 (Irregular RH for PV connections). *Let us consider the divisor $D^V := [0] + 2[1] + [\infty]$, and fix the set of jets $J_{t_0}^V = \{0, (1, t_0), \infty\}$ for a fixed complex number $t_0 \in \mathbb{C}^*$. The moduli space of PV connections on the rational irregular curve $(\mathbb{P}^1, D^V, J_{t_0}^V)$, with fixed generic spectral data Θ , is analytically isomorphic to the wild character variety χ_{Θ}^V for any choice of $t_0 \in \mathbb{C}^*$.*

Proof. See, for instance, Section 1.3.4 of [38]. □

Remark 39. The moduli space of connections on $(\mathbb{P}^1, D^V, J_{t_0}^V)$ is just the Okamoto space $\text{Con}_{\Theta}^V \cap \{t = t_0\}$. It corresponds to the space of initial conditions for the fifth Painlevé equation and it is isomorphic to a 8-blow-up of the Hirzebruch surface $\mathbb{F}_2$ as in Figure 2.4.

Remark 40. The fact that we consider spectral data $\exp(2i\pi\Theta)$ is the reason why we are interested to consider Θ just up to integer shifts.

In other words, the irregular Riemann-Hilbert correspondence establish an isomorphism between the Okamoto spaces and the wild character variety χ_{Θ}^V .

4.2 Painlevé V Foliation

One can wonder, given a generic class of PV connections, if we can deform it (that is, deform the parameters $(t, q, \hat{p})$) without changing the monodromy. The irregular Riemann-Hilbert correspondence and the Painlevé property suggest that we will never have a positive answer until we keep t constant. We indeed know, by Theorem 3.3.19, that the moduli space $\overline{\mathfrak{Con}}_{\Theta}^V$ comes with a natural fibration $\pi: \overline{\mathfrak{Con}}_{\Theta}^V \xrightarrow{t} \mathbb{P}^1$, decomposing it in an union of Okamoto spaces, that are the spaces of initial conditions for the Painlevé V equation, and the hypersurfaces corresponding to $t = 0, \infty$. Theorem 4.1.3 states that each Okamoto space is analytically isomorphic to χ_{Θ}^V . We would then like to patch these isomorphisms together to construct a "global" Riemann-Hilbert map $RH: \overline{\mathfrak{Con}}_{\Theta}^V \rightarrow \chi_{\Theta}^V$, but it turns out not to be possible for several reasons: first of all, we cannot define it on the boundary components $\pi^{-1}(0)$ and $\pi^{-1}(\infty)$. Moreover, even on $\mathfrak{Con}_{\Theta}^V := \pi^{-1}(\mathbb{C}^*)$, the Riemann-Hilbert map is not globally well defined. A monodromy action, coming from the topology of $\mathbb{C}^*$, arise, and therefore the Riemann-Hilbert map is well defined only as $RH: \pi^{-1}(U) \rightarrow \chi_{\Theta}^V$ for any simply connected open set $U \subseteq \mathbb{C}^*$.

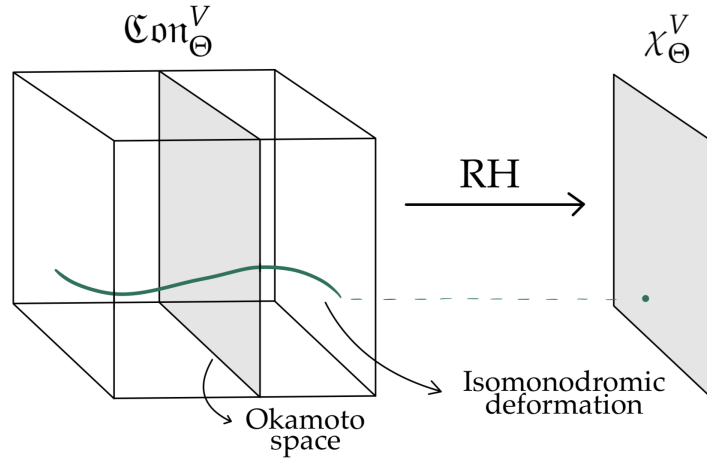


Figure 4.5 – Isomonodromic foliation induced by the Riemann-Hilbert map.

In 1981 Jimbo, Miwa and Ueno [31] shown that the fibers of the RH map are parametrised, in the dense open set $\mathcal{M} \times \mathbb{C}$, as solutions of a very famous differential equation: the

Painlevé V equation

$$\ddot{q}(t) = \left(\frac{1}{2q(t)} + \frac{1}{q(t)-1} \right) \dot{q}(t)^2 - \frac{1}{t} \dot{q}(t) + \frac{(q(t)-1)^2}{t^2} \left(\alpha q(t) + \frac{\beta}{q(t)} \right) + \gamma \frac{q(t)}{t} + \delta \frac{q(t)(q(t)+1)}{q(t)-1} \quad (4.1)$$

where

- (t, q) are the coordinates of $\mathcal{M} \subseteq \overline{\mathfrak{Con}}_{\Theta}^V$,
- the fixed parameters $\{\alpha, \beta, \gamma, \delta\}$ can be expressed in terms of $\Theta = \{\kappa_0, \kappa_1, \kappa_{\infty}\}$.

The symmetries of the moduli space $\overline{\mathfrak{Con}}_{\Theta}^V$ described in Section 3.3.4 can be found as symmetries of this differential equation.

4.2.1 Hamiltonian System

Associated to the fifth Painlevé equation, there is an Hamiltonian function

$$H^V := \frac{q(q-1)^2 p^2 - \left(\kappa_0 (q-1)^2 + (\kappa_1 - 1)q(q-1) - tq \right) p + \rho^V (q-1)}{t}$$

as described in [46], Section 4.1.

If we fix a basis $\{\partial_t, \partial_q, \partial_p\}$ of the tangent bundle $T(\mathcal{M} \times \mathbb{C})$, we can deduce the hamiltonian vector field

$$X^V := \partial_t + \frac{\partial H^V}{\partial p} \partial_q - \frac{\partial H^V}{\partial q} \partial_p$$

whose integral curves correspond to the solutions of the fifth Painlevé equation. We recall that a complex curve $\gamma: \mathbb{C} \rightarrow \mathcal{M} \times \mathbb{C}$, given by the expression $\gamma(t) = (t, q(t), p(t))$, is integral for X^V if it satisfies

$$\dot{\gamma}(t) = X^V(\gamma(t)) \quad \text{that is} \quad \begin{pmatrix} 1 \\ \dot{q}(t) \\ \dot{p}(t) \end{pmatrix} = \begin{pmatrix} 1 \\ \partial_p H^V(\gamma(t)) \\ -\partial_q H^V(\gamma(t)) \end{pmatrix}.$$

Remark 41. We can find back the fifth Painlevé equation via the following procedure: since $\partial_p H^V$ is linear in p , via the identity $\dot{q} = \partial_p H^V$ we can deduce a formula for p . Substituting this value in the identity $\dot{p} = -\partial_q H^V$ we get the fifth Painlevé equation as desired.

Remark 42. The relation of the parameter $\hat{p}$ appearing in [19, 45], and the parameter p appearing in [46] for the hamiltonian function is the following:

$$\hat{p} = -pq(q-1)^2.$$

Moreover, if we call $\varphi_p: \hat{p} \mapsto p$ this coordinate change, we get a strong relation between the hamiltonian H^V and the parameter $\hat{K}$ appearing in the normal form 2.1:

$$\hat{K} = t \cdot (H^V \circ \varphi_p) + \rho^V + \frac{\hat{p}}{q-1}.$$

We stress that the fifth Painlevé equation is defined by this formula only in the Zariski open set $\mathcal{M} \times \mathbb{C} \subseteq \overline{\mathfrak{Con}}_{\Theta}^V$. As said in Section 3.1, $\mathcal{M} \times \mathbb{C}$ corresponds to the moduli space of such PV connections with $t \neq 0, \infty$ and $q \neq 0, 1, \infty$. We remark that in $\mathcal{M} \times \mathbb{C}$ the foliation induced by the hamiltonian vector field is regular. The goal of this work is then to understand the behavior of the foliation in the boundary components of $\overline{\mathfrak{Con}}_{\Theta}^V$.

4.2.2 Extension of the Vector Field and First Integrals

We will refer to the boundary components of $\overline{\mathfrak{Con}}_{\Theta}^V$ as in Figure 3.16. Our goal is to extend the hamiltonian vector field in coordinates adapted to these boundary components and to identify first integrals that help describe the behavior of the foliation. The main idea can be resumed as follows: consider a leaf of the isomonodromic foliation and follow it as it approaches a boundary component. In this limit, the irregular curve degenerates to a nodal curve with two smooth components, and the corresponding PV connection splits accordingly into two connections, one on each component. Both connections acquire a pole in the node of the curve. The local monodromy at the node should, in some sense, encode the monodromy of the original connection and therefore remain invariant along the isomonodromic leaf. In other words, we expect that the residual spectral datum in the node (that we computed in Section 3.3.1) provides a first integral for the hamiltonian vector field. We recall that we will use the following coordinates:

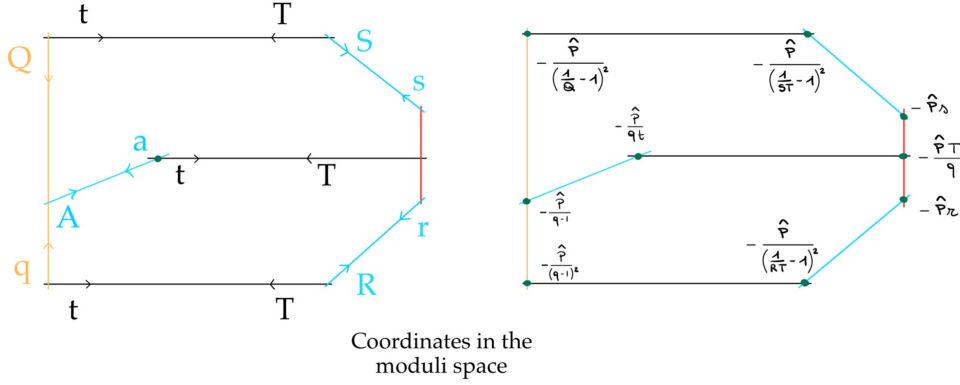


Figure 4.6 – Coordinates of the moduli space.

In particular:

$$T = \frac{1}{t}, \quad Q = \frac{1}{q}, \quad A = \frac{t}{q-1} = \frac{1}{a}, \quad S = \frac{T}{Q} = \frac{1}{s}, \quad R = \frac{q}{T} = \frac{1}{r}$$

For simplicity we call:

$$P^0 = -\frac{\hat{p}}{(q-1)^2} = -\frac{\hat{p}}{(RT-1)^2}, \quad P^1 = -\frac{\hat{p}}{qt} = -\frac{\hat{p}T}{q} = -\hat{p}s = -\hat{p}r,$$

$$P^\infty = -\frac{\hat{p}}{(\frac{1}{Q}-1)^2} = -\frac{\hat{p}}{(\frac{1}{ST}-1)^2}, \quad P_0^1 = -\frac{\hat{p}}{q-1}.$$

We proceed case by case.

We first study Q^0, Q^1 and Q^∞ . Recall that the line bundle $\mathcal{O}_{\widetilde{\mathcal{M}}}(D)$ is trivial when restricted to each of these components. From Okamoto [47] and from computations in Section 3.3.1 we deduce that no connections lie in the Q^i 's but rather on the special sections we blow up. Nevertheless, we wish to understand the behavior of the isomonodromic foliation along these components. We will resume in Theorem 4.2.1 the main results.

Vector Field on Q^0 . First of all we need to extend the hamiltonian vector field to Q^0 . Let us define f_{Q^0} the coordinate change $(t, q, p) \mapsto (t, q, P^0)$. We define

$$X_{Q^0}^V(t, P^0) := \lim_{q \rightarrow 0} (f_{Q^0*} X^V)(t, q, P^0) = P^0(P^0 - \kappa_0) \partial_{P^0},$$

the limit vector field restricted to the hypersurface Q^0 . It is a vertical vector field that is singular along the two special sections $P^0 = 0$ and $P^0 = \kappa_0$, as we expected. We recall that on such exceptional divisor, that we call $\mathcal{E}_\pm^0$, we set respectively the coordinates $\alpha_\pm^0$ given by the expression

$$\alpha_-^0 = \frac{q}{P^0} = -\frac{q(q-1)^2}{\hat{p}} = p \quad \text{and} \quad \alpha_+^0 = \frac{q}{P^0 - \kappa_0}.$$

Changing now coordinates of X^V by $(t, q, p) \rightarrow (t, q, \alpha_\pm^0)$ we get the following vector fields

$$\begin{aligned} X_{\mathcal{E}_-^0}^V(t, q, \alpha_-^0) &= t\partial_t \\ &+ [2\alpha_-^0 q^3 - (4\alpha_-^0 + \kappa_0 + \kappa_1 - 1)q^2 + (t + 2\alpha_-^0 + 2\kappa_0 + \kappa_1 - 1)q - \kappa_0]\partial_q \\ &+ [(-3q^2 + 4q - 1)(\alpha_-^0)^2 + (-2(\kappa_0 - \kappa_1 + 1)q - t - 2\kappa_0 - \kappa_1 + 1)\alpha_-^0 + \\ &+ (\kappa_1 - 1)\kappa_0 - \rho]\partial_{\alpha_-^0} \end{aligned}$$

$$\begin{aligned} X_{\mathcal{E}_+^0}^V(t, q, \alpha_+^0) &= t\partial_t \\ &+ [2\alpha_+^0 q^3 + (-4\alpha_+^0 + \kappa_0 - \kappa_1 + 1)q^2 + (t + 2\alpha_+^0 - 2\kappa_0 + \kappa_1 - 1)q + \kappa_0]\partial_q \\ &+ [(-3q^2 + 4q - 1)(\alpha_+^0)^2 + (-2(\kappa_0 - \kappa_1 + 1)q - t + 2\kappa_0 - \kappa_1 + 1)\alpha_+^0 + \\ &+ (\kappa_1 - 1)\kappa_0 - \rho]\partial_{\alpha_+^0} \end{aligned}$$

As computed in Section 3.3.1, the residual spectral datum in the node is, respectively $(\kappa_0 \pm 1)^2$, that, being constant, gives no information about the isomonodromic foliation. What it is important to notice is that the exceptional divisor is not invariant, since $X_{\mathcal{E}_\pm^0}^V|_{\mathcal{E}_\pm^0}$ has a non zero coefficient for ∂_q , and also that the coefficient of ∂_t is a unity for $t \in \mathbb{C}^*$, that is part of the Painlevé property.

Vector Field on Q^1 . Let us define f_{Q^1} the coordinate change $(t, q, p) \mapsto (t, a, P^1)$. The hamiltonian vector field restricted to Q^1 is then

$$X_{Q^1}^V(t, P^1) = \lim_{a \rightarrow 0} (f_{Q^1*} X^V)(t, q, P^1) = 2P^1(P^1 + 1)\partial_{P^1},$$

that, as expected, is vertical and presents singularities along the special sections $P^1 = 0, -1$.

As for Q^0 , we consider the vector field in the coordinates of the exceptional divisors. We recall that, as we can see in Figure 3.16, over Q^1 we have to blow-up twice the special sections. We call $\mathcal{E}_\pm^1$ the exceptional divisors that are part of the moduli space, that are arising from the second blow up. We call $\mathcal{F}_\pm^1$ the intermediate ones. We will call by $\beta_\pm^1$ the coordinates on $\mathcal{F}_\pm^1$ and by $\alpha_\pm^1$ the coordinates on $\mathcal{E}_\pm^1$. They are defined as follows:

$$P^1 = \beta_-^1(q-1) \quad \text{and} \quad P^1 = \frac{\alpha_-^1}{t^2}(q-1)^2,$$

that means that we blow-up the line $\beta_-^1 = 0$ to obtain $\mathcal{E}_-^1$. And

$$P^1 = \beta_+^1(q-1) - 1 \quad \text{and} \quad P^1 = \frac{\alpha_+^1}{t^2}(q-1)^2 + \frac{\kappa_1}{t} - 1,$$

that means that we blow up the line $\beta_-^1 = \kappa_1/t$ to obtain $\mathcal{E}_+^1$.

For the first exceptional divisor, we change the coordinates of X^V by $(t, q, p) \rightarrow (t, q, \beta_\pm^1)$ getting the following vector fields

$$(X_{\mathcal{F}_-^1}^V)_{|\mathcal{F}_-^1}(t, \beta_-^1) = \beta_-^1 t^2 \partial_{\beta_-^1}$$

$$(X_{\mathcal{F}_+^1}^V)_{|\mathcal{F}_+^1}(t, \beta_+^1) = t(-\beta_+^1 t + \kappa_1) \partial_{\beta_+^1}.$$

We remark that they are stationary and vanish on the lines $\beta_-^1 = 0$ and $\beta_+^1 = \kappa_1/t$, that are exactly the lines we want to blow-up. On the second exceptional divisor we get then to the vector fields, restricted on them, have the following expression

$$\begin{aligned} X_{\mathcal{E}_-^1}^V(t, q, \alpha_-^1) &= t^2 \partial_t \\ &+ [2\alpha_-^1 q^3 + ((-\kappa_0 - \kappa_1 + 1)t - 4\alpha_-^1)q^2 - (t^2 - (2\kappa_0 + \kappa_1 - 1)t - 2\alpha_-^1)q - \kappa_0 t] \partial_q \\ &+ [(-3q^2 + 4q - 1)(\alpha_-^1)^2 + (t + 2q(\kappa_0 + \kappa_1 - 1) - 2\kappa_0 - \kappa_1 + 2)t\alpha_-^1 - \\ &- \rho t^2] \partial_{\alpha_-^1} \end{aligned}$$

$$\begin{aligned} X_{\mathcal{E}_+^1}^V(t, q, \alpha_+^1) &= t^2 \partial_t \\ &+ [2\alpha_+^1 q^3 + ((\kappa_1 - \kappa_0 + 1)t - 4\alpha_+^1)q^2 - (t^2 - (2\kappa_0 - \kappa_1 - 1)t - 2\alpha_+^1)q - \kappa_0 t] \partial_q \\ &+ [(-3q^2 + 4q - 1)(\alpha_+^1)^2 + (t + 2q(\kappa_0 - \kappa_1 - 1) - 2\kappa_0 + \kappa_1 + 2)t\alpha_+^1 + \\ &+ ((\kappa_0 - 1)\kappa_1 - \rho)t^2] \partial_{\alpha_+^1} \end{aligned}$$

As computed in Section 3.3.1, the residual spectral data in the node are, respectively, $(\kappa_1 \pm 1)^2$, that, being constant, gives no information about the isomonodromic foliation. What it is important to notice is that the exceptional divisor is not invariant, since $X_{\mathcal{E}_{\pm}^1}^V|_{\mathcal{E}_{\pm}^1}$ has a non zero coefficient for ∂_q , and also that the coefficient of ∂_t is a unity for $t \in \mathbb{C}^*$, that is part of the Painlevé property.

Vector Field on Q^∞ . Thanks to the symmetries of the moduli space, and of the PV equation, we know that we get the same behavior as on Q^0 .

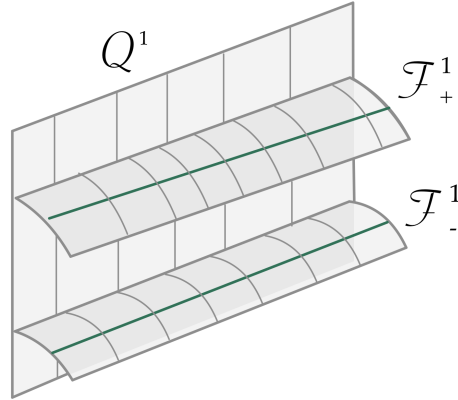


Figure 4.7 – Isomonodromic foliation on the total transform of Q^1 .

Before passing to the study of the isomonodromic foliation on the $\mathcal{B}_j^i$, let us resume what we found in the following result.

Theorem 4.2.1. *The Q^i s are invariant for the isomonodromic foliation, moreover the hamiltonian vector field is therein vertical, that is, parallel to ∂_{pi} . For $q = 1$, also the intermediate exceptional divisor $\mathcal{F}_{\pm}^1$ is invariant, and the foliation is vertical. On the other hand, the exceptional divisors $\mathcal{E}_{\pm}^i$ are, as expected [26], transverse to the foliation.*

Also the hypersurfaces $\mathcal{B}_j^i$ turn out to be invariant for the isomonodromic foliation. Since the line bundle is not trivial in restriction to them, we are not able to find a global vertical coordinate, as we did for the Q^i s. We remark that for any i , the intersection $\mathcal{B}_j^i \cap Q^i$ is a projective line. In particular we are interested in studying the behavior of the foliation near the intersection of $\mathcal{B}_j^i$ and the special sections of Q^i .

Vector Field and First integral on $\mathcal{B}_0^1$. Let us define $f_{\mathcal{B}_0^1}$ the coordinate change $(t, q, p) \mapsto (A, \tilde{q}, P_0^1)$, where $\tilde{q} = q - 1$, centered in $A_0 \cap B_0^1$, and $g_{\mathcal{B}_0^1}$ the coordinate

change $(t, q, p) \mapsto (t, a, P^1)$, centered in $Q^1 \cap B_0^1$. The hamiltonian vector field restricted to $\mathcal{B}_0^1$ is then

$$X_{\mathcal{B}_0^1}^V(A, P_0^1) = \lim_{\tilde{q} \rightarrow 0} (f_{\mathcal{B}_0^1*} X^V)(A, \tilde{q}, P_0^1) = -AP_0^1 \partial_{P_0^1} + (A + 2P_0^1 - \kappa_1) \partial_A,$$

or

$$X_{\mathcal{B}_0^1}^V(a, P^1) = \lim_{t \rightarrow 0} (g_{\mathcal{B}_0^1*} X^V)(t, a, P^1) = a(-\kappa_1 a + 2P^1 + 1) \partial_a - P^1(\kappa_1 a + 2P^1 + 2) \partial_{P^1}.$$

We remark that the vector field becomes vertical at the intersection with $Q^1 = \{a = 0\}$. The residual spectral data in the node are

$$(\kappa_1 + 1)^2 + 4P_0^1(A + P_0^1 - \kappa_1) \quad \text{and} \quad (\kappa_1 + 1)^2 - \frac{4P^1 \kappa_1}{a} + \frac{4P^1(P^1 + 1)}{a^2}.$$

Proposition 4.2.2. *The residual spectral data are first integrals for the respective vector fields.*

Proof. A computation shows

$$X_{\mathcal{B}_0^1}^V(A, P_0^1) \left((\kappa_1 + 1)^2 + 4P_0^1(A + P_0^1 - \kappa_1) \right) = 0,$$

and

$$X_{\mathcal{B}_0^1}^V(a, P^1) \left((\kappa_1 + 1)^2 - \frac{4P^1 \kappa_1}{a} + \frac{4P^1(P^1 + 1)}{a^2} \right) = 0.$$

By the definition of first integral, we conclude. $\square$

Proposition 4.2.3. *The isomonodromic leaves in $\mathcal{B}_0^1$ are contained in the level sets of the first integrals and they form a pencil of conics.*

Proof. The first part of the theorem is trivial. In order to state that the curves are of genus zero we use the Riemann-Hurwitz formula. We consider the level sets

$$C_b := \{(\kappa_1 + 1)^2 + 4P_0^1(A + P_0^1 - \kappa_1) = b\}$$

for $b \in \mathbb{C}$. They are curves in $\mathcal{B}_0^1 \cong \mathbb{F}_1$. The projection $\pi: \mathbb{F}_1 \rightarrow \mathbb{P}^1$ restricted to C_b is a ramified covering. The branching point are the roots of the discriminant

$$\frac{\Delta_b}{16} = (A - \kappa_1)^2 + b - (\kappa_1 + 1)^2 = A^2 - 2\kappa_1 A + b - 2\kappa_1 + 1.$$

For a generic value of b there are two roots defining the two branching points of the covering. We then know that $R = \infty$ is not a branching point since they must be even in number. Applying the Riemann-Hurwitz formula we get then

$$2g_{C_b} - 2 = -4 + \sum_{p \in C_a} (e_p - 1) = -4 + 2 = -2$$

deducing $g_{C_b} = 0$ as desired. $\square$

A study of the equation for the first integral around $B_0^1 \cap Q^1$ indeed shows that each conic is tangent at the two points $(t, a, P^1) = (0, 0, 0)$ and $(t, a, P^1) = (0, 0, -1)$, that coincide with the intersections between the two special sections in Q^1 and the hypersurface $\mathcal{B}_0^1$. Their tangent slopes at these points are 0 and κ_1 , respectively. These values correspond to the coordinates of the sections we blow up in the intermediate divisors $\mathcal{F}_\pm^1$ in Q^1 .

The foliation is shown in Figure 4.8

Vector Field and First integral on $\mathcal{B}_\infty^0$. Let us define $f_{\mathcal{B}_\infty^0}$ the coordinate change $(t, q, p) \mapsto (T, R, P^0)$ and $g_{\mathcal{B}_\infty^0}$ the coordinate change $(t, q, p) \mapsto (r, q, P^1)$. The hamiltonian vector field restricted to $\mathcal{B}_\infty^0$ is then

$$X_{\mathcal{B}_\infty^0}^V(R, P^0) = \lim_{T \rightarrow 0} (f_{\mathcal{B}_\infty^0}^* X^V)(T, R, P^0) = -R(2P^0 + R - \kappa_0)\partial_R - P^0(P^0 - \kappa_0)\partial_{P^0},$$

or

$$X_{\mathcal{B}_\infty^0}^V(r, P^1) = \lim_{q \rightarrow 0} (g_{\mathcal{B}_\infty^0}^* X^V)(r, q, P^1) = -r(2P^1 - \kappa_0 r + 1)\partial_r - P^1(P^1 + 1)\partial_{P^1}.$$

We remark that the vector field becomes vertical at the intersection with $Q_0 = \{R = 0\}$ and $\mathcal{A}_\infty = \{r = 0\}$ respectively.

In [45] we show that the residual spectral data in the node are not the same for the two curves, but their difference is κ_1 , that is a constant. They are the following:

$$2P^0 - \kappa_0 - 1 + \frac{2P^0(P^0 - \kappa_0)}{R} \quad \text{and} \quad -2P^1\kappa_0 - \kappa_0 - 1 + \frac{2P^1(P^1 + 1)}{r}.$$

We can therefore state the following proposition.

Proposition 4.2.4. *The residual spectral data are first integrals for the respective vector fields.*

Proof. We consider just one of the residual data. A computation shows

$$X_{\mathcal{B}_\infty^0}^V(R, P^0) \left(2P^0 - \kappa_0 - 1 + \frac{2P^0(P^0 - \kappa_0)}{R} \right) = 0,$$

and

$$X_{\mathcal{B}_\infty^0}^V(r, P^1) \left(-2P^1\kappa_0 - \kappa_0 - 1 + \frac{2P^1(P^1 + 1)}{r} \right) = 0.$$

By the definition of first integral, we conclude. $\square$

Proposition 4.2.5. *The isomonodromic leaves in $\mathcal{B}_\infty^0$ are contained in the level sets of the first integrals and they form a pencil of conics.*

Proof. The first part of the theorem is trivial. In order to state that the curves are of genus zero we use the Riemann-Hurwitz formula. We consider the level sets

$$C_a := \{R(2P^0 - \kappa_0 - 1 - a) + 2P^0(P^0 - \kappa_0) = 0\}$$

for $a \in \mathbb{C}$. They are curves in $\mathcal{B}_\infty^0 \cong \mathbb{F}_1$. The projection $\pi: \mathbb{F}_1 \rightarrow \mathbb{P}^1$ restricted to C_a is a ramified covering. The branching point are the roots of the discriminant

$$\frac{\Delta_a}{4} = R^2 - R(3\kappa_0 + 2 + 2a) + \kappa_0^2.$$

For a generic value of a there are two roots defining the two branching points of the covering. We then know that $R = \infty$ is not a branching point since they must be even in number. Applying the Riemann-Hurwitz formula we get then

$$2g_{C_a} - 2 = -4 + \sum_{p \in C_a} (e_p - 1) = -4 + 2 = -2$$

deducing $g_{C_a} = 0$ as desired. $\square$

A study of the equation for the first integral around $B_\infty^0 \cap Q^0$ shows that these conics pass through the two points $(T, R, P^0) = (0, 0, 0)$ and $(T, R, P^0) = (0, 0, \kappa_0)$, that coincide with the intersection points between the two special sections in Q^0 and the hypersurface $\mathcal{B}_\infty^0$.

Vector Field and First integral on $\mathcal{B}_\infty^0$. Thanks to the symmetries of the moduli space, and of the PV equation, we know that we get the same behavior than on $\mathcal{B}_\infty^0$.

We now study the behavior of the foliation in $\mathcal{A}_0$ and $\mathcal{A}_\infty$.

Vector Field and First integral on $\mathcal{A}_0$. Thanks to the symmetry exchanging Q^0 and Q^∞ , we will only study the foliation in a neighborhood of $\mathcal{A}_0 \cap Q^0$ and $\mathcal{A}_0 \cap \mathcal{B}_0^1$. On the first intersection we can apply the same coordinate change we used on Q_0 , that is $f_{Q_0}: (t, q, p) \mapsto (t, q, P^0)$. The hamiltonian vector field restricted to $\mathcal{A}_0$ is then

$$\begin{aligned} X_{\mathcal{A}_0}^V(q, P^0) &= \lim_{t \rightarrow 0} (f_{Q_0*} X^V)(t, q, P^0) = \\ &= 2q(q-1) \left(\left(P^0 - \frac{\kappa_0}{2} \right) (q-1) - \frac{\kappa_1-1}{2} q \right) \partial_q + \\ &\quad + \left((P^0)^2 - P^0(\kappa_0 + \kappa_1 - 1) + \rho \right) q^2 - P^0(-P^0 + \kappa_0) \partial_{P^0}. \end{aligned}$$

We remark that it becomes vertical, as expected, for $q = 0$ and moreover it vanishes for $P^0 = 0, \kappa_0$.

On the other intersection $\mathcal{A}_0 \cap \mathcal{B}_0^1$, we must define a new coordinate change $f_{\mathcal{A}_0}: (t, q, p) \mapsto (A, \tilde{q}, P_0^1)$. The vector field then becomes

$$\begin{aligned} X_{\mathcal{A}_0}^V(\tilde{q}, P_0^1) &= \lim_{A \rightarrow 0} (f_{\mathcal{A}_0*} X^V)(A, \tilde{q}, P_0^1) = \\ &= \tilde{q}(\tilde{q}+1) (\kappa_0 \tilde{q} + (\kappa_1 - 1)(\tilde{q}+1) - 2P_0^1) \partial_{\tilde{q}} + \tilde{q}((\tilde{q}+1)^2 \rho - P_0^1(P_0^1 + \kappa_0)) \partial_{P_0^1}. \end{aligned}$$

The residual spectral data in the node, in the new coordinates, are respectively

$$\begin{aligned} q \left((P^0)^2 - (\kappa_0 + \kappa_1 - 1)P^0 + \rho \right) - 2(P^0)^2 + (2\kappa_0 + \kappa_1 - 1)P^0 - \rho + \left(\frac{\kappa_1}{2} \right)^2 + \frac{P^0(P^0 - \kappa_0)}{q} \\ \frac{4P_0^1(P_0^1 + \kappa_0)}{\tilde{q} + 1} + 4(-\kappa_0 - \kappa_1 + 1)P_0^1 + 4\rho\tilde{q} + \kappa_1^2. \end{aligned}$$

Proposition 4.2.6. *The residual spectral data are first integral for the respective vector fields.*

Proof. By the definition of first integral, we conclude via a direct computation. $\square$

Proposition 4.2.7. *The isomonodromic leaves in $\mathcal{A}_0$ are contained in the level sets of the first integrals and they form a pencil of conics.*

Proof. The first part of the theorem is trivial. The Riemann-Hurwitz formula gives us the second part (as in Proposition 4.2.4). It is indeed easy to see that all the conics

pass through the points $(t, q, P^0) = (0, 0, 0)$ and $(t, q, P^0) = (0, 0, \kappa_0)$, that coincide with the intersection points between the two special sections in Q^0 and the hypersurface $\mathcal{A}_0$. $\square$

We represented in Figure 3.8 the isomonodromic foliation. Grey lines represent respectively the intersections $\mathcal{A}_0 \cap Q^0$ and $\mathcal{B}_0^1 \cap Q^1$.

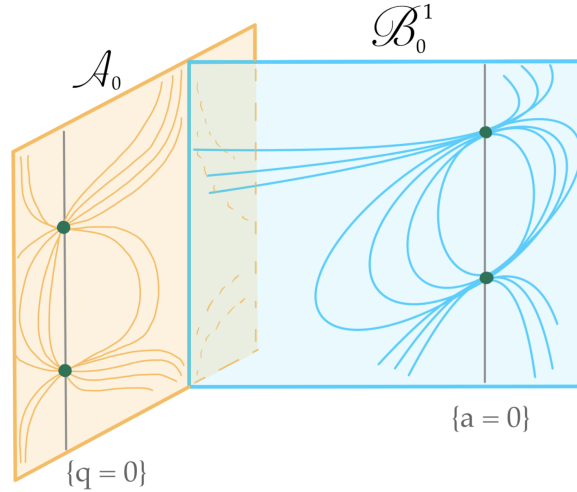


Figure 4.8 – Isomonodromic foliation on $\mathcal{A}_0 \cup \mathcal{B}_0^1$.

Vector Field and First integral on $\mathcal{A}_\infty$. As shown in Section 3.3.1, no PV connection lies on this boundary component, exactly as on the Q^i s. Let us indeed recall that on $\mathcal{A}_\infty$ we had a choice to do regarding the local coordinates.

- The coordinate P^1 , the same that we use on Q^1 , is the one that trivializes the line bundle $\mathcal{O}_{\widehat{\mathcal{M}}}(D)$ over $\mathcal{A}_\infty$. In Section 3.3.1 we have seen that, using P^1 as a parameter for the PV connections, the limit $t \rightarrow \infty$ does not lead to some well defined connection. Moreover, the limit curve that we get under the associated Moebius transformation is not stable.
- We introduced then coordinates $P_\infty^\pm$. Geometrically they correspond to the coordinates on the exceptional divisor arising by the blow up of the sections $P^1 = 0$ and $P^1 = -1$ in $\mathcal{A}_\infty$. With these coordinates, as computed in Section 3.3.1, we get well defined connections and well defined stable nodal curves.

Let us start studying the foliation in the coordinate P^1 in a neighborhood of $\mathcal{A}_\infty \cap \mathcal{B}_\infty^0$. We remark that, since $\mathcal{O}_{\widehat{\mathcal{M}}}(D)|_{\mathcal{A}_\infty}$ is trivial, the coordinate P^1 is globally defined.

Remark 43. Thanks to the symmetry that exchanges Q^0 and Q^∞ , the same behavior will occur near $A_\infty \cap B_\infty^\infty$.

Let us consider the coordinate change $: (t, q, p) \mapsto (r, q, P^1)$. We notice that $\mathcal{A}_\infty$ is defined by setting $r = 0$. We get then the following vector field in restriction

$$\begin{aligned} X_{\mathcal{A}_\infty}^V(q, P^1) &= \lim_{r \rightarrow 0} (g_{\mathcal{B}_\infty^0}^* X^V)(r, q, P^1) = \\ &= -q(q-1)(2P^1+1)\partial_q - (q+1)P^1(P^1+1)\partial_{P^1}. \end{aligned}$$

that is vertical, as expected, for $q = 0$.

Proposition 4.2.8. *A first integral for the isomonodromic foliation is given by*

$$\frac{qP^1(P^1+1)}{(q-1)^2},$$

that is the coefficient appearing in Proposition 3.3.15.

Proof. By the definition of first integral, we conclude via a direct computation. $\square$

Proposition 4.2.9. *The isomonodromic leaves in $\mathcal{A}_\infty$ are contained in the level sets of the first integrals and they form a pencil of elliptic curves.*

Proof. The first part of the theorem is trivial. The Riemann-Hurwitz formula gives us the second part (as in Proposition 4.2.4). It is indeed easy to see that each elliptic curve is tangent to the points $(T, q, P^1) = (0, 1, 0)$ and $(T, q, P^1) = (0, 1, -1)$, that coincide with the intersection points between the two special sections in Q^1 and the hypersurface $\mathcal{A}_\infty$. Their tangent slopes are 0 in both cases, meaning that the tangent lines are parallel to $P^1 = 0$, as shown in Figure 4.9 $\square$

Let us now consider the blow-up of the two sections $P^1 = 0$ and $P^1 = -1$, respectively with coordinates

$$P_\infty^+ = -\frac{\hat{p}}{(q-1)^2} = \frac{P^1 q}{T(q-1)^2} \quad \text{and} \quad P_\infty^- = -\frac{\hat{p}}{(q-1)^2} + \frac{T\kappa_1(q-1) - q}{T(q-1)^2} = \frac{q(P^1+1)}{T(q-1)^2} - \frac{\kappa_1}{q-1}.$$

To compute the vector field restricted to these exceptional divisors, corresponding to setting $T = 0$, we will use the set of coordinates $(T, q, P_\infty^\pm)$, getting to

$$X_{\mathcal{A}_\infty^+}^V(q, P_\infty^+) = -q^2\partial_q$$

$$X_{\mathcal{A}_{\infty}^{-}}^V(q, P_{\infty}^{-}) = q^2 \partial_q$$

that is hence horizontal, as shown in orange in Figure 4.9.

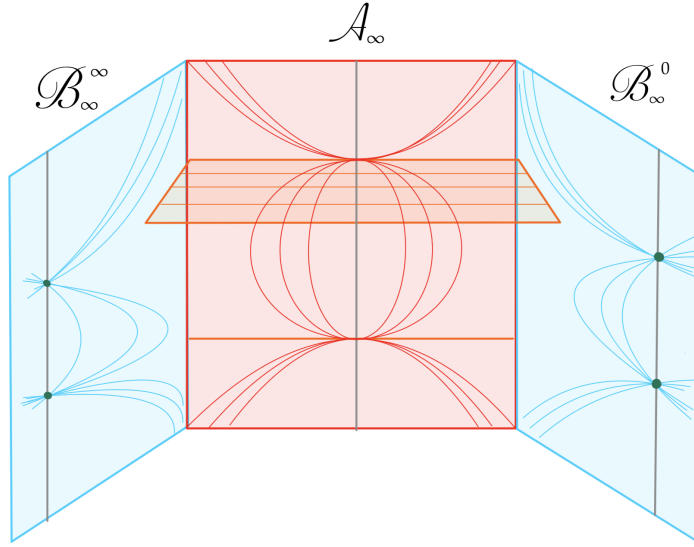


Figure 4.9 – Isomonodromic foliation on $\mathcal{B}_{\infty}^{\infty} \cup \mathcal{A}_{\infty} \cup \mathcal{B}_{\infty}^0$.

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Titre : Compactification du Feuilletage Painlevé V

Mot clés : Déformations isomonodromiques, Équations de Painlevé, Espace des modules des connexions

Résumé : Le but de cette thèse est de compactifier le feuilletage isomonodromique induit par l'équation Painlevé V. En particulier, nous analysons le comportement asymptotique des solutions de cette équation lorsque le paramètre de temps tend vers 0 ou l'infini. Pour ce faire, nous devons d'abord compactifier l'espace ambiant du feuilletage. Cet espace correspond à l'espace de modules d'une certaine classe de connexions méromorphes de rang deux sur la sphère de Riemann, présentant deux singularités régulières, ainsi qu'une singularité irrégulière d'ordre deux. La compactification que nous proposons étend et affine la compactification d'Okamoto en intégrant des

composantes de bord liées au paramètre temporel égal à 0 ou l'infini. Nous interprétons les connexions limites comme des connexions sur des courbes nodales rationnelles irrégulières stables. De telles connexions se décomposent en connexions méromorphes plus simples, une pour chaque composante lisse de la courbe. Les données spectrales résiduelles de ces connexions en chaque noeud fournissent un intégrale première pour le champ Hamiltonien associé à l'équation de Painlevé V. Grâce à ces données, nous pouvons déterminer le comportement du feuilletage sur les composantes de bord.

Title: Compactification of the Painlevé V foliation

Keywords: Isomonodromic deformations, Painlevé equations, Moduli space of connections

Abstract: The goal of this thesis is to compactify the leaves of the isomonodromic foliation induced by the fifth Painlevé equation. In particular, we analyze the asymptotic behavior of the solutions of the fifth Painlevé equation as the time parameter tends to 0 or infinity. To achieve this, we first need to compactify the ambient space of the foliation. This space corresponds to the moduli space of a certain class of rank-two meromorphic connections on the Riemann sphere, featuring two regular singularities, and an irregular singularity of order two. The compactification we present extends and refines the Okamoto compact-

ification by incorporating boundary components related to the time parameter equal to 0 or infinity. We interpret the limit connections as connections on irregular rational stable nodal curves. Such connections decompose into simpler, standard meromorphic connections, one for each smooth component of the curve. The residual spectral data of these connections at the nodes provides a first integral for the Hamiltonian vector field associated with the fifth Painlevé equation. Using this data, we are able to determine the behavior of the foliation on the boundary components.